

Rent Inflation and Monetary Policy

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Abstract

Rental prices enter the inflation indices that central banks target, yet rents behave unlike other consumer prices after a monetary policy shock. We build a Heterogeneous-Agent New Keynesian model where households rent, own, or become landlords, with endogenous house prices and rents, and sticky leases. A monetary tightening keeps would-be buyers renting, and rents rise until enough owners supply the extra rental housing. About three-quarters of the effect of higher interest rates on the rental market is supply-driven: higher financial returns and mortgage costs make supplying rental housing less attractive. We study how policy should treat rent inflation. A Taylor rule based on a price index that incorporates rents generates more consumption and inflation volatility than one which ignores rents. The welfare-optimal weight on rent inflation is negative under general demand, rental supply, and cost-push shocks: central banks should react to rent increases with accommodative policy.

JEL codes: D31, E21, E31, E52, E58, G51, R21, R31.

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1 Introduction

Shelter costs are a large part of the price indices that central banks target. In the euro area, the Harmonised Index of Consumer Prices (HICP) targeted by the European Central Bank (ECB) includes the actual rents paid by tenants, about 6 percent of the basket.¹ In the U.S., the weight of rents is larger because they are used to impute owner-occupied housing costs: they account for about 15 percent of headline PCE and 31 percent of headline CPI (Bolhuis et al., 2022). The post-pandemic episode showed how much these weights matter. Market rents surged from 2022; the shelter components of the indices were still climbing when central banks tightened, and they kept rising long afterwards, as existing leases caught up with market rents.

This paper studies how monetary policy should respond to rent inflation. Rents generate tradeoffs for monetary policy because, empirically, they tend to rise when policy tightens (e.g., Corsetti et al., 2022), unlike other consumer prices. We argue that understanding this empirical regularity requires considering the tenure choice and portfolios of different participants in rental markets. When a tightening lifts the return on financial assets, landlords require a higher rental yield. Higher rates also raise landlords' mortgage costs, a housing-specific form of the cost channel, and keep would-be buyers renting. We study these forces in a Heterogeneous-Agent New Keynesian (HANK) model in which households choose to rent, to own their home, or to become landlords, and can buy housing with a mortgage. Rental leases are sticky: only new contracts reprice at the market rent.

The model delivers three results. First, the supply side dominates: about three-quarters of the rent increase after a monetary tightening comes from higher returns on financial assets and higher mortgage costs for landlords. Second, a rule that targets a rent-inclusive index, as central banks effectively do, keeps reacting to rent pressure of its own making: the consumption contraction becomes deeper and more persistent, and both nondurable and rent inflation become more volatile. Third, within simple Taylor rules, the welfare-optimal response to rent inflation is neither to tighten against it nor to look through it, but to ease: under general demand, rental supply, and cost-push shocks, rent inflation should lower the policy rate relative to a rule that ignores rents. Accommodation lowers the cost of supplying rental housing and relieves poorer renters and mortgage borrowers, at the expense of landlords and savers.

¹ Following its 2021 strategy review, the ECB recommended a roadmap for incorporating owner-occupied housing costs into the index, on the grounds that doing so would better represent the inflation rate relevant for households (European Central Bank, 2021). The ECB's preferred approach is based on net acquisitions, in contrast to the rental-equivalence imputation used in the United States (Lane, 2022). The ECB's 2025 strategy assessment reiterates the Governing Council's support for including owner-occupied housing costs in the HICP (European Central Bank, 2025).

In the model, households facing uninsurable earnings risk decide each period whether to rent, own their home, or own an additional housing unit that they supply to the rental market as landlords. House prices, market rents, residential construction, and the real interest rate are equilibrium objects. Because only new contracts reprice, the model distinguishes the market rent on new leases from the average rent households actually pay, which is the object measured in consumer price indices. On the nominal side, unions set wages subject to Rotemberg adjustment costs, yielding a New Keynesian wage Phillips curve, and the central bank sets the nominal rate with a Taylor rule. Our policy experiments vary whether, and with what weight, rent inflation enters the policy rule, either through the inflation index the rule targets or as a separate term.

We calibrate the model to Spain at quarterly frequency and validate it against untargeted tenure-transition evidence from the Spanish survey of household finances. In Spain, although many mortgages originated between 2015 and 2022 are fixed-rate, most outstanding mortgages are still adjustable-rate ([Banco de España, 2025](#)).

The rent channel of monetary policy. We first characterize how a monetary tightening reaches the rental market. A contractionary monetary policy shock lowers consumption, output, and nondurable inflation, and at the same time raises market rents. The market rent on new leases rises, while the average rent, the component that enters the price index, inherits the increase only as leases turn over: it peaks at about 0.46 percent above its steady-state level eight quarters after a 100-basis-point annualized innovation to the monetary policy rule, long after the shock itself has faded. We show that roughly three-quarters of the effect on the rental market driven by higher interest rates comes from the supply side, with 42 percent explained by the rise in returns on alternative financial assets, which reduces the willingness of current landlords to invest in real estate, and a further 33 percent explained by the increase in mortgage costs for landlords, which also discourages them from providing homes for rent if rents do not increase. Displaced homebuyers who increase their rental demand because of the increased cost of originating mortgages account for only about a sixth. The model generates an impact house-price decline of about 0.4 percent, consistent with the empirical evidence that monetary tightenings lower house prices ([Dias and Duarte, 2022](#); [Lazarowicz and Richard, 2023](#)). This price adjustment dampens rent pressure in the first few quarters, while a quantitatively important reallocation of households across tenure states clears the rental market. On impact, the largest consumption losses fall on levered households, while renters' exposure builds later as leases expire and rents reset.

Rent in the inflation target. We next put measured rent inflation inside the policy rule. We give rent inflation its euro-area HICP weight of 6 percent in the inflation index the rule targets, and hold the economy and the monetary shock fixed, so the comparison with the rent-exclusive rule isolates the policy feedback from measured rent inflation to the path of interest rates. The feedback reshapes the timing of transmission and makes the impact contraction slightly larger. Because the rent component reaches the policy index only as leases turn over, the rule keeps responding to the second-round rent pressure of its own tightening long after the shock has faded; most of the additional contraction therefore accumulates after impact, making the response more persistent. Two years after the shock, the cumulative consumption contraction is about 10 percent larger than under the rent-exclusive rule, and nondurable-inflation volatility about 12 percent larger. The additional contraction does not stabilize rents: rent-inflation volatility is about 19 percent larger.

The optimal response to rent inflation. We let the central bank choose the weight on rent inflation in an additive Taylor rule, which adds a term $\phi_r \pi_t^{\text{rent}}$ to a conventional rule that responds to nondurable inflation; the weight ϕ_r determines the strength of the central bank's response to average rent inflation π_t^{rent} . We pick ϕ_r to maximize utilitarian welfare under general demand, rental availability, and cost-push shocks. The optimal weight is negative under all three: $\phi_r^* = -0.25$ under the demand shock, -0.10 under the rental availability shock, and -0.45 under the cost-push shock. A negative weight means that rent inflation lowers the policy rate relative to a rule that ignores it. The negative sign follows from the transmission mechanism: a lower policy rate reduces the cost of supplying rental housing and leans against the shock that raised rents, whether or not it is housing-specific. It also relieves mortgage borrowers and lowers the government's interest bill; under the optimized rule, the lower interest bill contributes more to the welfare gain than the rent relief itself.

The poorest renters, who hold the least financial wealth and have the highest marginal utility of consumption, benefit most from the lower rents and larger fiscal rebates, while landlords lose rental income and indebted homeowners gain from lower mortgage costs. While tenure is a key determinant of the welfare impact of the optimal policy, net asset positions also matter: rich renters incur losses because of the lower returns on their savings, and so do unlevered homeowners, who are not affected by the reduction in mortgage rates. The largest rent relief, and the sharpest transfer from landlords to renters, comes from a rule that responds to the level of real rents; under this rule, lower rental payments are the main source of renters' welfare gains.

These welfare gains may reflect both aggregate efficiency gains, arising from the stabilization of rental markets as well as from insurance, and redistributive gains, which arise whenever the (poorer) renters benefit from the monetary policy reaction. Both of these channels are specific to the rental market and therefore novel to our study. We use the [Dávila and Schaab \(2025\)](#) decomposition to establish that, in the case of an expansionary demand shock, it is the redistributive motive that pushes the monetary authority to respond negatively to rent inflation, while in the case of a rental availability shock, it is the efficiency motive that drives the response of monetary policy to rents. Under a cost-push shock, both motives favor a negative response.

Such gains cannot be achieved with a Taylor rule that targets only nondurable inflation but is less hawkish overall. In the case of a demand shock, such a rule would increase welfare, but less than under our optimal policy, and at the cost of significantly reducing the central bank’s reaction to inflation. In the case of a rental availability shock, such a rule would actually decrease welfare. In the case of a cost-push shock, a less hawkish rule also helps, but delivers only about two-thirds of the gain. Based on these observations, we conclude that central banks should pay particular attention to developments in rental markets and take these into account in their decisions, especially when movements in rents account for a substantial share of realized inflation.

Related literature. Our model builds on the Heterogeneous-Agent New Keynesian literature. [Kaplan et al. \(2018\)](#) and [Auclert et al. \(2024\)](#) study monetary transmission through household consumption under uninsurable risk and borrowing constraints, without a discrete tenure margin or rental market. On the normative side, [Bhandari et al. \(2021\)](#), [Acharya et al. \(2023\)](#), [Dávila and Schaab \(2023\)](#), and [Le Grand et al. \(2026\)](#) find that a utilitarian planner values redistribution toward indebted, high-marginal-utility households. Relative to this work, we add a discrete tenure choice and an endogenous landlord margin, so that a tightening transmits through tenure reallocation and the rents that clear the rental market, and we ask a narrower, implementable version of the normative question, characterizing an optimal simple rule ([Schmitt-Grohé and Uribe, 2007](#)) rather than the full Ramsey optimum. The redistributive motive then attaches to a specific price, the rent paid by renters.

On the housing side, we build on general-equilibrium models of housing and rental markets with endogenous tenure choice ([Castellanos et al., 2024](#); [Sommer et al., 2013](#); [Kaplan et al., 2020](#)). [Greenwald and Guren \(2025\)](#) show that the slope of the tenure-supply curve governs how a credit-driven increase in demand for homeownership translates into changes in the price–rent ratio and the homeownership rate; the slope reflects landlords’

willingness to sell rental properties to owner-occupiers. In our model this slope is the endogenous landlord margin. Lumpy landlord entry makes short-run rental supply inelastic, so a tightening is absorbed mainly by the market rent rather than by the value of the housing stock. These models are real; we add nominal rigidities and a Taylor rule, so the real rate, inflation, and rents become joint equilibrium responses to policy.

The paper also draws on empirical and measurement work on rents. [Dias and Duarte \(2019\)](#) and [Abramson et al. \(2025\)](#) document that contractionary monetary policy can raise housing rents in the United States, while the European evidence is more mixed ([Lazarowicz and Richard, 2023](#); [Groiss and Syrichas, 2026](#)). For the euro area as a whole, the estimated response of HICP rents to a contractionary ECB shock is positive ([Corsetti et al., 2022](#)). [Bolhuis et al. \(2022\)](#) and [Ball and Koh \(2025\)](#) show that recorded shelter inflation lags market rents because lease turnover is slow, and [Baker and Wroblewski \(2026\)](#) document that contract-rent setting is sticky and heterogeneous across landlords. We take these facts as inputs and add no new evidence. The tenure margin lets the sign and size of the rent response depend on market structure, in line with the cross-country heterogeneity, and the sticky-lease block reproduces the turnover lag, separating the market rent on new contracts from the average rent that enters the index.

The policy result speaks to the literature on which components of inflation central banks should stabilize. Since [Aoki \(2001\)](#) and [Woodford \(2003\)](#), the prescription has been to stabilize inflation in the sectors where prices are stickiest and supply is most inelastic.² Rents are sticky and housing supply is inelastic, so a mechanical application of this logic assigns rent inflation a larger weight than its expenditure share in the policy objective. The prescription fails here because the stickiness is of a different kind. Sticky prices in the tradition of [Calvo \(1983\)](#) are forward-looking and hold prices away from their flexible-price level, whereas a lease simply carries an old rent forward until it turns over. In our economy this backward-looking stickiness is a reason to respond less to rent inflation.

A recent literature asks directly how monetary policy should treat housing inflation. [Bianchi et al. \(2024\)](#) argue that housing services are rationed through search and inelastic short-run supply, so that tightening against a housing-demand shock mainly depresses the non-housing sector, and optimal policy stabilizes non-housing inflation while disregarding housing inflation. [Chodorow-Reich and Mehrotra \(2026\)](#) apply this logic to U.S. shelter inflation and argue for placing less weight on shelter than its expenditure share. [Dias and Duarte \(2022\)](#) show in a two-agent model with tenure choice that a rent-inclusive index creates a policy feedback loop. There, the rule responds to rent movements that monetary policy itself induced, and rents, house prices, and output all become more

² See also [Benigno \(2004\)](#) for currency areas, and [Eusepi et al. \(2011\)](#) and [Bodenstein et al. \(2008\)](#) for the implications for core-inflation measures.

volatile. [De Llanos \(2026\)](#) studies monetary transmission to rents in an estimated New Keynesian model of the United States with two representative household types and a firm-operated rental stock. In his model, a tightening reallocates constrained households toward renting and a shelter-inclusive rule overreacts to the resulting relative-price movement. He finds that a shelter-inclusive rule produces the largest output cost of disinflation among the rules he considers, and that excluding shelter from the target lowers that cost and improves welfare. When all policy coefficients are chosen jointly, his welfare search keeps a small positive response to real rent growth, with a negligible gain over targeting goods inflation. [Albuquerque et al. \(2025\)](#) study housing in a heterogeneous-agent economy with individual landlords and find that it can be optimal to look through housing supply shocks.

Our contribution relative to this work is threefold. First, we provide the first decomposition of the monetary rent response into distinct household-level channels. We are not aware of another paper on rent inflation that attributes the rent increase to named economic margins in this way: [Castellanos et al. \(2024\)](#) hold the real rate exogenous, [Dias and Duarte \(2022\)](#) work with two representative agents, and [De Llanos \(2026\)](#) focuses on the two elasticities that decide whether rents rise—tenure substitutability and rental-supply segmentation—rather than on the household-level channels through which the increase operates. Second, we ask about the sign of the response to measured rent inflation itself, where existing prescriptions stop at excluding, down-weighting, or separating shelter. The welfare-maximizing sign is negative under demand, rental availability, and cost-push shocks, and it obtains with fully rational landlords. Third, the full wealth distribution and the household-landlord margin let us trace the optimum across cohorts and identify who pulls it negative, which models with representative household types cannot.

We begin in [Section 2](#) by developing the model. Two features of the housing block matter throughout: the endogenous tenure margin, which makes rents rise when policy tightens, and sticky leases, which separate the market rent on new contracts from the average rent that price indices measure. [Section 3](#) calibrates the economy to Spain and validates it against untargeted tenure-transition evidence. [Section 4](#) isolates the rent channel of a monetary tightening, decomposes the rent response into channels, and quantifies its consumption incidence across households. [Section 5](#) gives rent inflation its euro-area weight in the policy index and quantifies the cost of the resulting policy feedback. [Section 6](#) characterizes the optimal weight on rent inflation and identifies who pulls it negative. [Section 7](#) concludes.

2 A HANK Model with Housing Tenure Choice

We study a discrete-time Heterogeneous-Agent New Keynesian economy with endogenous housing tenure choice, sticky rental leases, and housing supply. These margins determine how monetary shocks affect market rents, average rents, house prices, and households across tenure states. Housing services enter the household's utility function. Time is indexed by $t \in \{0, 1, 2, \dots\}$. The economy is populated by heterogeneous households, a representative consumption-good producer, a construction sector, labor unions that introduce nominal rigidities through sticky wages, a government, and a central bank. The final good's price acts as the numeraire.

2.1 Households

We consider a continuum of infinitely lived, ex-ante identical households that become heterogeneous ex-post along four dimensions: financial wealth, housing tenure status, idiosyncratic labor productivity, and a lease state. Each period, households consume a nondurable good, choose a financial asset position, and make a discrete housing tenure decision. The housing tenure choice determines whether a household rents, owns a single housing unit, or owns two housing units, in which case the household becomes a landlord and rents out one unit to the rental market, similarly to the framework in [Castellanos et al. \(2024\)](#).

Idiosyncratic risk. Households face two sources of idiosyncratic uncertainty. First, labor productivity follows a persistent stochastic process. Second, the discrete housing choice is subject to type-I extreme-value taste shocks, which smooth the tenure decision. There is no aggregate uncertainty. Aggregate shocks are modeled as one-time unanticipated MIT shocks: they are unexpected on impact, and the economy follows a deterministic transition thereafter.

Preferences. Households derive utility from nondurable consumption and housing services, and they dislike supplying labor. They value these flows according to the per-period utility function

$$\mathcal{U}(c, h, n) = u(c, h) - v(n) = \frac{(c f(h))^{1-\gamma}}{1-\gamma} - \psi_n \frac{n^{1+\eta}}{1+\eta}, \quad (1)$$

where γ is the coefficient of relative risk aversion, η is the inverse Frisch elasticity, ψ_n scales the disutility of labor, and $f(h)$ captures the housing services enjoyed by homeowners

relative to renters,

$$f(h) = \mathbb{1}_{h=0} + \mathbb{1}_{h>0} (\theta \tilde{h}_2 / \tilde{h}_1)^{\alpha_h}. \quad (2)$$

Here $\tilde{h}_1 < \tilde{h}_2$ denote the two housing-quality (equivalently, size) tiers, with renters occupying the low tier and homeowners (owner-occupiers and landlords) the high tier, while $\theta > 1$ raises the service flow from ownership relative to renting and $\alpha_h < 1$ captures diminishing returns to housing quality. Labor supply n is set by labor unions, as described in Section 2.3.

The marginal utility of consumption, $u_c(c, h) = f(h)^{1-\gamma} c^{-\gamma}$, depends on the household's tenure state through the housing-services shifter $f(h)$. Because $\gamma > 1$, the term $f(h)^{1-\gamma}$ is decreasing in $f(h)$: at a given level of consumption, a homeowner, who enjoys more housing services ($f(h) > 1$), values an additional unit of consumption less than a renter does. Consumption and housing services are therefore substitutes: a household is willing to accept lower nondurable consumption in exchange for the housing services of ownership.

Budget constraint. Households allocate their savings across housing h and financial assets a , which can be negative, reflecting the net debtor position of households with mortgages. Households consume the nondurable good, and renters pay rent. Thus, their budget constraint reads:

$$c + a' + \psi_t(h, h') \leq (1 + r_t(1 + \kappa \mathbb{1}_{a < 0}))a + w_t y n + T_t(y) + \hat{p}_t^r(h, s)(h - 1), \quad (3)$$

where a prime denotes a next-period variable, r_t is the real interest rate, κ is a mortgage wedge that makes the borrowing rate exceed the rate on savings, w_t is the real wage, $T_t(y)$ is a government transfer (Section 2.7), and $\hat{p}_t^r(h, s)$ is the rental price faced by a household with tenure state h and lease state s (see Section 2.2). The term $\hat{p}_t^r(h, s)(h - 1)$ captures the rental market: renters ($h = 0$) pay rent, owner-occupiers ($h = 1$) have no rental cash flow, and landlords ($h = 2$) receive rent. The function $\psi_t(h, h')$ summarizes all housing costs, as detailed below.

Collateral constraints. Borrowing is subject to collateral and income constraints. A household that upgrades its housing position ($h' > h$) faces loan-to-value (LTV) and loan-to-income (LTI) limits. Let p_t^h denote the real price per unit of housing quality. Following [Castellanos et al. \(2024\)](#), houses of different qualities can be freely converted into one another, so a housing unit of quality $\tilde{h}_i$ is worth $p_t^h \tilde{h}_i$. Let $q_t(h, h')$ denote the quantity of

housing quality acquired when moving from housing state h to h' ,

$$q_t(h, h') = \begin{cases} \tilde{h}_2, & \text{if } (h, h') = (0, 1), \\ \tilde{h}_1 + \tilde{h}_2, & \text{if } (h, h') = (0, 2), \\ \tilde{h}_1, & \text{if } (h, h') = (1, 2), \\ 0, & \text{otherwise.} \end{cases} \quad (4)$$

Thus, $p_t^h q_t(h, h')$ is the value of housing acquired in the transaction. An existing mortgage borrower who retains the same housing state must satisfy a minimum amortization requirement. Formally,

$$a' \geq \begin{cases} \max \{ -\lambda_{\text{LTV}} p_t^h q_t(h, h'), -\lambda_{\text{LTI}} w_t y_n \}, & \text{if } h' > h, \\ a(1 + r_t(1 + \kappa) - m), & \text{if } h' = h > 0 \text{ and } a < 0, \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

Households can borrow only when upgrading their housing position. The LTV constraint prevents them from borrowing more than a fraction λ_{LTV} of the value of the housing units purchased, while the LTI constraint caps borrowing at λ_{LTI} times current labor income. Existing borrowers who retain the same housing state may roll over their mortgage subject to the minimum amortization requirement, governed by m . Households who become renters or move to a lower housing state must repay their debt in full.

Housing costs. The housing cost function $\psi_t(h, h')$ captures depreciation, transaction taxes, and the cost of buying and selling housing units. An owner-occupier holds the single unit $\tilde{h}_2$, a landlord holds both $\tilde{h}_1$ and $\tilde{h}_2$, and a renter holds none. The cost matrix is

$$\psi_t(h, h') = \quad (6)$$

$$p_t^h \times \begin{bmatrix} 0 & (1 + \tau_h)\tilde{h}_2 & (1 + \tau_h)\tilde{h}_2 + (1 + \tau_h)\tilde{h}_1 \\ -(1 - \tau_h - \delta_h)\tilde{h}_2 & \delta_h\tilde{h}_2 & (1 + \tau_h)\tilde{h}_1 + \delta_h\tilde{h}_2 \\ -(1 - \tau_h - \delta_h)(\tilde{h}_2 + \tilde{h}_1) & \delta_h\tilde{h}_2 - (1 - \tau_h - \delta_h)\tilde{h}_1 & \delta_h(\tilde{h}_2 + \tilde{h}_1) \end{bmatrix},$$

where rows correspond to the current state h and columns to the next state h' , τ_h is the transaction tax rate, and δ_h is the depreciation rate. Each entry is the net housing outlay: it combines the cost of purchasing the new unit (including the transaction tax) with the proceeds from selling the current unit (net of tax and depreciation), so that a positive entry is a net cost to the household and a negative entry a net receipt. For example, an

owner-occupier who sells and becomes a renter ($h = 1, h' = 0$) collects the value of its $\tilde{h}_2$ unit, net of the depreciation accrued over the period and of the proportional transaction tax, so the entry is $-(1 - \tau_h - \delta_h) \tilde{h}_2 p_t^h$.

Recursive problem. We denote the individual state by (a, h, y, s) . At the beginning of each period, the household observes a vector of choice-specific taste shocks $\varepsilon = \{\varepsilon(h')\}_{h'}$ and solves

$$V_t^\varepsilon(a, h, y, s) = \max_{c, a', h'} \mathcal{U}(c, h, n) + \sigma_\varepsilon \varepsilon(h') + \beta_t \mathbb{E}_t[V_{t+1}(a', h', y', s') \mid y, h, h'] . \quad (7)$$

The shocks $\varepsilon(h')$ are drawn independently across households, housing choices, and dates from a type-I extreme-value distribution with unit scale and zero mean, and $\sigma_\varepsilon > 0$ scales the taste shocks. The continuation value $V_t(a, h, y, s) = \mathbb{E}_\varepsilon[V_t^\varepsilon(a, h, y, s)]$ integrates over these shocks, yielding logit transition probabilities across housing states. Appendix A.1 provides the corresponding formulas. The discount factor β_t may vary over time.

Distribution. We denote the distribution over the idiosyncratic state space (a, h, y, s) by $g_t(a, h, y, s)$ and the stationary distribution by $g_{ss}(a, h, y, s)$. The share of households with housing state h and lease state s is $\mu_t^{h,s} = \iint g_t(a, h, y, s) da dy$, and the share with housing state h is $\mu_t^h = \sum_s \mu_t^{h,s}$.

2.2 Sticky Rental Leases

In practice, the rents paid by households do not adjust instantaneously to market conditions, both because existing contracts take time to be updated and because of rental market regulations which limit the extent to which rents may change. We capture this plurality of frictions which generate stickiness in rental prices in a simple way with a binary state $s \in \{0, 1\}$ that tracks whether a renter or landlord is in the regulated ($s = 0$) or unregulated ($s = 1$) market.

New leases are signed at the market rent, i.e., households that change their rental tenure status and become landlords or renters in a given period participate in the unregulated rental market. For households that keep their rental tenure, both regulated and unregulated contracts reset at the market rent each quarter with probability $1 - \varphi$, and stay in or move to the regulated market with probability φ . This probability determines the stickiness of rental prices.

The rental price faced by a household of rental tenure h and lease state s is:

$$\hat{p}_t^r(h, s) = \tilde{p}_t^r(h) \mathbb{1}_{s=0} + p_t^r \mathbb{1}_{s=1}, \quad (8)$$

where p_t^r is the market rent and $\tilde{p}_t^r(h)$ is the regulated rent faced by households in rental tenure $h \in \{0, 2\}$. The market rent p_t^r is determined in equilibrium by equating rental supply and rental demand. We assume that the rent of the regulated market is a weighted average of last period's regulated rent and last period's market rent. In this way, it reflects the persistence of existing rental contracts and regulations, but also updates with recent market conditions, because every period a flow of previously unregulated contracts becomes regulated. Formally,

$$\tilde{p}_t^r(h) = (1 - \lambda_h) p_{t-1}^r + \lambda_h \tilde{p}_{t-1}^r(h), \quad (9)$$

where the weight λ_h represents the share of surviving regulated contracts, and $1 - \lambda_h$ is the contribution of previously unregulated contracts which now join the regulated pool. Our formulation implies that within a given rental tenure all regulated contracts pay the same price and allows us to incorporate rental price stickiness without having to explicitly model heterogeneity in rental contract origination dates. Let P_t^c denote the nominal price of the consumption good and $\pi_t^c = P_t^c / P_{t-1}^c - 1$ nondurable inflation. Written in nominal terms, (9) becomes $\tilde{P}_t^r(h) = (1 + \pi_t^c) [(1 - \lambda_h) P_{t-1}^r + \lambda_h \tilde{P}_{t-1}^r(h)]$: the lagged nominal weighted average is multiplied by the gross nondurable inflation factor $1 + \pi_t^c$. Here capital letters denote nominal variables. Regulated leases are therefore indexed contemporaneously to nondurable consumer prices.

The average rent in the economy weights the regulated and unregulated rents by the shares of regulated and unregulated renters, $\mu_t^{0,0}$ and $\mu_t^{0,1}$, normalized by the total share of renters μ_t^0 ,

$$\bar{p}_t^r = \frac{\mu_t^{0,0}}{\mu_t^0} \tilde{p}_t^r(0) + \frac{\mu_t^{0,1}}{\mu_t^0} p_t^r. \quad (10)$$

Rent inflation in the consumer price index is computed from the average rent $\bar{p}_t^r$, not from the market rent p_t^r . The average rent is the model counterpart of the rent index measured from observed leases and is the relevant object for a central bank that targets an HICP-based index (Section 5). Because $\bar{p}_t^r$ is expressed in units of the consumption good, the nominal average rent is $\bar{P}_t^r = P_t^c \bar{p}_t^r$, and its nominal inflation rate satisfies

$$1 + \pi_t^{\text{rent}} = (1 + \pi_t^c) \frac{\bar{p}_t^r}{\bar{p}_{t-1}^r}. \quad (11)$$

In practice, these assumptions mean that average rent is, to a close approximation, a geometrically weighted average of current and past market rents, as we detail in Appendix A.2. Therefore, average rents react more slowly to any given shock than market rents do, but eventually fully incorporate its effects.

2.3 Labor Market

As is standard in New Keynesian models with sticky wages, both without household heterogeneity (Erceg et al., 2000) and with it (Auclert et al., 2024), a continuum of labor unions sets nominal wages and determines the hours households work.³ Hours are rationed equally across households,

$$n_t = N_t, \quad (12)$$

where N_t is aggregate labor. Unions adjust nominal wages subject to a quadratic Rotemberg (Rotemberg, 1982) cost, which they pass through to their members as a utility loss. The flow payoff that the union internalizes augments the household utility (1) with this cost,

$$u(c, h) - v(n) - \frac{\theta_w}{2} (\pi_t^w)^2, \quad (13)$$

where $\pi_t^w = (W_t - W_{t-1})/W_{t-1}$ is nominal wage inflation and $\theta_w \geq 0$ governs the degree of wage rigidity.

Each union sets its wage to maximize the aggregate lifetime utility of the households it represents. Optimal nominal wage setting yields a nonlinear New Keynesian wage Phillips curve (see Appendix A.3 for the derivation),

$$\pi_t^w (1 + \pi_t^w) = \frac{\epsilon_t}{\theta_w} N_t \left(v'(N_t) - (1 + \tau_L) \frac{\epsilon_t - 1}{\epsilon_t} w_t \int y_{it} u_c(c_{it}, h_{it}) di \right) + \beta_t \pi_{t+1}^w (1 + \pi_{t+1}^w). \quad (14)$$

Here c_{it} and h_{it} denote the consumption and housing of household i , and $\int (\cdot) di$ averages over the continuum of households. Equivalently, one can integrate with respect to the distribution $g_t(a, h, y, s)$. The parameter ϵ_t is the elasticity of substitution across labor varieties and τ_L is an employment subsidy. With a subsidy that offsets the union's market

³ Placing the nominal rigidity in wages avoids the countercyclical profits that sticky-price heterogeneous-agent models generate and would have to assign to households (Broer et al., 2020; Auclert et al., 2025): with perfect competition and flexible prices, profits are zero at all dates. Symmetric rationing serves a separate purpose, shutting down the endogenous labor-supply margin so that labor income moves proportionally across households.

power in the steady state, namely $(1 + \tau_L)(\epsilon - 1)/\epsilon = 1$ at the steady-state elasticity ϵ , this equation shows that unions aim to equate the marginal disutility of labor to the marginal value of an additional hour of work, expressed in consumption units.⁴ The elasticity ϵ_t may vary over time.

2.4 Consumption-Good Producer

A representative firm produces the final nondurable consumption good using only labor.⁵ The technology is linear:

$$Y_t^c = A^c N_t, \quad (15)$$

where A^c is total factor productivity. Under perfect competition and flexible prices, profits from production are zero and the real wage equals the marginal product of labor,

$$w_t = A^c. \quad (16)$$

2.5 Construction Sector

The construction sector produces new housing units Y_t^h using land L_t and housing structures S_t (priced in units of the consumption good) according to a Cobb–Douglas technology,

$$Y_t^h = A^h L_t^{\alpha_L} S_t^{1-\alpha_L}, \quad (17)$$

where A^h is total factor productivity in construction and α_L is the land share. Under perfect competition, the construction firm solves

$$\max_{S_t, L_t} p_t^h A^h L_t^{\alpha_L} S_t^{1-\alpha_L} - p_t^L L_t - S_t, \quad (18)$$

where p_t^L is the price of land in consumption units. Land permits are in fixed supply, $L_t = \bar{L}$, and p_t^L adjusts to clear their market. The government issues the permits and absorbs their revenue $p_t^L \bar{L}$ (Appendix A.4). Because houses of different qualities can be freely converted into one another, we do not need prices that clear the market for each

⁴ We assume, as in [Erceg et al. \(2000\)](#), that employment in the steady state is subsidized to eliminate the monopolistic distortion associated with the wage-setting power of unions.

⁵ We adopt the simplest production structure, as in [Dávila and Schaab \(2023\)](#). An alternative is to assume firms produce differentiated varieties under monopolistic competition, as in [Auclert et al. \(2024\)](#). This generates positive profits that must be redistributed, adding a layer of complexity without affecting the main transmission channels we study.

individual housing type; it suffices to clear the market for the aggregate housing stock. Aggregate housing demand implied by the tenure distribution is $H_t = \mu_t^1 \tilde{h}_2 + \mu_t^2 (\tilde{h}_1 + \tilde{h}_2)$. The housing market clears when this demand equals the available aggregate stock. Since new construction augments the existing stock, which depreciates at rate δ_h , market clearing requires

$$H_{t+1} = (1 - \delta_h)H_t + Y_t^h. \quad (19)$$

2.6 Financial Market

We consider a closed economy in which the wealth of savers ($a_t > 0$) finances the mortgages of borrowers ($a_t < 0$). In addition, the government supplies a constant stock of public debt $\bar{B}$. Aggregate net financial wealth is $A_t \equiv \sum_{h,s} \int \int a g_t(a, h, y, s) da dy$. In each period t , the real interest rate r_t adjusts so that the financial market clears,

$$A_t = \bar{B}. \quad (20)$$

We adopt the following timing convention: the household's net position a_t is a one-period riskless claim rolled over each period. We fix the real payoff on the balance carried into t at the rate agreed when that balance was chosen, so monetary shocks realized at t affect the return on debt issued from t to $t + 1$ but do not revalue the balance brought into t .

Real and nominal returns are linked by the Fisher equation. Households trade a one-period nominal bond that pays one unit of currency next period, so the gross nominal rate is $1 + i_t$ and the real gross return in consumption units is $1 + r_{t+1} = (1 + i_t) P_t^c / P_{t+1}^c$. The nominal interest rate i_t , the real interest rate r_{t+1} , and nondurable inflation π_{t+1}^c satisfy

$$1 + i_t = (1 + r_{t+1})(1 + \pi_{t+1}^c). \quad (21)$$

Because each household's debt is a one-period claim rolled over every period, mortgages in the model are effectively adjustable-rate: the rate a borrower ($a_t < 0$) pays resets to the prevailing real rate as the debt rolls over. The mortgage rate on debt carried from t to $t + 1$ is $r_{t+1}(1 + \kappa)$, where κ is the mortgage wedge from the budget constraint (3). This ties mortgage costs directly to monetary policy. The central bank sets the nominal rate i_t , and the Fisher equation (21) maps it into the real rate r_{t+1} . A tightening that raises i_t affects r_{t+1} and, with it, the rate at which households service and originate mortgages.

2.7 Government and Monetary Authority

The role of fiscal policy is deliberately minimal. The government supplies a constant real stock of one-period nominal bonds. The fiscal authority pays an employment subsidy $\tau_L w_t y_t n_t$ to each household with labor productivity y_t . To finance this subsidy, it raises a tax proportional to labor productivity, equal to $\tau_L w_t y_t N_t$. In equilibrium this scheme does not redistribute resources across households, because hours are rationed symmetrically and $n_t = N_t$.⁶

The household transfer $T_t(y)$ also includes the fiscal adjustment that closes the government budget. In the stationary equilibrium, the housing transaction tax τ_h is calibrated so that the residual fiscal requirement is zero. Along the transition after a shock, τ_h and the stock of public debt $\bar{B}$ remain fixed, and the residual fiscal requirement is instead offset by a lump-sum component of $T_t(y)$.⁷ This component absorbs revenues from housing transaction taxes and the mortgage wedge, net of interest payments on government debt, as well as any rental market accounting residual that arises because renter and landlord masses need not coincide within each lease state.⁸

Because the model contains only two priced components of household consumption, nondurable consumption and rental housing services, we define the model counterpart of HICP inflation, π_t^{hicp} , as a weighted average of nondurable and rent inflation. House purchases are asset transactions and do not enter the consumption price index. The central bank sets the nominal interest rate using a Taylor rule that targets this inflation measure,

$$i_t = r_{ss} + \phi_\pi \pi_t^{\text{hicp}} + \varepsilon_t^m, \quad \pi_t^{\text{hicp}} = (1 - \omega_r) \pi_t^c + \omega_r \pi_t^{\text{rent}}, \quad (22)$$

where r_{ss} is the steady-state real interest rate, ϕ_π is the inflation response coefficient, ε_t^m is a monetary policy shock, ω_r is the weight on rents in the index, and π_t^{rent} is the nominal inflation rate associated with the *average rent* $\bar{p}_t^r$ defined in (10).

Nominal wage inflation π_t^w and nondurable inflation π_t^c are linked through the labor market. Since the real wage equals productivity, $w_t = A^c$, and the nominal wage is

⁶ Making both the subsidy and its financing proportional to labor productivity prevents this fiscal arrangement from redistributing income across households. This allows the employment subsidy to offset the steady-state wage markup without creating an additional fiscal transmission channel.

⁷ Kaplan (2025) emphasizes that fiscal assumptions are not innocuous in HANK models: the way monetary-policy-induced fiscal effects are financed can shape aggregate and distributional transmission. We use this closure to keep the fiscal block minimal.

⁸ Rental-market clearing equates the total masses of renters and landlords, $\mu_t^0 = \mu_t^2$, but not their composition across lease states. Because the two sides carry different regulated rents, aggregate rent paid by renters can differ from aggregate rent received by landlords. This residual is exactly zero in the stationary equilibrium and on impact, and small thereafter. Along the transition it is absorbed by the lump-sum component of T_t . Appendix A.4 reports the exact expression for the fiscal adjustment and the magnitude of this residual under each shock.

$W_t = w_t P_t^c$, nominal wage inflation satisfies $\pi_t^w = \pi_t^c$.

2.8 Equilibrium

We now define the competitive equilibrium for this economy. Aggregate consumption is $C_t \equiv \int c_{it} di$. The equilibrium requires that household and firm choices are individually optimal, that all markets clear, and that the distribution of households evolves consistently with individual decisions and aggregate prices.

Definition 1 (Competitive Equilibrium). *Given a sequence of shocks and a calibrated housing transaction tax τ_h , a competitive equilibrium consists of household value and policy functions $\{V_t, V_t^e, c_t, a_{t+1}, h_{t+1}\}$, prices $\{w_t, r_t, p_t^h, p_t^r, \tilde{p}_t^r(h), \bar{p}_t^r, p_t^L, i_t, \pi_t^c, \pi_t^{\text{hicp}}, \pi_t^w, \pi_t^{\text{rent}}\}$, aggregate quantities $\{N_t, C_t, Y_t^c, Y_t^h, H_t, A_t, L_t, S_t\}$, a transfer schedule T_t , and a distribution g_t such that:*

- (i) **Household optimality.** *Given prices, household policy functions solve the Bellman equation (7), with continuation value (A.1), subject to the budget constraint (3) and the borrowing constraints (5).*
- (ii) **Consumption-good firm optimality.** *The representative firm maximizes profits, yielding $w_t = A^c$.*
- (iii) **Construction firm optimality.** *The construction firm maximizes profits as in (18).*
- (iv) **Labor union optimality.** *Labor unions ration hours, so $n_t = N_t$, and nominal wage inflation π_t^w satisfies the New Keynesian wage Phillips curve (14).*
- (v) **Market clearing.** *Four markets clear:⁹*
 - (a) Housing market: $H_{t+1} = (1 - \delta_h)H_t + Y_t^h$.
 - (b) Rental market: *the total share of renters equals the total share of landlords, $\mu_t^0 = \mu_t^2$.*
 - (c) Financial market: $A_t = \bar{B}$.
 - (d) Land market: $L_t = \bar{L}$.
- (vi) **Government budget.** *The transfer schedule T_t contains the employment-subsidy financing described above and, along the transition, the lump-sum fiscal adjustment that balances the government budget.*
- (vii) **Law of motion for the distribution.** *The distribution g_t evolves according to household policy functions, the idiosyncratic income transition probabilities, and the lease-state transition rules of Section 2.2.*

⁹ Goods-market clearing follows from Walras' law given the other clearing conditions.

(viii) **Monetary policy.** The nominal interest rate satisfies the Taylor rule and HICP-index definition (22), and the Fisher equation (21) links the nominal and real interest rates.

2.9 Timing and Rent Measurement

The lease structure of Section 2.2 determines when aggregate shocks can move rents and how rents enter measured inflation. We collect these implications here, since the quantitative sections use them repeatedly.

Predetermination. We index the initial stationary equilibrium by $t = 0$ and the impact period by $t = 1$. Aggregate shocks are realized after period-1 leases have been signed. Every incumbent renter and landlord enters the impact period at the steady-state rent: the market rent satisfies $p_1^r = p_{ss}^r$, and by (9) so does the regulated rent. The average rent is then predetermined as well, $\bar{p}_1^r = p_{ss}^r$, regardless of the distribution of households across lease states.

Remark 1 (Impact equivalence). Since $\bar{p}_1^r = \bar{p}_0^r = p_{ss}^r$, the rent-inflation identity (11) implies that measured rent inflation coincides with nondurable inflation on impact,

$$\pi_1^{\text{rent}} = \pi_1^c \implies \pi_1^{\text{hicp}} = \pi_1^c \text{ for any } \omega_r. \quad (23)$$

The rent weight in the Taylor rule (22) is therefore irrelevant for the mapping from inflation to the policy rate on impact. Its effects operate only from $t = 2$ onward and, at $t = 1$, through expectations of the future path.

Rent-inflation decomposition. Solving the rent-inflation identity (11) for π_t^{rent} and subtracting nondurable inflation gives

$$\pi_t^{\text{rent}} - \pi_t^c = (1 + \pi_t^c) \left(\frac{\bar{p}_t^r}{\bar{p}_{t-1}^r} - 1 \right) \approx \text{real-rent growth}. \quad (24)$$

Measured rent inflation exceeds nondurable inflation exactly when the real average rent grows. A rule that responds separately to π_t^{rent} and π_t^c therefore responds to real average rent growth over and above its response to nondurable inflation. Section 6 uses this decomposition.

Turnover lag. Leases reset at rate $1 - \varphi$ per quarter, and a lease also ends when its household changes tenure, so a fraction λ_h of the regulated pool survives each quarter,

with λ_h defined in (A.4), and the average rent inherits movements in the market rent with a mean lag of $\lambda_0/(1 - \lambda_0)$ quarters. Market rents can move sharply while measured rent inflation responds only gradually.

3 Taking the Model to the Data

We take the model to Spanish data at quarterly frequency. Spain is particularly well suited to our analysis because it is an economy with a high homeownership rate, where the margin between renting and homeownership is particularly active, and because most outstanding mortgages and some newly originated ones are adjustable-rate, as in our model.

We divide the calibration between externally assigned parameters and five internally calibrated parameters. We choose (A^h, δ_h, θ) to reproduce the Spanish house-value-to-income ratio,¹⁰ rent-to-income ratio,¹¹ and renter share, choose β to reproduce a 2 percent annual real interest rate, and set the housing transaction tax τ_h so that the government budget balances. We then assess the model against moments it does not target. The most demanding of these is the quarterly tenure-transition matrix: nine conditional probabilities, estimated from the panel of the Spanish Survey of Household Finances (*Encuesta Financiera de las Familias*, EFF), that no calibrated parameter is chosen to reproduce.

3.1 Externally Calibrated Parameters

Table 1 reports the parameters assigned outside the internal calibration.

Housing and tenure. We adopt the housing qualities $(\tilde{h}_1, \tilde{h}_2) = (0.905, 1.095)$ and the housing preference parameter $\alpha_h = 0.5$ from Castellanos et al. (2024). Following Iskhakov et al. (2017), we introduce type-I extreme-value taste shocks and set their scale to $\sigma_\varepsilon = 0.15$ to smooth housing-choice probabilities. Finally, $\varphi = 11/12$ implies a mean interval of three years between exogenous rent resets, conditional on the household remaining in the same rental tenure. Effective lease spells are shorter, because a change of tenure also ends a contract.

¹⁰ The target is the mean value of homeowners' main residences divided by mean annual gross household income (ECB Household Finance and Consumption Survey, HFCS). Its model counterpart is the price of a reference dwelling of unit quality; see Appendix E.1.

¹¹ We set rent to 30 percent of mean household income, in line with Spanish survey evidence (Eurostat EU-SILC). Because renters' equilibrium income falls below the mean, the implied burden on renters' own income is about 38 percent, close to an untargeted Spanish affordability benchmark (Khametshin et al., 2024); see Appendix E.1.

Parameter	Description	Value	Source / Note
<i>Housing and tenure</i>			
$(\tilde{h}_1, \tilde{h}_2)$	Housing qualities	(0.905, 1.095)	Castellanos et al. (2024)
α_h	Housing preference parameter	0.5	Castellanos et al. (2024)
σ_ε	Taste-shock scale	0.15	Smoothing; Iskhakov et al. (2017)
φ	Quarterly lease-survival probability	11/12	Three-year mean rent reset time
<i>Mortgage and financial markets</i>			
λ_{LTV}	Loan-to-value limit	0.8	20% down payment; Bover et al. (2019)
λ_{LTI}	Loan-to-income limit	24	Max. six times annual income
κ	Mortgage wedge	0.75	Implies a 1.5 pp annual spread
m	Minimum amortization	0.0278	Approx. 30-year mortgage maturity
<i>Preferences and idiosyncratic income</i>			
γ	CRRA risk aversion	2	Standard
η	Inverse Frisch elasticity	2	Standard
ρ_y	Quarterly log-productivity persistence	0.95	External evidence; see text
σ_y	Std. of quarterly log-prod. innovation	0.1	External evidence; see text
<i>Production and market structure</i>			
α_L	Land share in housing production	0.33	Castellanos et al. (2024)
ϵ	Elasticity across labor varieties	10	Standard
τ_L	Employment subsidy	1/9	Offsets the steady-state wage markup
θ_w	Wage Rotemberg coefficient	500	Wage Phillips-curve slope of 0.02
<i>Government and monetary policy</i>			
ϕ_π	Inflation coefficient	1.5	Standard
B	Government debt	4	Debt equal to 100% of annual GDP

Table 1. Externally Calibrated Parameters

Mortgage and financial parameters. We set the loan-to-value limit to $\lambda_{LTV} = 0.8$, consistent with the Spanish evidence in Bover et al. (2019). The quarterly loan-to-income limit is $\lambda_{LTI} = 24$, equivalent to six times annual income, the threshold that IMF (2026) uses to classify high-LTI mortgages in Spanish loan-level data. The mortgage wedge $\kappa = 0.75$ implies an annualized spread of 1.5 percentage points, slightly above the sample average of 1.38 percentage points.¹² We set the minimum amortization requirement to $m = 0.0278$.¹³

Preferences and earnings process. We set risk aversion to $\gamma = 2$ and the inverse Frisch elasticity to $\eta = 2$. Idiosyncratic labor productivity follows

$$z_{t+1} = \rho_y z_t + \sigma_y \zeta_{t+1}, \quad z_t \equiv \log y_t, \quad \zeta_{t+1} \sim \mathcal{N}(0, 1).$$

¹² Authors' calculations using the ECB Data Portal. The spread is the average, from May 2001 to April 2026, of the difference between the interest rate on new Spanish variable-rate mortgages with an initial rate fixation of up to one year and the one-year Euribor.

¹³ Because the minimum-amortization constraint generates a geometric repayment path, we choose m so that a borrower who retains the same housing state repays 90 percent of the initial mortgage balance over 30 years at the steady-state mortgage rate.

Our calibration is guided by the Swedish estimates of [Flodén and Lindé \(2001\)](#), who estimate an annual persistence of 0.8139 and an annual innovation variance of 0.0326. Their quarterly equivalents are $\rho_y = 0.9498$ and $\sigma_y = 0.0972$. We use Sweden as a European benchmark with less persistent and less dispersed earnings risk than in the United States.

Production and wage rigidity. On the production side, we adopt a land share of $\alpha_L = 0.33$ from [Castellanos et al. \(2024\)](#). We set the elasticity across labor varieties to $\epsilon = 10$ and the employment subsidy to $\tau_L = 1/9$, which offsets the steady-state wage markup. The wage Rotemberg coefficient $\theta_w = 500$ implies a slope of the wage Phillips curve of 0.02, in line with [Auclert et al. \(2024\)](#).

Government, monetary policy, and normalizations. We set government debt to $\bar{B} = 4$, corresponding to 100 percent of annual GDP.¹⁴ We follow [Auclert et al. \(2025\)](#) in setting the inflation coefficient in the monetary policy rule to $\phi_\pi = 1.5$. We normalize consumption-sector productivity, land supply, and stationary mean labor productivity so that $A^c = 1$, $\bar{L} = 1$, and $\mathbb{E}[y] = 1$. The scale of labor disutility ψ_n is chosen so that aggregate employment also equals one in the stationary equilibrium.

3.2 Internally Calibrated Parameters and Model Fit

Five parameters are calibrated internally. Four of them, $(A^h, \delta_h, \theta, \beta)$, are chosen to hit the three Spanish moments and the 2 percent real rate introduced above. The housing transaction tax τ_h is solved jointly with them, since it enters the household problem directly, so that the government budget balances in the stationary equilibrium. Table 2 reports their values.

Parameter	Description	Value
A^h	Housing-sector productivity	0.0344
δ_h	Housing depreciation rate	0.00745
θ	Homeownership preference shifter	1.205
β	Discount factor	0.989
τ_h	Housing transaction tax	0.0246

Table 2. Internally Calibrated Parameters

The internally calibrated parameters pin down distinct margins of the stationary equilibrium. The housing parameters (A^h, δ_h, θ) jointly govern housing supply, the user

¹⁴ Spain’s general government gross debt stood at 100.7 percent of GDP at the end of 2025, according to Eurostat.

cost of housing, and tenure choices. The discount factor β governs desired saving and is chosen to deliver a quarterly real interest rate of 0.5 percent, corresponding to 2 percent annualized.

Table 3 compares the resulting targeted and untargeted moments with their Spanish counterparts; Appendix E documents the data. The calibrated model reproduces the targeted moments in Table 3 by construction.

Moment	Model	Spain
<i>Targeted moments</i>		
House value / annual income	5.25	5.25
Rent / income	30.0%	30.0%
Renter share	21.7%	21.7%
Real interest rate (annualized)	2.0%	—
<i>Untargeted moments</i>		
Residential construction / GDP	11.0%	6.9%
Household net worth / annual income	6.53	7.33
Owner-occupier share	56.6%	68.8%
Landlord share	21.7%	9.5%

Table 3. Targeted and Untargeted Model Moments

Note. The 2 percent annual real interest rate is imposed as a calibration restriction. Spanish entries are sample averages. In the model, the house value refers to a dwelling of unit quality, and rent is expressed relative to mean household income. Sources and construction are documented in Appendix E.

Untargeted moments in Table 3. Residential construction accounts for 11.0 percent of GDP in the model, compared with 6.9 percent in Spain.¹⁵ We construct the model counterpart as S_{ss}/Y_{ss}^c , the value of new housing net of land over consumption-good output, which equals model GDP; as in the national accounts, land is excluded from both. This overstatement is consistent with the model’s asset structure. In the model, housing is the only produced illiquid asset, whereas in Spain nonresidential investment averages 15.6 percent of GDP.¹⁶ Introducing productive capital would provide households with an alternative saving vehicle and could redirect resources away from residential construction.

Turning to household balance sheets, the model generates household net worth equal

¹⁵ Residential construction over GDP is gross fixed capital formation in dwellings divided by GDP at market prices (Eurostat national accounts).

¹⁶ Authors’ calculations using Eurostat national accounts data. Nonresidential gross fixed capital formation is defined as total gross fixed capital formation less investment in dwellings; the reported value is the average between 1999 and 2025.

to 6.53 times annual income, compared with 7.33 in Spain.¹⁷ We measure the model’s ratio of net worth to annual income as $(A_{ss} + p_{ss}^h H_{ss}) / (4\mathbb{E}[y])$, where A_{ss} denotes aggregate net financial wealth and H_{ss} aggregate housing in dwelling-size units. The corresponding euro-area ratio is 6.13, close to the model. Spain’s higher ratio is driven by housing wealth relative to income, which equals 5.66 in Spain and 4.37 in the euro area.

The model matches the overall share of property-owning households, although its composition differs from the data. Owner-occupiers account for 56.6 percent of households and landlords for 21.7 percent, compared with 68.8 and 9.5 percent in Spain. The landlord share equals the targeted renter share mechanically because each landlord supplies one rental unit and each renter occupies one unit. In the data, by contrast, a landlord household may supply several rental properties. The discrepancy concerns the number of landlord households; the size of the rental market itself is matched through the targeted renter share.

Wealth distribution. The upper tail of the model’s wealth distribution is far less concentrated than in the data: the top decile holds 18.7 percent of net worth in the model against 53.2 percent in Spain (HFCS 2021 wave). Because households share a common discount factor, the model generates limited persistent heterogeneity in desired saving. Matching the upper tail more closely would likely require heterogeneity in patience, with low- and high- β households, as in [Auclert et al. \(2025\)](#). [Castellanos et al. \(2024\)](#) show that allowing households to own a third property also improves the fit of the wealth distribution. [Kaplan et al. \(2014\)](#) document for Spain that hand-to-mouth households are predominantly wealthy rather than poor: 15.2 percent of Spanish households hold illiquid wealth but little liquid wealth, and only 4.4 percent hold neither. Our model reproduces that ordering but exaggerates it. Owner-occupiers and landlords who hold housing wealth but no positive net financial wealth are 29.7 percent of households, while renters holding no more than a quarter of mean annual income in financial wealth are 1.7 percent. The shortfall at the poor end is structural rather than a matter of calibration, since renters cannot borrow in our model. One feature of the Spanish data is favorable for our purposes: essentially all wealthy hand-to-mouth households there own housing, so the illiquid asset that defines the group is the one the model has.

Tenure-transition fit. We construct tenure-transition probabilities using the panel component of the EFF, the Banco de España’s survey of household finances, for the 2014, 2017, 2020, and 2022 waves. We classify households as renters if they neither own their main

¹⁷ Household net worth over income is financial assets plus housing wealth minus liabilities, divided by annual gross disposable household income (ECB Distributional Wealth Accounts and Quarterly Sector Accounts).

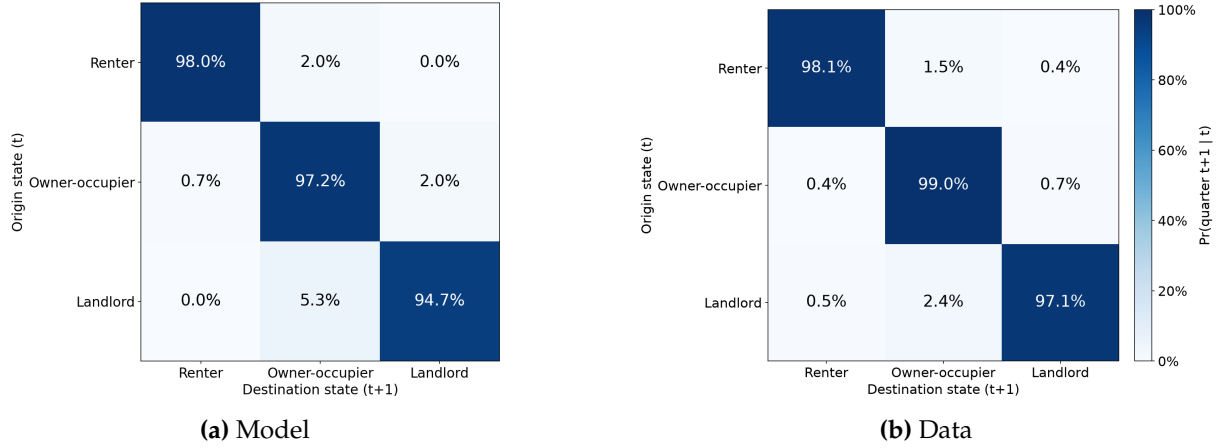


Figure 1. Quarterly Tenure-Transition Matrices

Note. Rows denote a household's current tenure state, columns denote its tenure state in the following quarter, and entries report the corresponding conditional transition probabilities. Panel (a) reports the model matrix, computed from stationary joint tenure flows. Panel (b) reports the data matrix, constructed from households observed in consecutive EFF waves between 2014 and 2022. Each wave-pair matrix is converted to quarterly frequency, and the resulting matrices are pooled using origin-state household counts. Both panels use the same color scale, shown in Panel (b).

residence nor receive rental income, as owner-occupiers if they own their main residence but receive no rental income, and as landlords if they receive real-estate rental income, regardless of main-residence ownership. We compute transition matrices for households observed in consecutive waves, convert each matrix from its two- or three-year survey horizon to quarterly frequency, and pool the resulting quarterly matrices. We use the pooled matrix as an untargeted test of the model's tenure persistence and mobility.¹⁸

Figure 1 shows that the model closely reproduces the high quarterly persistence of all three tenure states, without using any of the matrix's nine cells as a calibration target. It also captures the main economically relevant mobility patterns: renters who change tenure move primarily into owner-occupation, while mobility between owner-occupation and landlord status occurs in both directions. The main discrepancy concerns direct transitions between renting and landlord status, which are negligible in the model but occur with small positive probability in the data. Two cells of the model matrix become load-bearing later: the renter and landlord diagonals, 0.9804 and 0.9473, are the tenure-persistence probabilities that set the two lease-pool survival shares λ_0 and λ_2 in (A.4), and hence the mean lag of the measured rent index and the accounting residual of Section 2.7. They are untargeted moments, so the two pool shares add no parameter beyond φ . The model somewhat overstates mobility across tenure states relative to the data. Raising τ_h would

¹⁸ Authors' calculations using the EFF panel; see Appendix E.5.

reduce these off-diagonal transition probabilities, and at $\tau_h = 3$ percent the renter-to-owner and owner-to-landlord flows move substantially toward their EFF values. We do not take this route, because τ_h is the instrument that balances the steady-state government budget, and fixing it at a higher level would require a second fiscal instrument. Re-solving the economy with a higher τ_h and such an instrument leaves the aggregate impact responses nearly unchanged but raises the peak average rent.

4 Monetary Tightening and the Rent Channel

We begin with the benchmark monetary experiment to highlight the transmission mechanism of monetary policy to rent inflation in our model. Throughout this section, the central bank responds only to nondurable inflation, so that $\omega_r = 0$ in the Taylor rule (22). In this environment, a monetary tightening raises the real rate and increases mortgage costs, making ownership less accessible for marginal households. Some households delay their move into ownership or return to renting, shifting demand toward rental housing and raising market rents even as aggregate consumption and output fall. We first describe the aggregate response to the monetary shock, then study the housing and rental market mechanism, decompose the rent response into its channels, and finally quantify the consumption incidence of the tightening across households.¹⁹

4.1 A Contractionary Monetary Policy Shock

The experiment starts from the calibrated stationary equilibrium and applies a one-percentage-point annualized innovation to ε_t^m in the Taylor rule (22), equivalent to 25 basis points at quarterly frequency. The shock follows an AR(1) process with persistence $\rho_m = 0.7$, which implies a half-life of about two quarters. Figure 2 plots the policy shock, nondurable inflation, the real interest rate, and aggregate real activity.

The shock contracts real activity and lowers nondurable inflation. On impact, aggregate consumption falls by 0.24 percent and nondurable output by 0.39 percent, while nondurable inflation declines by about 0.63 annualized percentage points. The real interest rate rises by 53 annualized basis points on impact. The real-rate response is quantitatively

¹⁹ Castellanos et al. (2024) establish this cross-sectional rent channel (displaced buyers crowding into the rental market and marginal landlords requiring higher compensation) in a life-cycle model with an exogenous real interest rate, and show that a rate rise lowers homeownership, raises rents, and shifts welfare from young renters toward landlords. We inherit this channel and contribute by studying how monetary policy propagates through it and by decomposing the rent response into the channels through which the tightening operates.

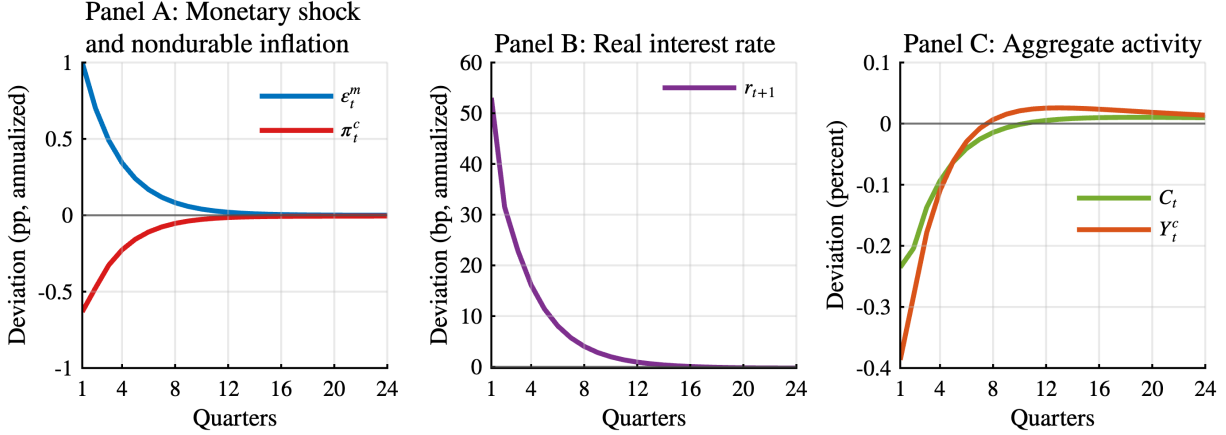


Figure 2. Aggregate Response to a Contractionary Monetary Shock

Note. Panel A reports the monetary policy innovation and nondurable inflation in annualized percentage points. Panel B displays the real interest rate in annualized basis points. Panel C reports aggregate consumption and nondurable output as percent deviations from steady state.

close to standard HANK benchmarks. On impact, the real rate absorbs 53 percent of the monetary innovation, compared with roughly 60 percent in [Kaplan et al. \(2018\)](#).

4.2 The Housing and Rental Market Response

The housing-market response is concentrated in rents. Panel A in Figure 3 shows that the real market rent is predetermined on impact and rises thereafter, while the real average rent responds gradually as leases reset (Section 2.9). Because only a fraction of leases reset each quarter, the increase in market rents reaches tenants late but in full: the average rent peaks at 0.46 percent above steady state eight quarters after the shock, long after the innovation itself has faded, and is still 0.11 percent above it after six years.²⁰ Panel B shows that the house price falls by 0.38 percent on impact. With land and construction-sector productivity fixed, the housing-supply block maps this price decline into a 0.77 percent fall in new housing production.²¹ The tenure distribution is predetermined on impact, so the house-price decline is forward-looking: it reflects the anticipated reallocation shown in Panel C. Because rental units are smaller than owner-occupied homes, the anticipated

²⁰ The increase in average rents also obtains with alternative lease-survival probabilities. Greater lease survival lowers the peak response, while the cumulative rent response varies much less.

²¹ Combining the construction technology (17) with the construction firm's problem (18) and fixed land supply gives

$$Y_t^h = (A^h)^{1/\alpha_L} ((1 - \alpha_L)p_t^h)^{(1-\alpha_L)/\alpha_L} \bar{L}.$$

Hence, conditional on construction-sector productivity, the elasticity of new housing production with respect to the house price is $(1 - \alpha_L)/\alpha_L$.

shift out of owner-occupation lowers desired holdings of the housing stock. With the stock itself nearly fixed in the short run, this shift appears almost entirely in the price: aggregate housing falls by less than 0.02 percent over the transition, compared with a 0.38 percent decline in the house price. As we will show later, the drop in house prices mitigates the increase in rents.

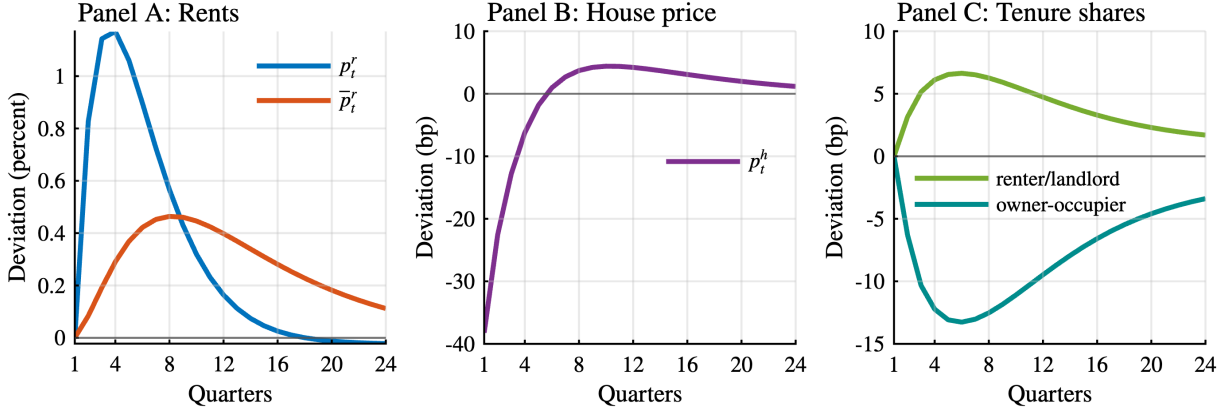


Figure 3. Rental and Housing Market Response to a Contractionary Monetary Shock

Note. Panel A reports real market and average rents as percent deviations from steady state. Panel B reports relative deviations of the real house price in basis points. Panel C reports tenure-share deviations in basis points.

Panel C shows the reallocation that generates the rent response. Because mortgages reprice every quarter, the higher real rate raises the cost of servicing existing debt and financing new purchases in the budget constraint (3). The decline in house prices and labor income also tightens the collateral and income limits in (5). Households close to the ownership margin therefore become less likely to enter owner-occupation, raising the renter share and lowering the owner-occupier share. Rental-market clearing requires an equal increase in landlords. Because constrained households cannot take these positions, the market rent rises until households with sufficient balance-sheet capacity are willing to become landlords.

The model reproduces two qualitative features of the evidence in [Dias and Duarte \(2022\)](#): lower homeownership and a higher rent-to-price ratio following a monetary tightening. In the model, tighter borrowing conditions shift households toward renting, while rents rise relative to property prices to induce additional rental supply.²²

As summarized by [De Llanos \(2026\)](#), existing evidence places the rental-supply elas-

²² For a monetary tightening normalized to a 25-basis-point increase in the federal funds rate, their Figure 1 shows nominal rents approximately 0.16–0.23 percent higher, nominal house prices 0.7–1.2 percent lower, and homeownership 0.05–0.08 percentage points lower after eight to twelve quarters, across their two identification instruments.

ticity at 0.4–0.7 in the short run and around 1.4 in the long run. When we perturb the market-rent path around the calibrated steady state, the model’s rental-supply elasticity is 0.57 over the impact quarter and the subsequent three quarters and rises to 1.58 over the following four quarters, placing the initial response within the short-run range and the subsequent response close to, though above, the long-run benchmark.²³

4.3 Rent Decomposition

In our economy, a tightening affects rents through several channels. On the renter side, it makes it harder for households to get a mortgage, which increases their rental demand; on the landlord side, the increase in mortgage costs reduces rental supply, which is potentially exacerbated by the increase in the returns on financial assets, which are alternative investments for potential property owners. A tightening also reduces labor income through the standard HANK general-equilibrium channels, which may also impact rental demand.

To separate these channels, we compute excess rental supply in a set of counterfactual scenarios in which we keep all prices constant at their steady-state levels after the monetary shock, with the exception of one, which we vary in turn. For example, in one of the experiments, we change the mortgage rate paid by indebted landlords, while keeping the mortgage rate paid by owner-occupiers, the return on financial savings, labor income, taxes and transfers, the house price, and rents constant. By inputting the mortgage rate’s equilibrium path with all other prices unchanged, we measure the contribution of this channel to rental-market excess supply, i.e., the difference between the amount of housing that landlords are willing to rent out and the amount of rental housing that renters demand. We then do the same with all other prices.²⁴ Although there are interactions between channels, to first order these single-input responses sum to the equilibrium response. For clarity, we also report the effect of rents in our decomposition. Naturally, rents adjust to clear the disequilibrium in the rental market generated by the combined pressure of all the other channels, so that the total effect reported in the decomposition adds up to zero excess supply and the rental market clears. A channel that contracts excess supply therefore contributes to raising rents.

Figure 4 shows that the dominant interest-rate channels operate through rental supply.

²³ The elasticity is a partial-equilibrium object: we move only the market rent, with persistence 0.97, close to that of the housing-demand shifter in De Llanos (2026), and hold other prices, labor income, and taxes at steady state. Quarters 5–8 are his year-one horizon.

²⁴ In equilibrium the three interest-rate inputs comove through the real rate r_t and the mortgage wedge κ in the budget constraint (3); the decomposition perturbs them separately to attribute the rent response across the savings, owner-mortgage, and landlord-mortgage margins. Appendix B states the decomposition formally, defines each input, and reports the resulting contributions to the rent. The equilibrium paths fed into the household problem are those of Figures 2 and 3.

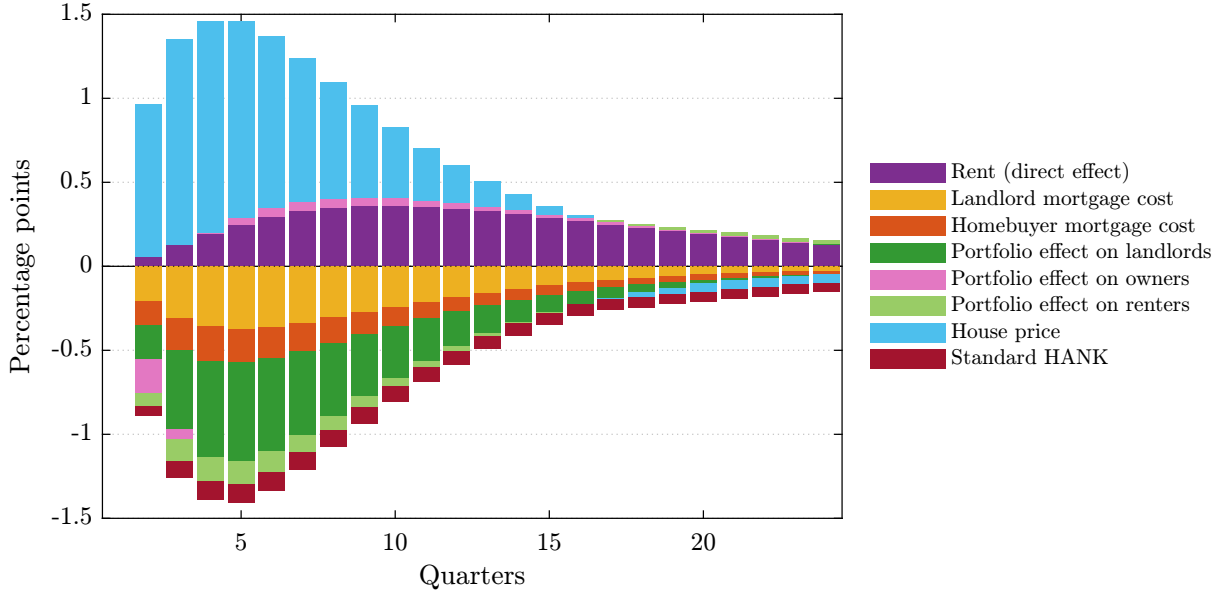


Figure 4. Decomposition of the Rent Response

Note. Stacked bars report each channel’s contribution to rental-market excess supply (the landlord share minus the renter share) in percentage points. Each channel feeds one equilibrium input path through the household problem, holding the remaining inputs at their steady-state values: the mortgage rate on landlord debt (landlord mortgage cost), the mortgage rate on owner-occupier debt (homebuyer mortgage cost), the return on financial savings, read separately by the tenure state of the household it reaches (the three portfolio effects), labor income and taxes (standard HANK), the house price, and the rent itself (direct effect). Appendix B defines the inputs and maps each channel’s excess-supply contribution into its contribution to the rent path.

At unchanged rents, a higher return on financial savings raises the opportunity cost of holding rental property (portfolio effect on landlords, dark green bars), so incumbent landlords lean toward exit. Higher mortgage costs make it more expensive to hold a rental unit with debt (landlord mortgage cost, yellow bars), so levered landlords are more willing to exit and owners who would need a mortgage to become landlords stay out. Together, these two landlord channels account for about three-quarters of the cumulative contraction in excess rental supply generated by the interest-rate channels over the first 24 quarters. Higher rents (purple) are therefore needed to compensate households for the increase in the financing and opportunity costs of supplying rental housing.

The demand side has secondary but still quantitatively relevant effects. A higher rate on owner-occupier mortgage debt raises the cost of being a levered owner and lowers the value of becoming one (homebuyer mortgage cost, orange bars), so would-be buyers stay renters and the pool of renters grows. The standard HANK channel of lower labor income and higher taxes (dark red) adds to this pressure. The portfolio effect on renters (light

green) also raises rents, while the net portfolio effect on owners (pink) is modest.²⁵

The fall in house prices (light blue) provides the main offset. Housing becomes cheaper to buy for prospective landlords and homebuyers alike (house price channel), encouraging rental supply and easing access to homeownership. Over the same horizon, this channel offsets about two-thirds of the upward rent pressure generated by the other channels. Table B.1 reports the cumulative contributions underlying these comparisons.

The rental market clears through a rise in rents that induces additional supply. On impact, owners become more likely to enter landlordship, while incumbent landlords become less likely to exit. The additional entrants are high-asset households, typically unlevered before entry: expected higher future rents compensate them for the forgone return on financial savings. The equilibrium expansion of rental supply thus depends on rents rising enough to attract households with the balance-sheet capacity to meet the additional rental demand. Figure F.1 in Appendix F.1 reports the underlying tenure-transition responses.

4.4 Consumption Incidence across Households

The consumption effects of the tightening depend on both mortgage debt and rental exposure. We measure them by following cohorts defined by tenure and net financial assets at the date of the shock, relative to the same initial cohorts in a no-shock counterfactual.

Mortgagor-owners experience the largest impact contraction as they anticipate higher debt service; levered landlords also cut consumption more than their unlevered counterparts. Renters face an additional housing-cost burden which reduces the resources available for nondurable consumption—the renter cohort’s consumption falls more on impact and recovers less strongly than that of unlevered owners and landlords. Sticky leases delay the increase in rental payments, but households already adjust consumption in anticipation of rising rents. We report detailed consumption responses in Appendix F.2.

5 Rent in the Inflation Target

So far, we have focused on the case in which the central bank does not react to rent inflation. We now move to the case in which the Taylor rule includes rent inflation with weight

²⁵ The portfolio effect on renters is the demand counterpart of the effect on landlords: a higher return on savings makes remaining a renter, with wealth in the liquid asset, more attractive than buying a house. The portfolio effect on owners nets two opposing margins of the same rate—the opportunity cost that deters owners from becoming landlords against the wealth effect of higher interest income, which enriches owners and, over time, makes becoming a landlord more affordable—and the net effect on the rent is modest.

$\omega_r = 0.06$, which approximates the expenditure weight of rents in the euro-area HICP.²⁶ The rent component is nominal average rent inflation, π_t^{rent} , which reflects payments on continuing and newly reset leases. Appendix E.6 documents the empirical weights and compares them with the model’s rent expenditure share.

5.1 Policy Feedback under a Rent-Inclusive Rule

Including rent inflation in the policy index deepens the consumption contraction after a monetary policy shock while delivering only a transitory gain in nondurable disinflation. It also raises rents, so the additional contraction does not stabilize the rental market. Figure 5 reports the responses under the two rules.

The feedback operates through the gradual transmission of market rents into rental payments. As the initial increase in market rents passes into average rents, nominal rent inflation exceeds nondurable inflation and raises the policy index. The resulting monetary response increases both the financing cost of rental property and the return on alternative financial investments. At unchanged rents, these forces discourage rental supply through the landlord mortgage cost and portfolio channels identified in Section 4.3. Higher rents are then needed to induce households to supply rental housing. Responding to rent inflation therefore reinforces the rental pressure generated by the initial tightening and deepens the consumption contraction.²⁷

5.2 The Costs of Responding to Rent Inflation

In the spirit of Mountford and Uhlig (2009), we summarize the response paths with cumulative statistics at fixed horizons. Table 4 reports the percentage differences in these statistics between the rent-inclusive rule and the benchmark.

Slow rent resetting sustains the policy response to earlier rent increases. Leases reset exogenously at a rate of one in twelve per quarter, with additional turnover when households change tenure. Average rents therefore continue to reflect earlier market-rent increases after the monetary innovation has faded. By responding to this delayed rent inflation, the rule prolongs the tightening generated by its own earlier policy response. At eight quarters, the cumulative consumption contraction is 10.5 percent larger. The

²⁶ HICP weights are revised once a year from past expenditure, so over the horizon of a shock the weight the central bank faces is effectively fixed; Appendix C shows that every statistic is monotone in the weight.

²⁷ On impact, real rents are predetermined, so the policy index coincides with nondurable inflation for any rent weight (Remark 1). Consumption nevertheless contracts more under the rent-inclusive rule, even though its initial real rate is about three annualized basis points lower. This reflects the forward-looking response to the additional future policy tightening: households adjust spending before real rents begin to rise.

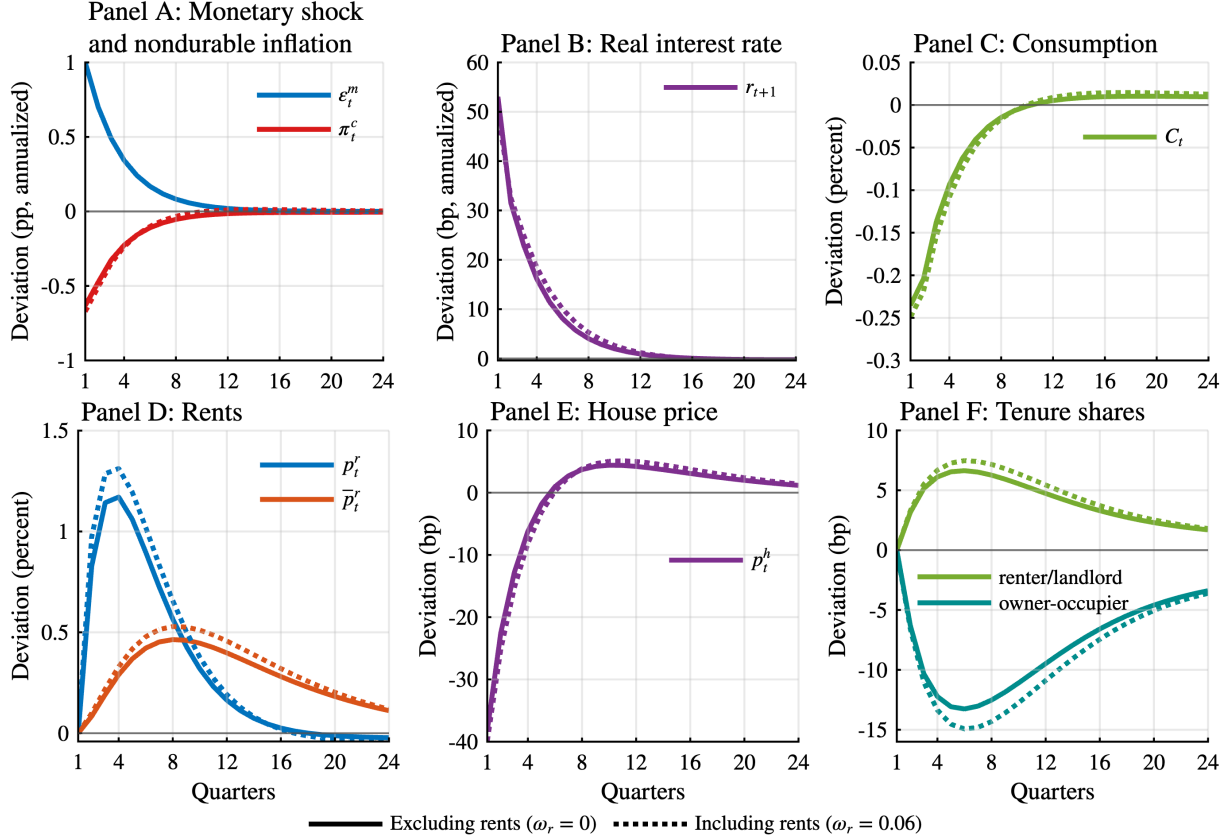


Figure 5. Monetary Policy Shock under Alternative Policy Rules

Note. Responses under $\omega_r = 0$ (solid lines) and $\omega_r = 0.06$ (dotted lines). Panel A reports the monetary policy innovation and nondurable inflation in annualized percentage points; Panel B reports the real interest rate in annualized basis points; Panel C reports consumption in percent deviations from steady state. Panel D reports real market and average rents in percent deviations; Panel E reports relative house-price deviations in basis points; Panel F reports tenure-share deviations in basis points.

initial gain in nondurable disinflation from responding to rent inflation eventually reverses, leaving a larger cumulative consumption loss and less cumulative disinflation than under the benchmark.

The additional tightening also raises tenants' rental burden. Higher interest rates strengthen the financing and portfolio incentives that put upward pressure on market rents (Section 4.3), and lease turnover gradually passes these increases into rents paid by tenants. The cumulative real average rent deviation is about 14 percent larger at twenty quarters. Responding to rent inflation therefore reinforces the rental-market pressure generated by the monetary shock.²⁸

The quadratic statistics show that the additional contraction does not improve stabi-

²⁸ Appendix C extends the comparison across rent weights at the same horizons.

	Horizon (quarters)			
	1	4	8	20
<i>Cumulative deviations</i>				
Consumption, C_t	5.83	8.88	10.45	5.57
Nondurable inflation, π_t^c	6.39	6.33	2.85	−8.61
Real average rent, $\bar{p}_t^r$	—	14.48	13.79	13.99
<i>Quadratic deviations</i>				
Consumption, C_t	11.99	16.54	17.66	18.23
Nondurable inflation, π_t^c	13.20	13.26	11.81	11.53
Average rent inflation, π_t^{rent}	13.20	14.40	19.05	13.06

Table 4. Aggregate Effects of a Rent-Inclusive Rule across Horizons

Note. The two rules face the same monetary policy shock. For a deviation $x_t^{\omega_r}$ of a variable under rule ω_r , define $S_H^{\omega_r} = \sum_{t=1}^H x_t^{\omega_r}$ and $Q_H^{\omega_r} = \sum_{t=1}^H (x_t^{\omega_r})^2$, where $t = 1$ is the impact quarter. Entries report $100(S_H^{0.06}/S_H^0 - 1)$ in the upper panel and $100(Q_H^{0.06}/Q_H^0 - 1)$ in the lower panel. A positive entry means the statistic is larger in magnitude under the rent-inclusive rule. In other words, the accumulated deviations until horizon H of the rent-inclusive rule dominate those of the other rule. Consumption and real average rents are measured in percent deviations from steady state; inflation is measured in annualized percentage points.

lization of either inflation measure. The rent-inclusive rule increases squared deviations of consumption, nondurable inflation, and nominal average rent inflation at every reported horizon. These statistics connect the experiment to the stabilization objectives represented by standard quadratic loss functions in representative-agent New Keynesian models (Galí, 2015). Nondurable-inflation fluctuations are also costly in our model, through the quadratic wage-adjustment term in the welfare objective (26). These aggregate statistics describe stabilization performance, while welfare also depends on how consumption and housing are distributed across households and on labor costs.

Mortgagor-owners experience the largest additional consumption contraction, reflecting their exposure to the interest-rate path through mortgage debt. Measured relative to each cohort’s no-shock consumption, their additional cumulative contraction is about four times as large as that of renters (Table F.2 in Appendix F.3). Renters also bear additional costs through higher rental payments and the broader contraction.

6 The Optimal Response to Rent Inflation

The previous section showed that putting rent inflation directly into the policy index changes the timing and incidence of monetary transmission. We now turn to the normative question: how should a central bank that maximizes welfare treat rent inflation?

To do so, we let the central bank choose the weight on nominal rent inflation in an additive Taylor rule,

$$i_t = r_{ss} + \phi_\pi \pi_t^c + \phi_r \pi_t^{\text{rent}}, \quad (25)$$

and report gains relative to $\phi_r = 0$, the rule that responds only to nondurable inflation. The rule incorporates nominal average rent inflation π_t^{rent} , the object observed in a consumer price index. Section 6.2.3 introduces an alternative rule based on real average rents.

We evaluate welfare after three shocks that raise rents: a demand shock, a rental supply shock, and a cost-push shock. For all three, the welfare-optimal response to rent inflation is negative: rent inflation should lower the policy rate relative to a rule that ignores rents. We first introduce the structural shocks used to evaluate policy, then characterize optimal simple rules, and finally decompose their welfare gains across cohorts and channels.

6.1 Structural Shocks

We evaluate policy under three shocks that raise rents through different mechanisms. A demand shock generates a broad expansion and an endogenous monetary tightening that contributes to rent pressure. A rental availability shock raises the cost of supplying rental housing while generating little nondurable inflation. A cost-push shock is an inflationary shock outside housing that contracts activity and raises rents through the rule's own tightening. Together, they allow us to assess whether a negative response to rent inflation is desirable both when monetary policy contributes to the rent increase, in a demand expansion or after an inflationary supply shock, and when rent pressure originates primarily in the rental market. Appendices F.4, F.5, and F.6 report the full impulse responses.

6.1.1 Demand Shock

A transitory fall in β_t makes households more impatient and raises current demand. We study an innovation that lowers β_t by 0.05 on impact, after which β_t progressively returns to its steady-state level following $\beta_t = \beta + \Delta\beta \rho_\beta^{t-1}$ for $t \geq 1$, with $\Delta\beta = -0.05$ and $\rho_\beta = 0.5$.

As in a standard demand expansion, households bring consumption forward, raising output, labor, and nondurable inflation (Clarida et al., 1999; Galí, 2015). The benchmark Taylor rule responds by raising the policy rate, transmitting the expansion to housing decisions through borrowing costs and returns on savings.

The expansion shifts households toward renting despite the rise in labor income. Lower patience reduces the incentive to save for a down payment and lowers the (dis-

counted) benefits of being a homeowner in the future, while the endogenous tightening raises mortgage payments and the cost of new borrowing. Both forces discourage homeownership. Higher labor income works in the opposite direction by relaxing the loan-to-income limit in (5) and easing access to ownership. The rise in the renter share indicates that lower patience and higher borrowing costs jointly outweigh this improvement in access to credit.

6.1.2 Rental Availability Shock

Our demand shock is broad-based and impacts both housing and non-housing consumption. We now turn to a shock that directly affects the rental market. Specifically, we study a reduction in rental supply in the form of a transitory increase in the mortgage costs borne by landlords. Namely, households in the landlord state ($h = 2$) who carry mortgage debt pay a wedge κ_t^L in place of the common wedge κ , where $\kappa_t^L = \kappa + \Delta\kappa \rho_\kappa^{t-1}$, $\Delta\kappa = 0.60$, and $\rho_\kappa = 0.7$. The wedge rises from 0.75 to 1.35 on impact and then decays, without directly changing owner-occupiers' borrowing costs or the return on savings. At the steady-state real interest rate, this increase corresponds to an annualized mortgage spread rising from 1.5 to 2.7 percentage points. The additional wedge revenue is rebated lump sum to households through (A.9).

At unchanged rents, higher financing costs discourage owners from becoming landlords and encourage indebted landlords to exit. The resulting contraction in rental supply raises rents and shifts households toward owner-occupation. Under the benchmark rule, the real average rent peaks at about 1.5 percent above steady state, while nondurable inflation and the policy rate move little. The experiment therefore generates substantial rent pressure with only a small endogenous monetary response.

6.1.3 Cost-Push Shock

Our third shock raises inflation outside housing. We study a transitory fall in the elasticity across labor varieties in the wage Phillips curve (14), $\ln \epsilon_t = \ln \epsilon + \Delta_\epsilon \rho_\epsilon^{t-1}$, with $\Delta_\epsilon = -0.20$ and $\rho_\epsilon = 0.7$. A lower elasticity raises the desired wage markup, and with it wage and nondurable inflation, without directly changing housing costs or borrowing conditions.

Under the benchmark rule, nondurable inflation rises by about 0.4 annualized percentage points on impact, and the policy response raises the real interest rate by about 35 annualized basis points. Consumption and the house price fall. As after a monetary tightening, higher rates raise rents through the financing and portfolio margins of Section 4.3

and shift households from owner-occupation toward renting: the real average rent peaks at about 0.3 percent above steady state after two years. The rent pressure therefore originates in the rule's own response to inflation outside housing.

6.2 Optimal Simple Rules

We evaluate welfare conditional on a one-time, unanticipated MIT innovation to each structural shock and choose ϕ_r to maximize welfare over the resulting transition, in the spirit of the optimal simple rules of [Schmitt-Grohé and Uribe \(2007\)](#),

$$\phi_r^* \in \arg \max_{\phi_r} \mathcal{W}(\phi_r), \quad \mathcal{W}(\phi_r) = \sum_{t=1}^{\infty} \tilde{\beta}_t \left[\int u(c_{it}, h_{it}) di - v(N_t) - \frac{\theta_w}{2} (\pi_t^w)^2 \right]. \quad (26)$$

Whenever we evaluate welfare, we adopt an equal-weighted utilitarian criterion.²⁹ The discount factor satisfies $\tilde{\beta}_1 = 1$ and $\tilde{\beta}_t = \prod_{\ell=1}^{t-1} \beta_{\ell}$.³⁰

The benchmark is $\phi_r = 0$, under which policy responds only to nondurable inflation. Thus, the exercise asks whether nominal rent inflation should raise or lower the policy rate relative to that benchmark, holding fixed the response to nondurable inflation. We compute the optimum separately for the demand, rental availability, and cost-push shocks³¹ and report welfare differences as consumption equivalents, in percent of permanent consumption. Finally, we also report welfare changes from the benchmark to the rent-inclusive (HICP-style) rule of Section 5.

6.2.1 The Optimal Weight on Rent Inflation

We find that the welfare-optimal coefficient on rent inflation is negative under all three shocks: $\phi_r^* = -0.25$ under the demand shock, $\phi_r^* = -0.10$ under the rental availability shock, and $\phi_r^* = -0.45$ under the cost-push shock. Table 5 reports the associated welfare gains relative to the rule that responds only to nondurable inflation.³² These gains are conditional on a single unanticipated shock, so they are not comparable to unconditional

²⁹ The objective excludes the type-I extreme-value taste shocks $\sigma_{\varepsilon} \varepsilon(h')$ that smooth the discrete tenure choice ([Iskhakov et al., 2017](#)). Their scale is not calibrated to any preference moment; we treat them as a smoothing device of the solution method, with no welfare content.

³⁰ The planner discounts with the households' realized discount-factor path, so, under the demand shock, welfare inherits the shocked path—a form of time-dependent discounting in the spirit of [Dávila and Schaab \(2023\)](#).

³¹ We do not optimize ϕ_r against the monetary shock itself. Optimizing a rule's response to a shock that originates in the rule conflates responding to rent inflation with reversing the policy innovation.

³² All three coefficients also improve welfare relative to $\phi_r = 0$ when lease survival rises from 11/12 to 23/24, which doubles the mean interval between exogenous rent resets from three to six years, conditional on remaining in the same rental tenure. This comparison holds the policy coefficients at their baseline values.

welfare gains in models with aggregate risk, where shocks recur every period. A natural benchmark is the welfare cost of the shock itself: under the benchmark rule, the demand shock costs households 0.05 percent of permanent consumption, and the rental availability and cost-push shocks 0.008 percent each. The optimized rule offsets 3.5, 7.2, and 43.2 percent of these costs.³³

	Demand	Rental availability	Cost push
<i>Rule on nominal average rent inflation, π_t^{rent}</i>			
Optimal weight, ϕ_r^*	-0.25	-0.10	-0.45
Welfare gain at ϕ_r^*	0.0018	0.0006	0.0035
Effective coefficient, $\phi_\pi + \phi_r^*$	1.25	1.40	1.05
<i>HICP-style rule, $\omega_r = 0.06$</i>			
Welfare change	-0.0013	-0.0012	-0.0007
<i>Rule on the real average rent level, $(\bar{p}_t^r - p_{ss}^r) / p_{ss}^r$</i>			
Optimal weight, ϕ_r^*	-0.20	-1.00	< -2.00
Welfare gain at ϕ_r^*	0.0024	0.0113	0.0050
<i>Cumulative average-rent deviation</i>			
Benchmark, $\phi_r = 0$	17.66	7.04	4.43
HICP-style rule	18.19	7.19	4.91
Rent-inflation rule, at ϕ_r^*	16.31	7.12	2.50
Rent-level rule, at ϕ_r^*	4.78	-0.11	0.19

Table 5. Optimal Rent Responses under Alternative Targets

Note. The first block optimizes the additive rule (25) and the third block the rule on the rent level (28), against one shock per column and holding $\phi_\pi = 1.5$, while the second block reports the fixed HICP-style rule of Section 5, equivalent to $(\phi_\pi, \phi_r) = (1.41, +0.09)$. Welfare changes are consumption equivalents relative to the $\phi_r = 0$ benchmark under the same shock, in percent of permanent consumption; Appendix D.2 defines the consumption equivalent and the discounting convention behind it. The lower block reports the sum of percent deviations of the real average rent from steady state over the transition. Under the cost-push shock, welfare under the rule on the rent level still rises at $\phi_r = -2.00$; the table reports the gain and the cumulative rent deviation at that coefficient.

A negative coefficient makes the rent component of the policy rule prescribe accommodation when measured rent inflation is positive. This response works through the rental-supply margins identified in Section 4.3: lower interest rates reduce both mortgage costs and the return available on alternative financial investments. At unchanged rents, supplying rental housing becomes more attractive, so a smaller increase in rents is needed

³³ The welfare cost of a shock is the consumption equivalent, defined as in Appendix D.2, that makes the benchmark transition as good as remaining in the stationary equilibrium. For the demand shock, it uses the same realized discount factors as the welfare criterion. The HICP-style rule of Section 5 instead raises these costs by 2.6, 15.4, and 9.4 percent.

to induce households to provide it. Under the demand shock, this accommodation offsets part of the rental pressure generated by the endogenous monetary tightening. Under the rental availability shock, it partly offsets the increase in landlords' financing costs. Under the cost-push shock, it offsets part of the rental pressure generated by the rule's own response to higher inflation. The common sign therefore extends the case for accommodation beyond rent pressure generated by monetary policy itself.

Rent relief benefits tenants by lowering their housing expenditures, and is particularly valuable to households with low consumption and high marginal utility. The rent-inflation rule provides some of this relief by moderating the initial increase in rents. The welfare consequences of accommodation also extend beyond rental payments. Lower mortgage costs benefit indebted households, while lower returns on savings reduce interest income. The government's interest savings are returned through the lump-sum rebate, changing how financial income is distributed across households. The optimum reflects these effects alongside the transfer from landlords to tenants through lower rents, and the changes in labor supply and wage-inflation costs.

The HICP-style rule, by contrast, lowers welfare under all three shocks. In these experiments, the welfare criterion favors a negative response to rent inflation over both excluding rents and incorporating them with a positive expenditure weight. This raises the question of whether responding to rents is essential to the improvement, or whether a generally less hawkish rule could achieve the same gains.

6.2.2 Is This Just a Less Hawkish Rule?

This negative weight admits two readings. The additive rule (25) can be rewritten identically as

$$i_t = r_{ss} + (\phi_\pi + \phi_r) \pi_t^c + \phi_r (\pi_t^{\text{rent}} - \pi_t^c), \quad (27)$$

so the optimized rule amounts to a lower effective coefficient on nondurable inflation plus a negative response to real rent growth. We now study whether the welfare improvement just comes from the first component rather than from a meaningful response to rents. To do so, we compare the optimized rule with rules that have no rent term ($\phi_r = 0$) but a different coefficient on nondurable inflation.

We report the result of the comparison in Figure 6. Under the demand shock, the welfare criterion favors a more accommodative response than the benchmark: lowering ϕ_π is welfare-improving. However, even at $\phi_\pi = 1.1$, which would substantially weaken the central bank's response to inflation, the rule without a rent term delivers only about

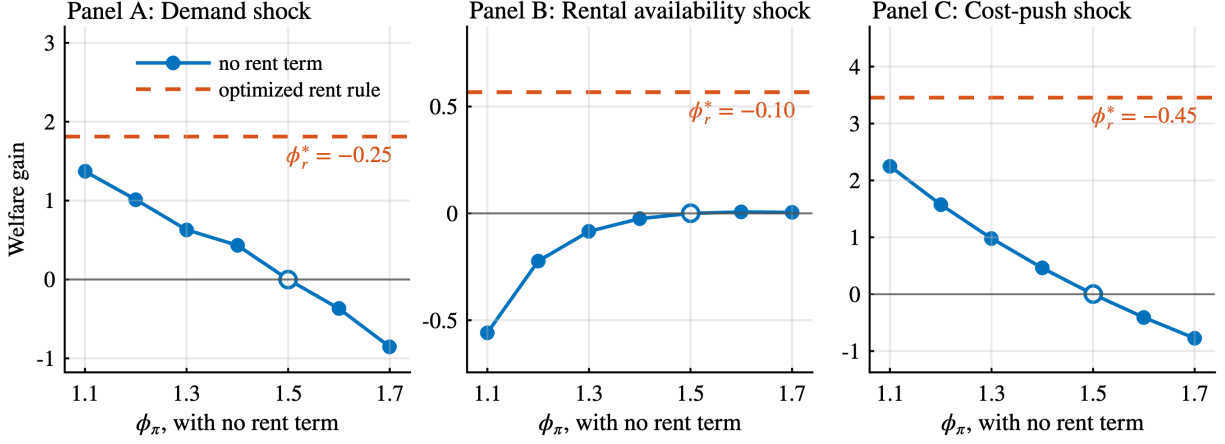


Figure 6. Welfare under Alternative Monetary Policy Rules

Note. The solid blue lines report welfare relative to the shock-specific benchmark $(\phi_\pi, \phi_r) = (1.5, 0)$ for rules without a rent term, varying ϕ_π over $[1.1, 1.7]$ with $\phi_r = 0$. The dashed orange lines report the welfare gain from optimizing the rent coefficient while holding $\phi_\pi = 1.5$. Both are consumption-equivalent gains, measured in thousandths of a percent of permanent consumption.

three-quarters of the welfare gain obtained with the optimized rent coefficient.

In the case of the rental availability shock, lowering ϕ_π is welfare-decreasing, and raising it does not deliver much improvement either. When shocks directly affect the rental and housing markets, the central bank can substantially improve welfare by reacting negatively to rents, which cannot be achieved with a more (or less) hawkish rule on nondurable inflation alone.

Under the cost-push shock, the welfare criterion again favors a more accommodative response, and lowering ϕ_π improves welfare. Even at $\phi_\pi = 1.1$, however, the rule without a rent term delivers only about two-thirds of the gain from the optimized rent coefficient.

Put together, these results highlight that our main welfare results are driven by the housing and rental supply and demand mechanisms we have described in Sections 4 and 5, and cannot be replicated simply with a less hawkish rule.

6.2.3 Rent Inflation versus the Rent Level

The rule so far responds to nominal average rent inflation, the growth rate of the rent component in the consumer price index. However, what enters renters' budgets, and most directly affects their welfare, is the cost of rental payments relative to prices of other goods. We now explore the possibility that the central bank directly targets this measure by solving the same economy under a rule on the deviation of the real average rent from

steady state, holding the remaining features of the policy rule fixed,

$$i_t = r_{ss} + \phi_\pi \pi_t^c + \phi_r \frac{\bar{p}_t^r - p_{ss}^r}{p_{ss}^r}. \quad (28)$$

Figure 7 shows that the choice of target substantially changes the rental burden under the demand shock. At their respective optima, while the inflation rule reduces the cumulative deviation of the real average rent by only about 8 percent, the level rule achieves a 73 percent reduction. Under the rental availability shock, the inflation rule leaves the cumulative deviation approximately unchanged, while the level rule eliminates it (see Figure F.8 in Appendix F.5). Under the cost-push shock, the inflation rule reduces it by about 43 percent and the level rule, at $\phi_r = -2.00$, by 96 percent (see Figure F.12 in Appendix F.6). The level rule also delivers a larger welfare gain under all three shocks, as reported in the third block of Table 5.

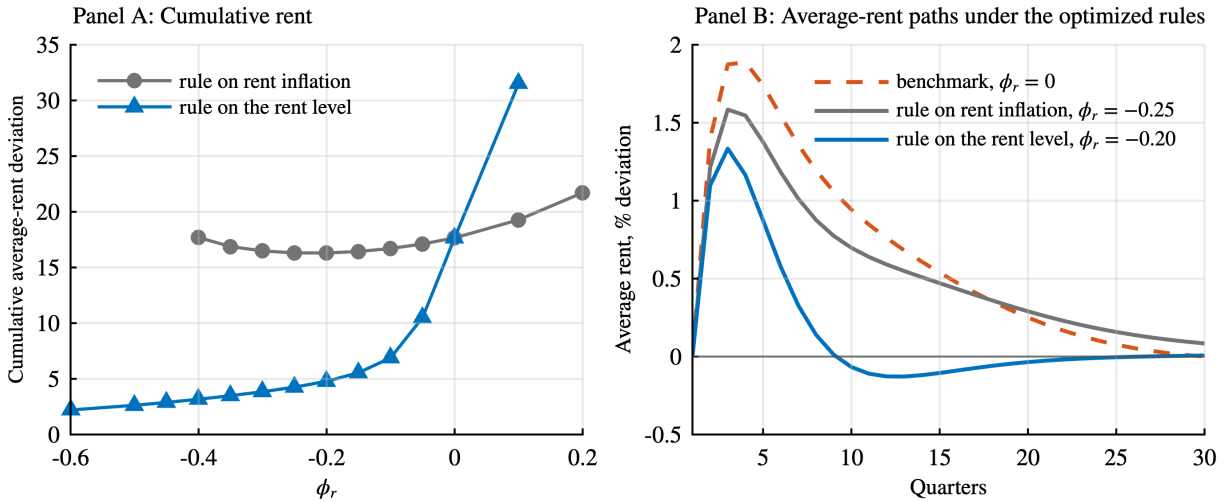


Figure 7. Average Rents under Alternative Policy Rules: Demand Shock

Note. All rules hold $\phi_\pi = 1.5$. Panel A plots the sum of percentage deviations of the real average rent from steady state over the quarterly transition against the rent coefficient, for rules on nominal rent inflation and on the real rent level. Panel B reports the real average rent in percent deviations from steady state under the benchmark ($\phi_r = 0$) and each of the two optimized rules.

The difference comes from the duration of the response implied by the rent term. As rent inflation weakens, this term provides less accommodation; when rent inflation turns negative, its contribution to the policy rate reverses even if real rents remain above steady state. The level rule continues to prescribe lower interest rates while the real rent gap is positive. This more persistent response acts on the financing and portfolio margins described in Section 4.3, reducing the compensation households require to supply rental

housing. Slow lease adjustment delays the transmission of lower market rents to the rents paid by tenants. Both rules operate through this adjustment process, and contracts eventually reset under either rule. The greater cumulative relief under the level rule therefore reflects a larger and more persistent reduction in market rents.

In practice, while rent inflation as a component of the price index is observable, deviations of real rents from their (trend) steady-state level are not. Therefore, the implementation of this policy requires measurement or modeling assumptions. However, the larger welfare gains it implies suggest that even approximating it might be welfare-improving.

6.3 Distributional Decomposition across Cohorts and Channels

We now examine who benefits from the reduction in the rental burden and through which channels. We focus on the optimized rent-level rule of Section 6.2.3 under the demand shock.³⁴ We track cohorts defined at the date of the shock and decompose their welfare changes following the decomposition based on different prices, interest rates, and taxes introduced in Section 4.3. Each channel replaces one input path under the benchmark rule with its path under the optimized rule, holding the remaining inputs at their benchmark values. Figure 8 reports these contributions together with the net welfare change for each cohort.

Lower rents (purple) under the optimal policy transfer resources from landlords to tenants. The rent channel raises welfare in every initial renter cohort and reduces it among landlords. Its welfare value depends on the households on each side of the market: a reduction in rental payments is particularly valuable to tenants with low consumption and high marginal utility. This gives a utilitarian planner a distributional reason to value rent relief.

The net welfare gain is progressive within renters. Poor renters benefit most from lower housing costs and have little financial wealth exposed to the decline in returns on savings. As financial wealth rises, the loss of interest income (light green) becomes more important and eventually outweighs the benefit of lower rents. The wealthiest renter cohorts therefore lose overall despite gaining through the rent channel.

The rent channel is the largest source of gains for renter cohorts, contributing between two and three times as much as the fiscal channel (taxes and transfers, in rose).³⁵ However,

³⁴ Appendix F.4 reports the decomposition under the demand-shock rent-inflation rule. Appendices F.5 and F.6 report the decompositions under both targets for the rental availability and cost-push shocks.

³⁵ The fiscal channel mainly reflects the redistribution associated with public debt service (Appendix F.7). Lower interest rates reduce government interest payments and increase the lump-sum rebate, while the corresponding losses of interest income enter the portfolio effects. Because public debt is held by households, this mechanism redistributes resources rather than creating them. Lower mortgage-wedge revenues partly

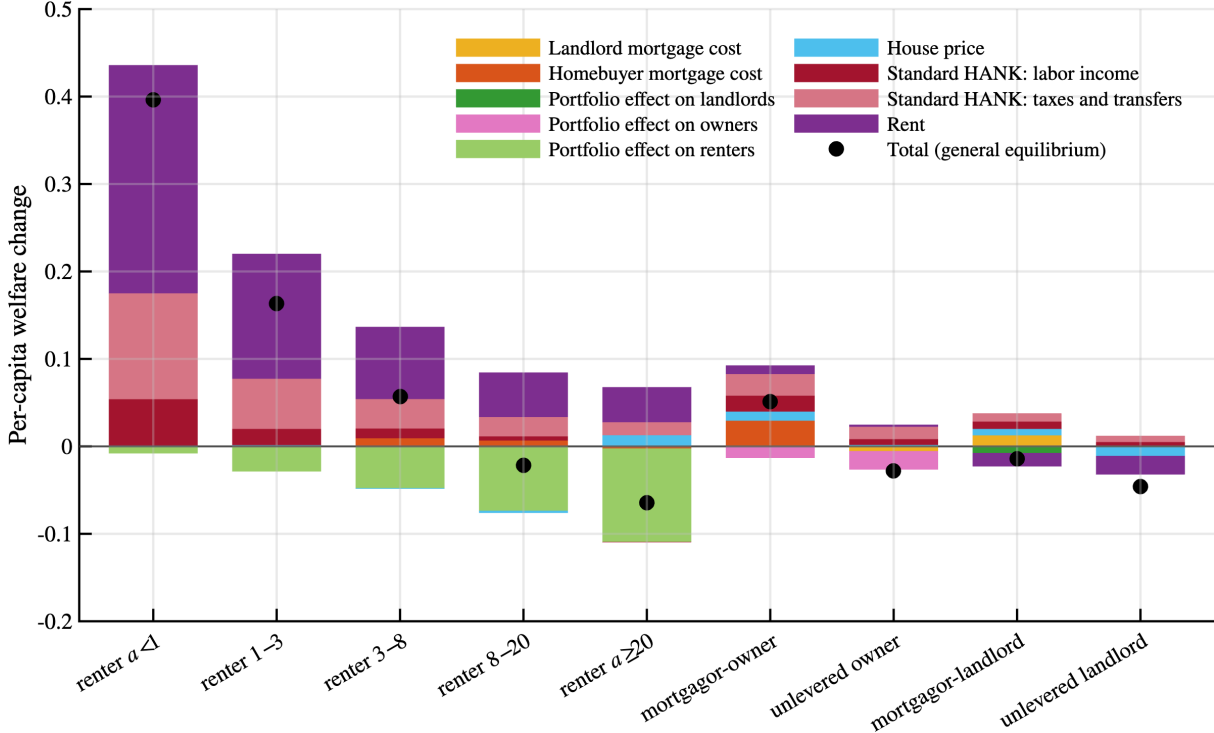


Figure 8. Distributional Decomposition of the Welfare Gain

Note. Per-capita welfare change from moving ϕ_r from 0 to -0.20 under the rent-level rule, demand shock, by date-of-shock cohort. Welfare changes are measured in utility units. Net financial assets a are measured in units of quarterly steady-state income. Each channel holds the benchmark equilibrium and replaces one input with its path under the optimized rule; the rent channel moves the rent tenants pay and the rent landlords receive together. Channels and colors follow Figure 4, and the standard HANK channel is split into its labor-income and fiscal (taxes and transfers) legs. Black dots are the general-equilibrium total. The channel experiments hold aggregate labor and wage inflation at their benchmark paths. The difference between the sum of the channel effects and the general-equilibrium total includes changes in labor disutility and wage-adjustment costs, together with interaction effects across channels.

mortgagor-owners also benefit from the optimal policy, mostly through the reduction in their mortgage costs implied by a lower interest-rate path than under the benchmark rule.

Efficiency versus redistribution. Following [Dávila and Schaab \(2025\)](#), we split the marginal welfare gain from the rent coefficient into efficiency, the sum of households' willingness to pay, and redistribution, which captures whether the change favors the households the planner weighs more (Figure 9). Under the demand shock, efficiency favors a less negative coefficient: accommodation relieves households whose marginal utility is temporarily high, but it also raises hours and wage inflation during a boom, and

offset the increase in the rebate.

these costs dominate. The negative optimum rests on redistribution. Under the rental availability shock the ranking reverses. The shock barely moves aggregate activity, while accommodation relieves levered landlords and mortgagors precisely when their financing costs have risen; efficiency alone favors a coefficient of about -0.15 , and redistribution pulls it back to the optimum of -0.10 . Under the cost-push shock, both components favor a negative coefficient, with redistribution the larger of the two. As under the rental availability shock, efficiency alone favors a coefficient of about -0.49 , and redistribution pulls it back to the optimum of -0.45 . Rent relief itself is mostly redistribution, but part of it is insurance: it moves resources toward the dates and circumstances in which tenants value them most.

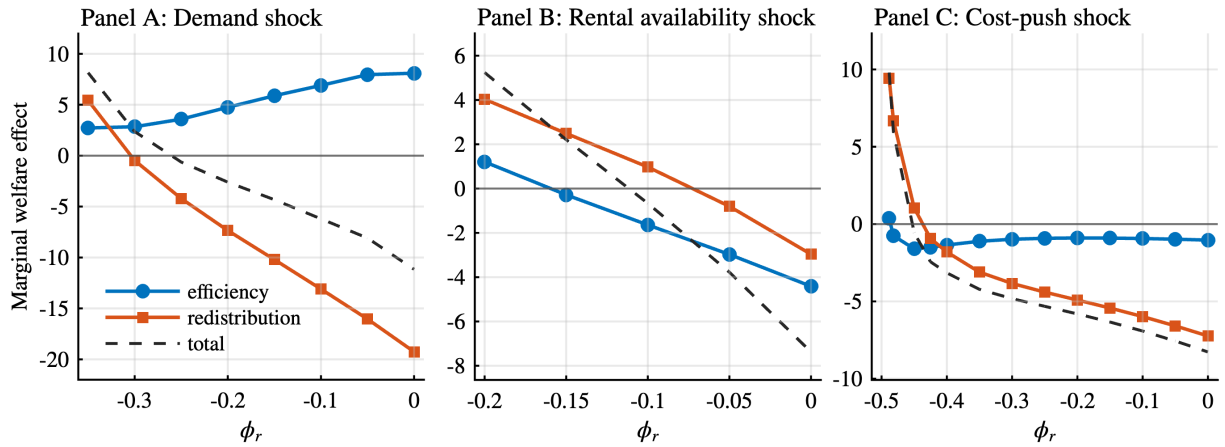


Figure 9. Efficiency and Redistribution

Note. Marginal welfare effect of the coefficient on rent inflation in rule (25), $dW/d\phi_r$, holding $\phi_\pi = 1.5$, split into efficiency and redistribution components following [Dávila and Schaab \(2025\)](#). The dashed line is their sum. Positive values favor a higher coefficient. Effects are measured in 10^{-5} units of perpetual consumption, a unit of consumption to every household at every date. Appendix D.2 contains the derivations.

Robustness to assumptions about fiscal rebates. The residual fiscal requirement is rebated lump sum in our baseline, which makes it progressive. A natural question is whether the optimal negative sign is an artifact of that assumption and of the relatively large size of the taxes and transfers channel in Figure 8. For robustness, we re-solve the model under the demand shock with the residual rebated in proportion to labor productivity, allocating a larger share of the rebate to households with higher labor incomes. We find that the optimal weight on rent inflation in rule (25) remains negative, at $\phi_r^* = -0.20$, and it remains negative under the other two shocks (Appendix F.7). The gain is smaller, about three-fifths of its value under the lump-sum closure, reflecting that the proportional rebate

does not distribute fiscal residuals progressively and therefore forgoes potential welfare gains. However, this result highlights that our welfare results are not a direct artifact of fiscal redistribution, but rather reflect meaningful mechanisms related to the housing and rental markets.³⁶

7 Conclusion

Rent inflation poses a distinct problem for monetary policy because interest rates affect both the demand for rental housing and the return required to supply it. In our model, a monetary tightening keeps would-be buyers renting while raising landlords' financing and opportunity costs. Rents rise until households are willing to supply the additional rental housing. About three-quarters of the effect of higher interest rates on the rental market comes from landlords' financing and portfolio incentives. Sticky leases delay the response of measured rents.

This combination changes the consequences of including rents in the inflation target. A rule that responds positively to measured rent inflation keeps reacting to rental pressure generated by its own earlier tightening. The additional response raises the cost of supplying rental housing and prolongs the consumption contraction. Two years after the monetary shock, the cumulative consumption loss is about 10 percent larger than under a rule that ignores rents, while both nondurable and rent inflation are more volatile. Giving rents their expenditure weight in the policy index therefore worsens stabilization.

The welfare-optimal response is accommodative under the general demand, rental supply, and cost-push shocks we consider: rent inflation lowers the policy rate relative to a rule that ignores it. Accommodation reduces the financing and opportunity costs of supplying rental housing, relieves mortgage borrowers, and lowers the government's interest bill. Returning these interest savings to households changes the distribution of financial income, while savers receive lower returns. The welfare gain reflects these household exposures together with changes in employment and inflation costs. Within the simple rules we study, responding negatively to rent inflation delivers gains that cannot be replicated by changing the response to nondurable inflation alone.

The choice of rent target also matters. Responding to the level of real rents delivers more sustained reductions in rental payments than responding to rent inflation, producing

³⁶ In a separate experiment, we hold government spending, the land revenue G_t^L absorbed by the government (Appendix A.4), at its steady-state level and rebate deviations of land income lump sum to households. Under this closure, the baseline rent-inflation coefficients, $\phi_r = -0.25$ under the demand shock, $\phi_r = -0.10$ under the rental availability shock, and $\phi_r = -0.45$ under the cost-push shock, continue to improve welfare relative to $\phi_r = 0$. These comparisons hold the policy coefficients fixed.

larger welfare gains under all three shocks. This rent relief has a redistributive effect: landlords receive less rental income, while tenants face lower housing costs. It particularly benefits renters with little financial wealth and high marginal utility of consumption. Wealthier renters can nevertheless lose overall because lower interest income outweighs their rent relief, while indebted homeowners benefit through lower mortgage payments. The appropriate treatment of rent inflation therefore depends on rental-supply incentives and households' exposure to changes in rents and interest rates.

Our framework abstracts from institutional features that shape monetary transmission in housing and rental markets. Mortgages in our model are effectively adjustable-rate. With fixed-rate debt, a tightening would not immediately reprice existing mortgages, weakening the direct mortgage-cost channel. Higher rates would still affect new borrowing and raise the return on alternative financial investments, so the landlord portfolio channel would remain operative.

In our welfare analysis, we optimize a simple rule and stop short of solving the full Ramsey problem. Therefore, our results should be interpreted as conditional on the particular shocks that we study, with the additional caveat that these are unanticipated by households. Both solving the full Ramsey problem and incorporating our framework in a macro model with aggregate risk are very interesting, but computationally intensive, avenues for future research.

Taken broadly, the policy implications of our work suggest that central banks should consider the specific features of rent inflation when making policy decisions.

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APPENDIX

A Derivations and Model Details

A.1 Logit Housing Tenure Choice

At the beginning of each period, the household observes the vector of choice-specific taste shocks $\varepsilon = \{\varepsilon(h')\}_{h'}$ in the household problem (7) and then chooses its next-period housing state. The formulas below are ex-ante with respect to these shocks: they condition only on the state (a, h, y, s) .

The continuation value depends on whether the household retains its rental tenure. Conditional on remaining a renter or landlord, the lease survives with probability φ and enters the regulated pool; otherwise it resets at the market rent. Households entering a new rental tenure begin with an unregulated lease. Thus,

$$\begin{aligned} \mathbb{E}_t[V_{t+1}(a', h', y', s') \mid y, h, h'] \\ = \begin{cases} \mathbb{E}_{y'|y}[\varphi V_{t+1}(a', h', y', 0) + (1 - \varphi)V_{t+1}(a', h', y', 1)], & h' = h \in \{0, 2\}, \\ \mathbb{E}_{y'|y}[V_{t+1}(a', h', y', 1)], & h' = 1 \text{ or } h' \neq h. \end{cases} \end{aligned} \quad (\text{A.1})$$

For owner-occupiers, the lease state has no effect on payoffs and is normalized to $s' = 1$.

For each next-period housing state $h' \in \{0, 1, 2\}$, we define the deterministic choice-specific value

$$\tilde{V}_t(a, h, y, s; h') = \max_{c, a'} \mathcal{U}(c, h, n) + \beta_t \mathbb{E}_t[V_{t+1}(a', h', y', s') \mid y, h, h'],$$

where the maximization is subject to the budget constraint (3) and the borrowing constraints (5), and the expectation over the next lease state s' follows the transition rule in equation (A.1). The value $\tilde{V}_t$ collects everything the household controls once the tenure decision is fixed: it is the value of choosing consumption and savings optimally, conditional on moving to housing state h' .

After the taste shocks are observed, the household problem (7) reduces to a deterministic discrete choice across the three shock-augmented values,

$$V_t^\varepsilon(a, h, y, s) = \max_{h' \in \{0, 1, 2\}} \tilde{V}_t(a, h, y, s; h') + \sigma_\varepsilon \varepsilon(h').$$

Because the shocks $\varepsilon(h')$ are drawn independently from a type-I extreme-value distribution with unit scale and zero mean, integrating over them yields closed-form expressions (Iskhakov et al., 2017). The ex-ante probability that a household in state (a, h, y, s) chooses housing state h' is

$$\mathbb{P}_t(h' \mid a, h, y, s) = \frac{\exp(\tilde{V}_t(a, h, y, s; h')/\sigma_\varepsilon)}{\sum_{j \in \{0,1,2\}} \exp(\tilde{V}_t(a, h, y, s; j)/\sigma_\varepsilon)}, \quad (\text{A.2})$$

and the expected value before the shocks are realized is the log-sum

$$V_t(a, h, y, s) = \mathbb{E}_\varepsilon[V_t^\varepsilon(a, h, y, s)] = \sigma_\varepsilon \log \left[\sum_{j \in \{0,1,2\}} \exp(\tilde{V}_t(a, h, y, s; j)/\sigma_\varepsilon) \right]. \quad (\text{A.3})$$

Equation (A.3) is the continuation value that enters the Bellman equation (7) one period earlier, and the probabilities in equation (A.2) govern the evolution of the distribution $g_t(a, h, y, s)$ across housing states. Castellanos et al. (2024) and Albuquerque et al. (2025) use the same extreme-value structure for the housing tenure decision.

The scale parameter σ_ε governs how sensitive tenure choices are to differences in the choice-specific values. As $\sigma_\varepsilon \rightarrow 0$, the probabilities in equation (A.2) concentrate on the housing state with the highest deterministic value, and the tenure decision reduces to choosing the h' that maximizes $\tilde{V}_t$. A larger σ_ε smooths the transition probabilities across the renter, owner-occupier, and landlord states: households whose choice-specific values are close transition gradually rather than all at once. The shocks are idiosyncratic taste shocks over tenure states, capturing motives for moving that are orthogonal to the household's financial state.

A.2 Evolution of Average Rents

The weight λ_h in (9) is the share of renters or landlords who effectively stay in the regulated market each period, which takes into account both the exogenously set φ and the stationary probabilities that renters and landlords choose to stay in their respective states:

$$\lambda_h \equiv \varphi \cdot \mathbb{P}_{ss}(h_{t+1} = h \mid h_t = h), \quad h \in \{0, 2\}. \quad (\text{A.4})$$

The tenure-persistence probability is itself an equilibrium object: households choose tenure, so its realized value $\mathbb{P}_t(h_{t+1} = h \mid h_t = h)$ moves along the transition as prices do. We hold λ_h at its stationary value (A.4) throughout. Households therefore form expectations and choose tenure under a constant λ_h , while the realized tenure and lease

shares respond endogenously to the shock. The approximation is accurate because tenure is highly persistent and its persistence barely moves: across the four shocks we study, the realized renter tenure-persistence probability departs from its stationary value by at most 0.43 percent and the landlord probability by at most 1.31 percent. Feeding the realized path of λ_h , built from the realized tenure-persistence probabilities, through (9) in place of the constant λ_h changes the average rent by less than 0.35 percent of its peak deviation. This bounds the measurement cost of holding λ_h fixed, not its effect on decisions, which households take under the constant λ_h . Fixing λ_h adds no parameter of its own: φ is set outside the lease block and the persistence probabilities are the model's own stationary tenure transitions.

Landlord tenure is less persistent than renter tenure in the stationary equilibrium, so $\lambda_2 < \lambda_0$, and both are held at their stationary values throughout the transition. The two sides of the rental market therefore carry different regulated rents, and the aggregate rent paid by renters need not equal the aggregate rent received by landlords. Section 2.7 and Appendix A.4 discuss this accounting residual.

Iterating (9) backward, using $\frac{\mu_{t,0}^{0,0}}{\mu_t^0} = \lambda_0 \frac{\mu_{t-1}^0}{\mu_t^0} \approx \lambda_0$, and substituting the result into (10) gives the measured rent index a simple form. The average rent is, to a close approximation, a geometrically weighted average of current and past market rents,

$$\bar{p}_t^r = (1 - \lambda_0) \sum_{k=0}^{\infty} \lambda_0^k p_{t-k}^r, \quad \text{with } p_{t-k}^r \equiv p_{ss}^r \text{ whenever } t - k \leq 1. \quad (\text{A.5})$$

Two properties follow. The weight on a market rent k quarters old falls by the constant factor λ_0 each quarter, so the index lags the market rent by a mean of $\lambda_0 / (1 - \lambda_0)$ quarters. Over the transition, a movement in market rents passes into the average rent essentially one-for-one: lease turnover delays the movement without shrinking it. The approximation is tight: across the four shocks studied in the paper, the geometric filter (A.5) tracks the solved average rent to within 0.8 percent of its peak deviation. Figure A.1 illustrates both properties.

A.3 Derivation of the Nonlinear New Keynesian Wage Phillips Curve

Each union introduced in Section 2.3 chooses its nominal wage to maximize the discounted utility of its members, internalizing the effect of its wage on their labor income and hours, and passing the Rotemberg adjustment cost through to them as a utility loss. Optimal wage setting delivers equation (14).

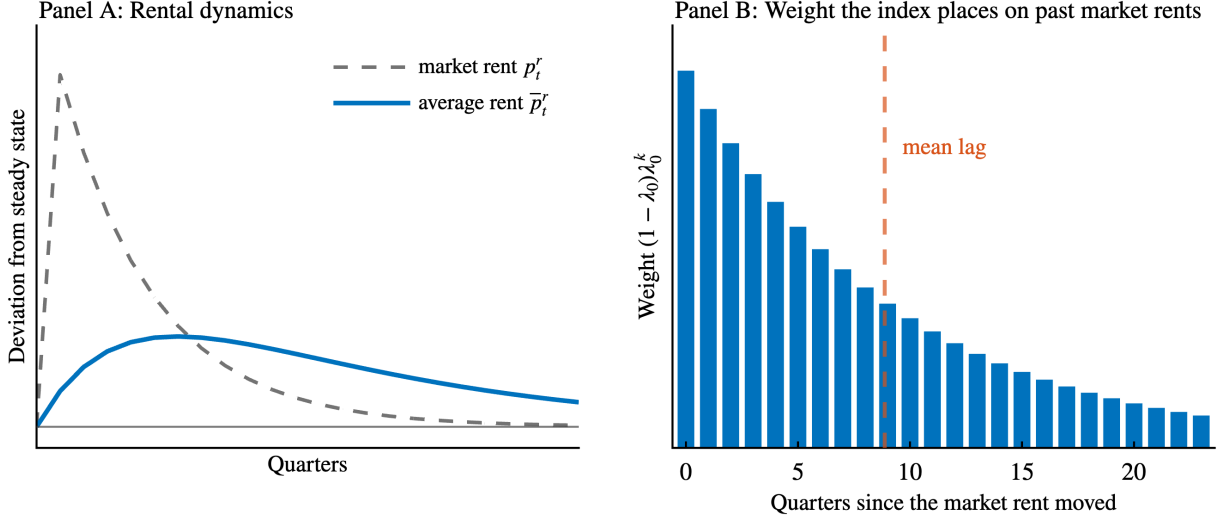


Figure A.1. Adjustment of Average Rents

Note. Panel A plots illustrative rental dynamics, both the average rent and the market rent. Panel B shows the weight $(1 - \lambda_0)\lambda_0^k$ that the index places on the market rent of k quarters ago, with the implied mean lag marked.

Labor packer and union labor demand. A continuum of unions indexed by $k \in [0, 1]$ transforms household hours into differentiated labor services. Each household supplies hours to every union, so a household's total hours aggregate the services supplied through all unions, and each union rations hours equally across its members. A competitive labor packer combines the union services N_{kt} into aggregate labor according to the CES technology

$$N_t = \left(\int_0^1 N_{kt}^{\frac{\epsilon_t - 1}{\epsilon_t}} dk \right)^{\frac{\epsilon_t}{\epsilon_t - 1}},$$

where ϵ_t is the elasticity of substitution across labor varieties, and sells the bundle to the consumption-good producer at the nominal wage W_t . Cost minimization by the packer yields the demand for union k 's service and the wage index,

$$N_{kt} = \left(\frac{W_{kt}}{W_t} \right)^{-\epsilon_t} N_t, \quad W_t = \left(\int_0^1 W_{kt}^{1 - \epsilon_t} dk \right)^{\frac{1}{1 - \epsilon_t}},$$

where W_{kt} is the nominal wage set by union k , so that $\partial N_{kt} / \partial W_{kt} = -\epsilon_t N_{kt} / W_{kt}$. Consistent with the definition of aggregate wage inflation in Section 2.3, union-specific gross wage inflation is $1 + \pi_{kt}^w = W_{kt} / W_{k,t-1}$. When varying its own wage, the union takes all

aggregate objects as given, including W_t , N_t , prices, and the household distribution.

Union's wage-setting problem. Because the discount factor may vary over time, we define the compound discount $D_{t,j} = \prod_{\ell=t}^{j-1} \beta_\ell$, with $D_{t,t} = 1$. At date t , union k chooses the sequence of its wages $\{W_{kj}\}_{j \geq t}$ to maximize the discounted utility of its members, augmented with the Rotemberg cost as in (13),

$$\sum_{j=t}^{\infty} D_{t,j} \left[\int \left(u(c_{ij}, h_{ij}) - v(n_{ij}) \right) di - \frac{\theta_w}{2} (\pi_{kj}^w)^2 \right], \quad (\text{A.6})$$

where i indexes households. Union k internalizes that its wage changes both its members' labor income and, through its labor demand, their total hours: $\partial n_{ij} / \partial W_{kj} = \partial N_{kj} / \partial W_{kj}$. In a symmetric equilibrium, hours rationing delivers $n_{ij} = N_j$, as in equation (12).

A member with productivity y_{it} receives from union k the labor income $W_{kt} y_{it} N_{kt} / P_t^c$, where P_t^c is the nominal price of the consumption good, together with the proportional employment subsidy $\tau_L W_{kt} y_{it} N_{kt} / P_t^c$, which the union internalizes when setting its wage. The subsidy and the tax that finances it, $\tau_L w_t y_{it} N_t$, both enter the transfer $T_t(y)$ described in Section 2.7; because the tax depends on aggregate employment, the individual union takes it as given. The derivative of the wage bill with respect to the union's wage is

$$\frac{\partial(W_{kt} N_{kt})}{\partial W_{kt}} = N_{kt} + W_{kt} \frac{\partial N_{kt}}{\partial W_{kt}} = (1 - \epsilon_t) N_{kt},$$

so the direct change in member i 's resources is

$$\frac{\partial}{\partial W_{kt}} \left[(1 + \tau_L) \frac{W_{kt} y_{it} N_{kt}}{P_t^c} \right] = (1 + \tau_L)(1 - \epsilon_t) y_{it} \frac{N_{kt}}{P_t^c}.$$

By the envelope theorem, the effect on member i 's value is its marginal utility of consumption times this change in resources, $u_c(c_{it}, h_{it}) (1 + \tau_L)(1 - \epsilon_t) y_{it} N_{kt} / P_t^c$.

First-order condition. The date- t wage enters the objective (A.6) through members' date- t resources, their date- t hours, and the wage-adjustment costs at t and $t + 1$. From the definition of union wage inflation,

$$\frac{\partial \pi_{kt}^w}{\partial W_{kt}} = \frac{1}{W_{k,t-1}}, \quad \frac{\partial \pi_{k,t+1}^w}{\partial W_{kt}} = -\frac{W_{k,t+1}}{W_{kt}^2},$$

so the first-order condition with respect to W_{kt} is

$$\int u_c(c_{it}, h_{it})(1 + \tau_L)(1 - \epsilon_t)y_{it} \frac{N_{kt}}{P_t^c} di - v'(n_t) \frac{\partial N_{kt}}{\partial W_{kt}} - \theta_w \pi_{kt}^w \frac{1}{W_{k,t-1}} + \beta_t \theta_w \pi_{k,t+1}^w \frac{W_{k,t+1}}{W_{kt}^2} = 0. \quad (\text{A.7})$$

Symmetry and the aggregate equation. All unions face the same problem, so we impose the symmetric equilibrium $W_{kt} = W_t$, $N_{kt} = N_t$, and $\pi_{kt}^w = \pi_t^w$. Multiplying equation (A.7) by W_t , using $W_t/W_{t-1} = 1 + \pi_t^w$, $W_{t+1}/W_t = 1 + \pi_{t+1}^w$, the real wage $w_t = W_t/P_t^c$, and $\partial N_{kt}/\partial W_{kt} = -\epsilon_t N_t/W_t$, and rearranging yields

$$\pi_t^w (1 + \pi_t^w) = \frac{\epsilon_t}{\theta_w} N_t \left(v'(N_t) - (1 + \tau_L) \frac{\epsilon_t - 1}{\epsilon_t} w_t \int y_{it} u_c(c_{it}, h_{it}) di \right) + \beta_t \pi_{t+1}^w (1 + \pi_{t+1}^w),$$

which is equation (14). Because consumption is conditional on the realized housing tenure choice, the integral is shorthand for the probability-weighted cross-sectional object

$$\int y_{it} u_c(c_{it}, h_{it}) di = \sum_s \sum_{h', h} \iint y u_c(c_t(a, h, y, s; h'), h) \mathbb{P}_t(h' | a, h, y, s) g_t(a, h, y, s) da dy,$$

where $c_t(a, h, y, s; h')$ is the consumption policy conditional on choosing housing state h' and $\mathbb{P}_t(h' | a, h, y, s)$ is the logit probability (A.2) of Appendix A.1.

A.4 Aggregation, Fiscal Closure, and Goods-Market Clearing

We aggregate the household budget constraint, derive the rental cash-flow residual generated by sticky leases, and state the fiscal adjustment under which the goods-market clearing condition (A.10) holds.

Aggregate objects. Let $A_t^- = \int a \mathbb{1}_{a < 0} dg_t < 0$ denote the aggregate borrowing position, $M_t(h, h')$ the mass of households moving from housing state h to h' , and $\bar{T}_t \equiv \int T_t(y) dg_t$ the aggregate transfer. Because the employment subsidy and its financing are both proportional to productivity and $n_t = N_t$, they cancel household by household, so $\bar{T}_t$ equals the lump-sum fiscal adjustment of Section 2.7.

Aggregate housing outlays. Direct verification of the nine entries of the housing-cost matrix (6) shows that each entry decomposes into a change in the value of the housing position and a transaction-tax payment on the units transacted,

$$\psi_t(h, h') = p_t(h') - (1 - \delta_h) p_t(h) + \tau_h p_t^h [q_t(h, h') + q_t(h', h)],$$

where $p_t(h) = p_t^h \tilde{h}_2 \mathbb{1}_{h=1} + p_t^h (\tilde{h}_1 + \tilde{h}_2) \mathbb{1}_{h=2}$ is the market value of the housing position and $q_t(h, h')$ is the quantity of housing quality acquired in the transition (4), so that $q_t(h, h') + q_t(h', h)$ is the gross quantity transacted, purchases plus sales. Summing over the transition masses, using $\sum_{h'} M_t(h, h') = \mu_t^h$ and $\sum_h M_t(h, h') = \mu_{t+1}^{h'}$, and recalling that the aggregate stock satisfies $H_t = \mu_t^1 \tilde{h}_2 + \mu_t^2 (\tilde{h}_1 + \tilde{h}_2)$, we can write aggregate housing outlays as

$$\sum_{h, h'} M_t(h, h') \psi_t(h, h') = p_t^h [H_{t+1} - (1 - \delta_h) H_t] + \mathcal{T}_t^h = p_t^h Y_t^h + \mathcal{T}_t^h, \quad (\text{A.8})$$

where $\mathcal{T}_t^h \equiv \tau_h p_t^h \sum_{h, h'} M_t(h, h') [q_t(h, h') + q_t(h', h)]$ is transaction-tax revenue and the second equality uses the housing law of motion (19).

Rental cash flows under sticky leases. From the budget constraint (3), the rental cash flow of a household in state (h, s) is $\tilde{p}_t^r(h, s)(h - 1)$. Aggregate rent received by landlords net of rent paid by renters is therefore

$$R_t^{\text{net}} = \left(\mu_t^{2,0} \tilde{p}_t^r(2) - \mu_t^{0,0} \tilde{p}_t^r(0) \right) + (\mu_t^{2,1} - \mu_t^{0,1}) p_t^r.$$

Rental-market clearing equates renters and landlords only in total, $\mu_t^{0,0} + \mu_t^{0,1} = \mu_t^{2,0} + \mu_t^{2,1}$, but the two sides' regulated pools do not carry the same regulated rent: by (9) and (A.4) they survive at different rates, so that $\tilde{p}_t^r(0) \neq \tilde{p}_t^r(2)$ and the expression does not collapse further. This residual is zero in the stationary equilibrium and on impact, where $p_1^r = \tilde{p}_1^r(0) = \tilde{p}_1^r(2) = p_{ss}^r$, but is generally nonzero along the transition because the model does not clear the rental market lease state by lease state. From about the eighth quarter onward, the share of renters on regulated leases exceeds the share of landlords on regulated leases by $\lambda_0 - \lambda_2 = 0.030$, as implied by (A.4); measured along the monetary transition, $|R_t^{\text{net}}|$ peaks at 0.007 percent of steady-state consumption, along the demand transition of Section 6.1.1 at 0.047 percent, along the rental availability transition of Section 6.1.2 at 0.030 percent, and along the cost-push transition of Section 6.1.3 at 0.005 percent.

Aggregate budget and fiscal adjustment. Integrating the budget constraint (3), which holds with equality at the optimum, using the housing-outlay identity (A.8), the unit mean of labor productivity, and $w_t N_t = Y_t^c$ from equations (15) and (16), gives

$$C_t + A_{t+1} + p_t^h Y_t^h + \mathcal{T}_t^h = (1 + r_t) A_t + r_t \kappa A_t^- + Y_t^c + \bar{T}_t + R_t^{\text{net}}.$$

Financial-market clearing (20) imposes $A_t = A_{t+1} = \bar{B}$, so

$$Y_t^c - C_t - p_t^h Y_t^h = \mathcal{T}_t^h - r_t \bar{B} - r_t \kappa A_t^- - \bar{T}_t - R_t^{\text{net}},$$

where the right-hand side collects transaction-tax revenue, interest paid on public debt, mortgage-wedge revenue ($-r_t \kappa A_t^- > 0$ because $A_t^- < 0$), the lump-sum fiscal adjustment, and the rental residual. The government chooses the aggregate transfer to rebate these revenues,

$$\bar{T}_t = \mathcal{T}_t^h - r_t \bar{B} - r_t \kappa A_t^- - R_t^{\text{net}}, \quad (\text{A.9})$$

which delivers the goods-market clearing condition (A.10) below. In the stationary equilibrium, the transaction-tax rate τ_h is calibrated so that the right-hand side of equation (A.9) is zero and the lump-sum adjustment vanishes. Along the transition, τ_h and $\bar{B}$ remain fixed, and $\bar{T}_t$ adjusts period by period according to equation (A.9). Under the rental availability shock, replace κA_t^- throughout these aggregate budget identities by $\kappa A_t^- + (\kappa_t^L - \kappa) A_{t,L}^-$, where $A_{t,L}^- = \int a \mathbb{1}_{a < 0} \mathbb{1}_{h=2} dg_t$ is aggregate landlord mortgage debt. The replacement adds the landlord-specific wedge revenue to the fiscal rebate.

Construction input and land. The first-order conditions of the construction problem (18) imply $S_t = (1 - \alpha_L) p_t^h Y_t^h$ and $p_t^L \bar{L} = \alpha_L p_t^h Y_t^h$, so that payments to the construction input and to land exhaust construction revenue,

$$p_t^h Y_t^h = S_t + p_t^L \bar{L}.$$

Following Kaplan et al. (2020), the government issues the fixed supply of land permits $\bar{L}$ and absorbs the revenue $p_t^L \bar{L}$ one-for-one rather than rebating it to households, so this revenue and the associated absorption offset within the government budget and leave the fiscal adjustment (A.9) unchanged.

Goods-market clearing reads

$$Y_t^c = C_t + S_t + G_t^L, \quad (\text{A.10})$$

where $G_t^L \equiv p_t^L \bar{L}$ is the land revenue absorbed by the government and plays the role of government spending.

B Rent Decomposition: Derivations and Additional Results

Section 4.3 decomposes the rent response by asking how each equilibrium input changes rental-market excess supply, the gap between the mass of landlords willing to supply a rental unit and the mass of households demanding one. This appendix formalizes that construction and reports the cumulative contributions. Recall that $h \in \{0, 1, 2\}$ indexes renters, owner-occupiers, and landlords. Given a path of household inputs $\mathbf{X}$, excess supply at $t + 1$ is

$$\begin{aligned} \text{ES}_{t+1}^r(\mathbf{X}) &= \mathcal{S}_{t+1}^r(\mathbf{X}) - \mathcal{D}_{t+1}^r(\mathbf{X}) \\ &= \sum_{s,h} \iint [\mathbb{P}_t(2 \mid a, h, y, s; \mathbf{X}) - \mathbb{P}_t(0 \mid a, h, y, s; \mathbf{X})] g_t(a, h, y, s; \mathbf{X}) da dy. \end{aligned} \quad (\text{B.1})$$

Thus, an input can change excess supply both by changing current tenure choices and by changing the distribution inherited from earlier choices. We keep both margins inside the economic channel generated by that input rather than reporting separate extensive and distribution channels.

The household block takes as inputs the paths

$$\mathbf{X}_t = \left\{ r_t^{\text{save}}, r_t^{\text{debt,own}}, r_t^{\text{debt,ll}}, w_t N_t, \tau_t, p_t^r, p_t^h \right\}. \quad (\text{B.2})$$

Here τ_t denotes the path of taxes and transfers, $T_t(\cdot)$. The three rates comove in equilibrium. We separate them only in the decomposition: r_t^{save} applies to positive financial positions, while $r_t^{\text{debt,own}}$ and $r_t^{\text{debt,ll}}$ apply to the mortgage debt of owner-occupiers and landlords, respectively. Let $\mathbf{X}^{\text{ss}}$ and $\mathbf{X}^{\text{eq}}$ denote the steady-state and equilibrium paths. For input j , let $\mathbb{1}_j$ select its path and define the one-input experiment

$$\mathbf{X}^j = \mathbf{X}^{\text{ss}} + \mathbb{1}_j(\mathbf{X}^{\text{eq}} - \mathbf{X}^{\text{ss}}), \quad d^j \mathbf{ES}^r = \mathbf{ES}^r(\mathbf{X}^j) - \mathbf{ES}^r(\mathbf{X}^{\text{ss}}). \quad (\text{B.3})$$

The remaining discrepancy from exact additivity is reported as a first-order residual.

Proposition B.1 (Rental-market decomposition). *To first order around the steady-state path, the equilibrium change in rental excess supply satisfies*

$$d\mathbf{ES}^r = d^{p^r} \mathbf{ES}^r + \sum_{k=1}^7 \chi^k + \epsilon^{\text{ES}}. \quad (\text{B.4})$$

The seven indirect channels are

$$\begin{aligned}\chi^{\text{ll mortg.}} &= d^{r^{\text{debt,ll}}} \mathbf{ES}^r, & \chi^{\text{buyer mortg.}} &= d^{r^{\text{debt,own}}} \mathbf{ES}^r, \\ \chi^{\text{portfolio},h} &= d^{r^{\text{save},h}} \mathbf{ES}^r, & h &\in \{2, 1, 0\}, \\ \chi^{\text{house price}} &= d^{p^h} \mathbf{ES}^r, & \chi^{\text{standard HANK}} &= d^{wN} \mathbf{ES}^r + d^r \mathbf{ES}^r.\end{aligned}$$

Here $d^{r^{\text{save},h}} \mathbf{ES}^r$ partitions the effect of the return on savings by the household's current tenure. The rent path is represented by the six-column basis Ψ used to solve the transition (Appendix D.1). The vector $\mathbf{c}$ collects the six weights on these basis functions and therefore parameterizes the rent path:

$$\mathbf{p}_{2:T}^r = p_{ss}^r (\mathbf{1} + \Psi \mathbf{c}), \quad J_c = \left. \frac{\partial \mathbf{ES}_{2:T}^r}{\partial \mathbf{c}'} \right|_{\mathbf{x}^{ss}}. \quad (\text{B.5})$$

The rent contribution of channel k is therefore

$$d\mathbf{p}_{2:T}^{r,k} = -p_{ss}^r \Psi J_c^+ \chi_{2:T}^k, \quad dp_1^{r,k} = 0, \quad (\text{B.6})$$

where J_c^+ denotes the Moore–Penrose pseudoinverse, which gives the least-squares solution in the restricted rent basis. Consequently, $d\mathbf{p}^r = \sum_k d\mathbf{p}^{r,k} + \epsilon^p$.

Proof. Differentiating equation (B.1) attributes the first-order change in tenure probabilities and in the inherited distribution to the input path that generates them. First-order additivity across the one-input experiments gives equation (B.4). Rental-market clearing then requires the direct rent effect to offset the combined indirect pressure. Linearizing the clearing conditions with respect to the rent-basis weights gives $J_c d\mathbf{c}^k = -\chi_{2:T}^k$; the least-squares solution and equation (B.5) yield equation (B.6). ■

Cumulative contributions. Figure 4 plots the direct rent effect and the seven indirect channels in excess-supply units. A negative bar means that, at the steady-state rent, the channel contracts rental supply relative to demand and therefore places upward pressure on the clearing rent. Table B.1 reports the corresponding cumulative effects. Because the rent mapping is dynamic, it uses the entire time profile of each excess-supply path, so the sign of a channel's cumulative excess-supply effect need not determine the sign of its cumulative rent contribution when the channel changes sign over the transition.

Extending the horizon from 24 to 100 quarters leaves the main pattern of the decomposition unchanged. The portfolio effect on landlords and the landlord mortgage-cost channel remain the two largest positive contributions to the rent, while the house-price response remains the main offset. The standard-HANK contribution strengthens from 2.62

Channel	Notation	Excess supply (pp-quarters)	Rent contribution (rent-response units)
Rent (direct effect)	$d^{p^r} \mathbf{ES}^r$	5.71	—
Landlord mortgage cost	$\chi^{\text{ll mortg.}}$	−4.08	5.93
Homebuyer mortgage cost	$\chi^{\text{buyer mortg.}}$	−2.15	3.25
Portfolio effect on landlords	$\chi^{\text{portfolio},2}$	−5.13	7.60
Portfolio effect on owners	$\chi^{\text{portfolio},1}$	0.26	−0.08
Portfolio effect on renters	$\chi^{\text{portfolio},0}$	−0.88	1.41
House price	$\chi^{\text{house price}}$	8.60	−13.96
Standard HANK	$\chi^{\text{standard HANK}}$	−1.79	2.62
First-order residual	ϵ^{ES}	−0.55	0.94
Realized total	$d\mathbf{ES}^r$	0.00	7.71

Table B.1. Cumulative Rental-Market Decomposition, Quarters 1–24

Note. Excess supply is the landlord share minus the renter share. Its cumulative effects are measured in percentage-point-quarters (pp-quarters) of household mass. One rent-response unit is a one-percentage-point deviation of the market rent from p_{ss}^r for one quarter. The rent column reports the cumulative $d\mathbf{p}^{r,k}$ of (B.6); its first-order residual, ϵ^p , is 0.94 rent-response units, or 12.19 percent of the realized cumulative rent increase. In each column, the entries above the line sum to the realized total.

to 3.92 rent-response units. The portfolio effect on renters unwinds from 1.41 to 0.35 units; in excess-supply units, its cumulative effect reverses from a 0.88 percentage-point-quarter contraction to a 0.63 increase. The realized cumulative rent response falls from 7.71 to 6.64 units, since the cumulative market-rent deviation is negative over quarters 25–100, and the residual share rises from 12.19 to 14.55 percent.

C The Cost of the Rent-Inclusive Rule across Weights

Figure C.1 extends the two-rule comparison in Table 4 across rent weights, reporting the same statistics at the same horizons for each ω_r on the grid $\omega_r \in \{0, 0.01, \dots, 0.15\}$. Every statistic is monotone in the rent weight at every horizon. The costs of the rule scale with the weight given to rents, and the 6 percent experiment is conservative in this sense.

The horizon separates two kinds of cost. The additional rental burden and the additional quadratic deviations of nondurable inflation barely depend on it, and the lines nearly coincide in Panels C and E. The additional consumption contraction and the additional quadratic deviations of average rent inflation do depend on it, and both are largest at the eight-quarter horizon, by which about half of all leases have reset. The effect on cumulative nondurable inflation changes sign with the horizon: a larger rent weight buys more disinflation at horizons of up to eight quarters and less at twenty quarters, where cumulative disinflation falls monotonically as the weight rises. A larger rent weight therefore buys

disproportionately little disinflation for the additional contraction it produces.

Discounting each path with $\prod_{s \leq t} (1 + r_s)^{-1}$, computed under each rule's own rate path, leaves the comparison unchanged.

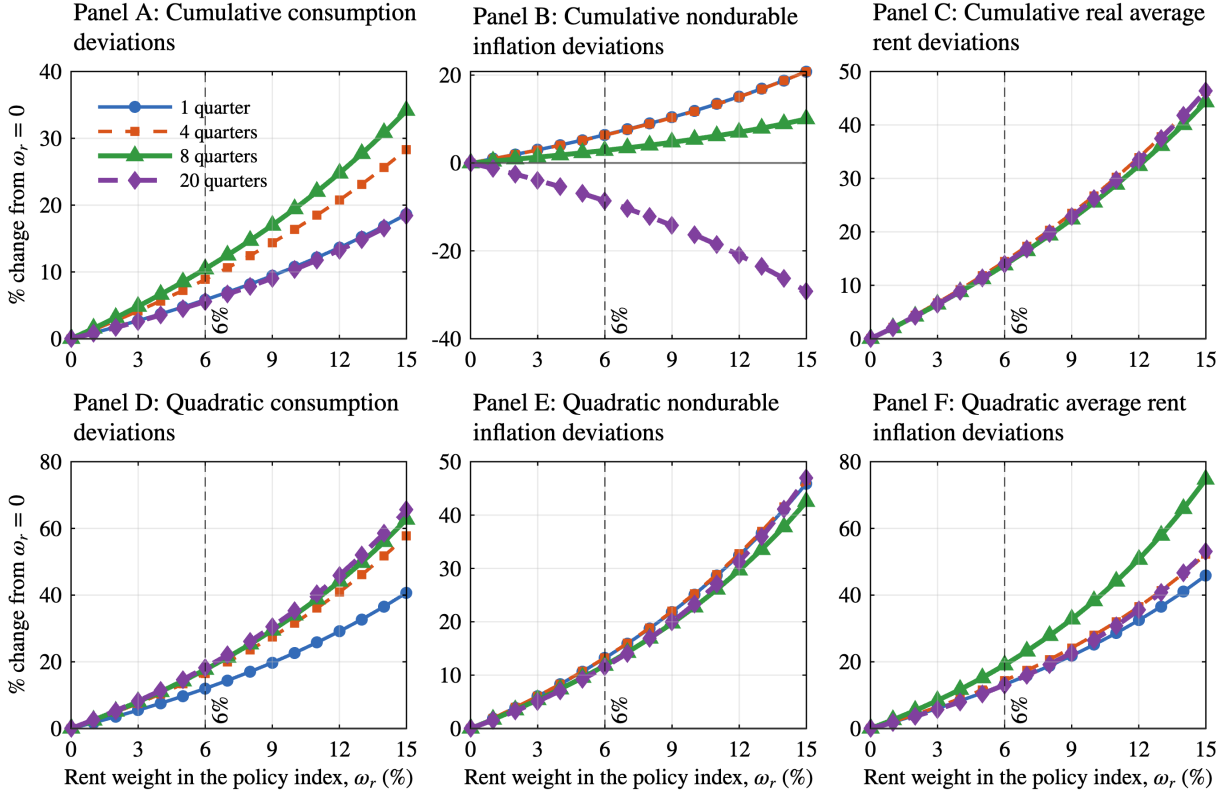


Figure C.1. Aggregate Costs across Rent Weights and Horizons

Note. The reported weights are $\omega_r \in \{0, 0.01, \dots, 0.15\}$. At each weight, we hold the economy and the monetary policy shock fixed and re-solve the transition. Each panel reports the percent change in the magnitude of the corresponding statistic, cumulated to the indicated horizon, relative to the rent-exclusive rule $\omega_r = 0$; the statistics are defined as in Table 4. The dashed vertical line marks the main experiment, $\omega_r = 0.06$, and the horizontal line marks the rent-exclusive benchmark. Panel C omits the one-quarter horizon, at which real average rents are unchanged under both rules.

D Numerical Solution

We solve for the full nonlinear transition after each shock: at every iteration, the household problem is solved on its complete state space, including the discrete tenure choice and the collateral constraints, and the equilibrium conditions are solved without linearizing around the steady state. The welfare comparisons of Section 6 require this: the welfare differences across rules are small, so even modest approximation error in the allocation could change their ranking.

D.1 Smooth Approximation of the Rent Path

Treating each rent p_t^r , $t = 2, \dots, T$, as a separate unknown and clearing the rental market exactly at every date delivers a rent path that alternates from quarter to quarter rather than tracing a smooth hump. Raising p_t^r attracts landlords and reduces the mass of renters, but the induced change in the tenure distribution persists into subsequent periods. A rent adjustment that closes the market at t therefore generates excess rental supply at $t + 1$, calling for an adjustment in the opposite direction. The alternation is a property of the equilibrium system rather than a failure of the solver. In this exact-clearing solution the residual falls to 2×10^{-10} , and the oscillation does not shrink as the solver tolerance tightens; it grows.

The source is the persistence of the landlord stock. The tenure margin is highly persistent at quarterly frequency, with a landlord-to-landlord transition probability of 0.947 in the calibrated stationary equilibrium (Figure 1a), so the mapping from the rent path into rental supply is a discounted average whose weights decline by a factor of about 0.94 per quarter, a half-life of about eleven quarters. Each date's clearing condition therefore responds to rents over a long window of neighboring dates. Requiring the market to clear exactly at every date amounts to inverting this averaging operator, and inverting an averaging operator amplifies precisely the high-frequency components that the operator suppresses.

Restricted rent path. Approximating a solution within a low-dimensional span of basis functions is standard in computational economics; see Judd (1998). We apply this approach to the market-rent component of the transition. For $t = 2, \dots, T$, we parameterize the rent as

$$p_t^r(\mathbf{c}) = p_{ss}^r \left[1 + \sum_{j=1}^J \sum_{k=1}^K c_{jk} \frac{\xi_{jk}(t-1)}{\max_{u \in \{1, \dots, T-1\}} \xi_{jk}(u)} \right], \quad \xi_{jk}(v) = \left(\frac{v}{\ell_j} \right)^{k-1} \exp \left(-\frac{v}{\ell_j} \right),$$

with $p_1^r = p_{ss}^r$. Here $\mathbf{c}$ collects the JK basis weights, ℓ_j denotes the decay horizon of basis block j , and $T = 100$ quarters is the length of the transition. The baseline uses $J = 2$ horizons with $K = 3$ terms each; stacking the normalized functions as columns gives the six-column basis Ψ of Appendix B. The denominator normalizes every basis function to have a unit peak over the transition, which puts the weights c_{jk} and the corresponding Jacobian columns on comparable scales. Each function is smooth and converges to zero as t grows, so the market rent converges back to its stationary value. Figure D.1 compares the transition after the monetary policy shock of Section 4.1 under alternative parameterizations of the

basis. The six parameterizations in Figure D.1 deliver nearly indistinguishable transitions: the peak average rent ranges from 0.461 to 0.467 percent across them.

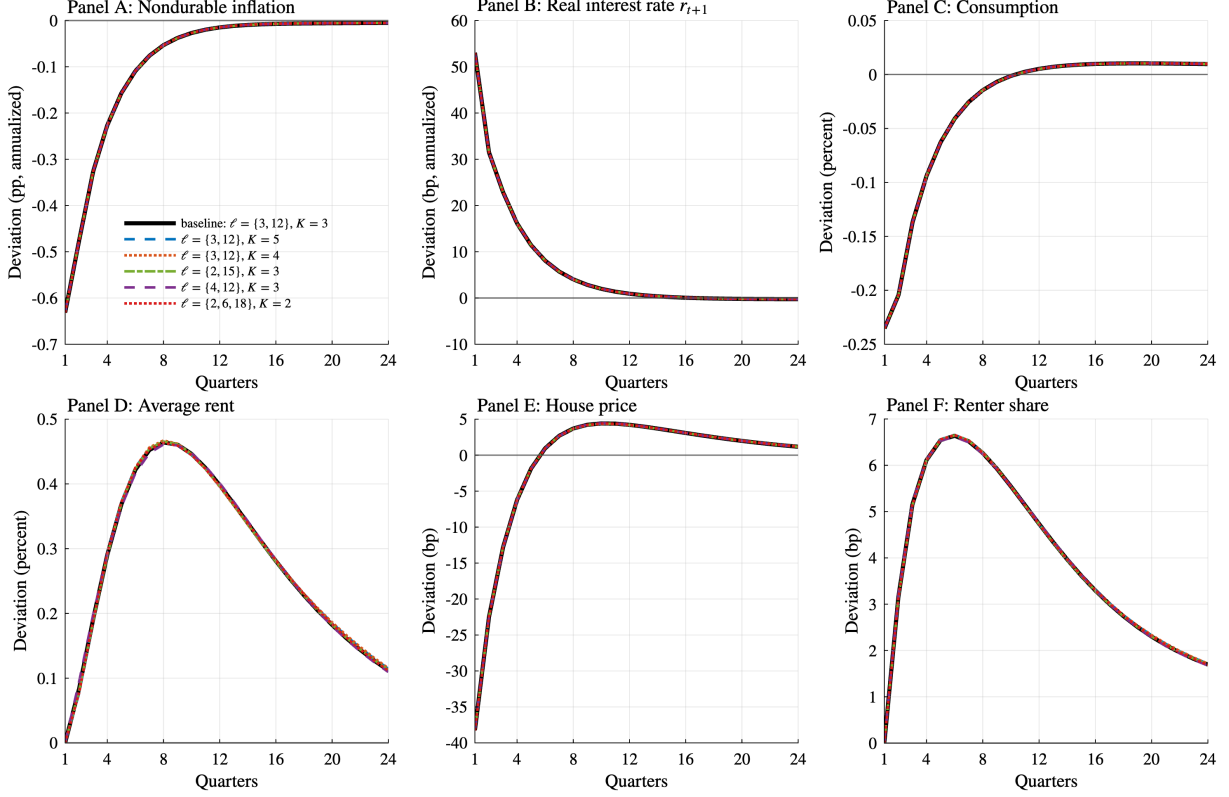


Figure D.1. Robustness to the Parameterization of the Rent Basis

Note. The figure compares the transition under six smooth rent bases. The solid black line is the baseline parameterization, $\ell = \{3, 12\}$ with three terms per horizon, which is the solution reported throughout the paper. The remaining specifications vary the number of terms, the decay horizons, or the number of horizons, holding the stationary equilibrium, the monetary policy shock of Section 4.1, and the initialization fixed. Deviations are reported in annualized percentage points for inflation, in annualized basis points for the real rate, in basis points for the house price and the renter share, and in percent for consumption and the average rent.

D.2 Welfare Computation

Consumption-equivalent units. We express welfare differences as consumption equivalents. Let λ be the permanent proportional change in consumption under the benchmark rule ($\phi_r = 0$) that delivers the same aggregate utilitarian welfare (26) as the alternative rule. Split the benchmark objective into its consumption-utility part $\mathcal{W}_0^u = \sum_{t \geq 1} \tilde{\beta}_t \int u(c_{it}, h_{it}) di$ and the remainder $\mathcal{W}_0^n = \sum_{t \geq 1} \tilde{\beta}_t [-\nu(N_t) - \frac{\theta_w}{2} (\pi_t^w)^2]$, so that benchmark welfare is $\mathcal{W}_0 = \mathcal{W}_0^u + \mathcal{W}_0^n$, both evaluated at $\phi_r = 0$ and summed over the transition plus a

steady-state continuation tail. Because $u(c, h) = (c f(h))^{1-\gamma} / (1 - \gamma)$ is CRRA, scaling every household's consumption by $(1 + \lambda)$ multiplies $\mathcal{W}_0^u$ by $(1 + \lambda)^{1-\gamma}$ and leaves $\mathcal{W}_0^n$ unchanged. The consumption equivalent of a rule with welfare $\mathcal{W}$ then solves $(1 + \lambda)^{1-\gamma} \mathcal{W}_0^u + \mathcal{W}_0^n = \mathcal{W}$, that is, $\lambda = [(\mathcal{W} - \mathcal{W}_0^n) / \mathcal{W}_0^u]^{1/(1-\gamma)} - 1$, and we report 100λ .

The ϕ_r grids. We choose the rent coefficient by comparing welfare across a grid of policy rules. For each coefficient, we solve the transition dynamics from the same stationary equilibrium, changing only the central bank's response ϕ_r . Figure D.2 reports the resulting welfare comparisons.

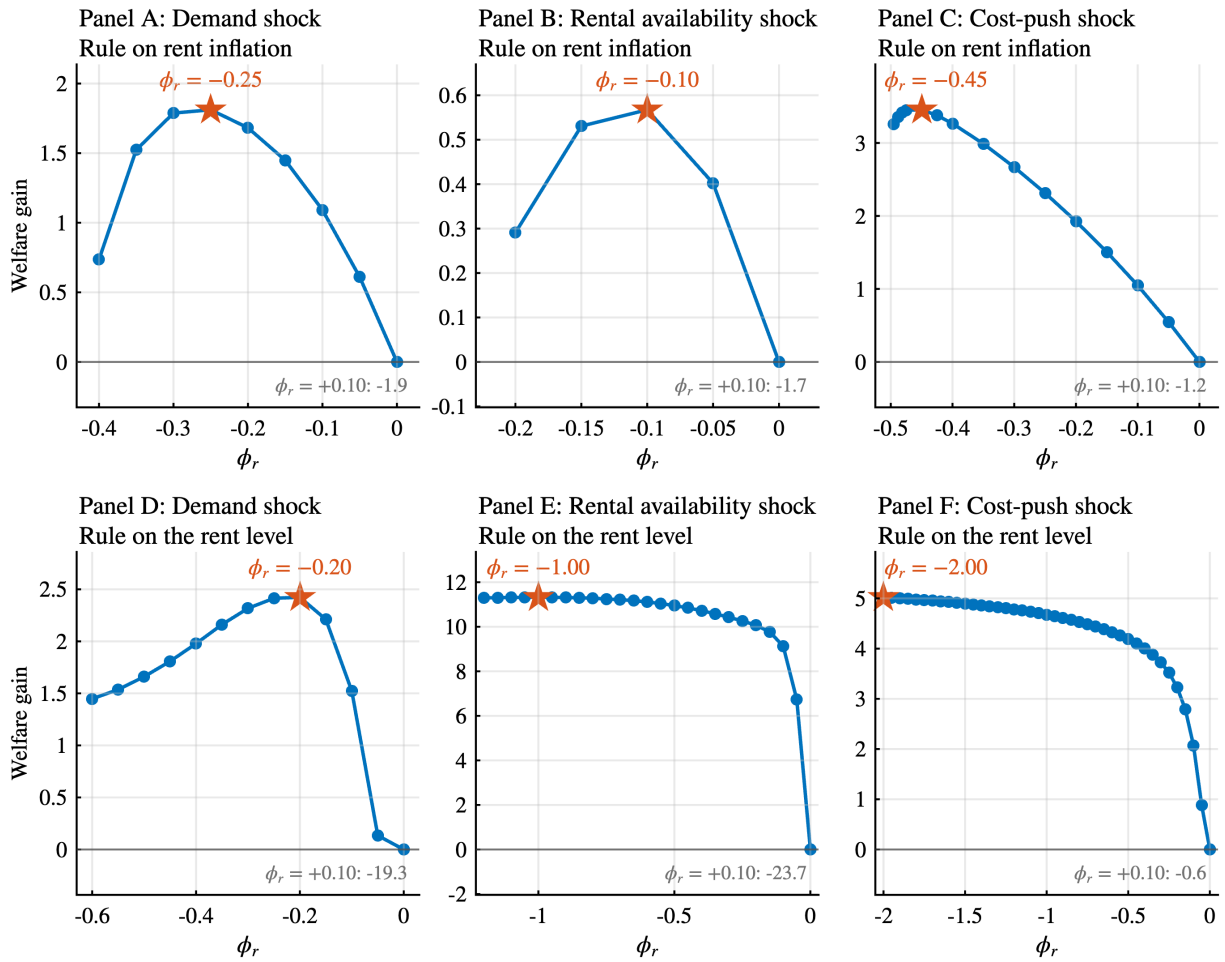


Figure D.2. Welfare Gains across Rent Coefficients

Note. Welfare gains relative to $\phi_r = 0$ under the same shock, with $\phi_\pi = 1.5$ throughout, measured in thousandths of a percent of permanent consumption. Each point represents a separately computed equilibrium transition. Welfare at $\phi_r = 0.10$ is reported by the annotations because it falls outside the plotted range.

Distributional decomposition of welfare gains. Figure 8 splits the welfare change from moving to the optimized rule across initial cohorts and across the channels through which it reaches them.

Consider a partition of the (a, h) state space into groups indexed by g , with masses m_g under the distribution at the date of the shock. Tracking each group forward under the transition's own policies gives cohort welfare $\mathcal{W}_g(\phi_r)$, where the aggregate labor disutility and wage-adjustment cost in (26) are allocated across cohorts pro rata by mass and the steady-state continuation value by date-of-shock mass. The cohorts then add up, $\sum_g \mathcal{W}_g(\phi_r) = \mathcal{W}(\phi_r)$, so at an interior optimum their marginal effects offset,

$$\sum_g \left. \frac{d\mathcal{W}_g}{d\phi_r} \right|_{\phi_r = \phi_r^*} = 0. \quad (\text{D.1})$$

Figure 8 reports the level counterpart, the per-capita change $[\mathcal{W}_g(\phi_r^*) - \mathcal{W}_g(0)]/m_g$.

The channels are the one-input experiments of Appendix B, applied to welfare rather than to rental-market excess supply. Write $\mathcal{W}_g(\mathbf{X})$ for cohort g 's welfare when the household block is solved at the input path $\mathbf{X}$ of (B.2), and let $\mathbf{X}^0$ and $\mathbf{X}^*$ be the equilibrium paths under $\phi_r = 0$ and $\phi_r = \phi_r^*$. With the selector $\mathbb{1}_j$ of (B.3), channel j 's contribution to cohort g is

$$\Delta_g^j = \frac{\mathcal{W}_g(\mathbf{X}^0 + \mathbb{1}_j(\mathbf{X}^* - \mathbf{X}^0)) - \mathcal{W}_g(\mathbf{X}^0)}{m_g}.$$

Figure 8 also reports the general-equilibrium total; the difference between this total and the sum of the channel contributions is the residual.

Welfare assessments (Dávila and Schaab, 2025). Figure 9 splits the marginal welfare effect of the rent coefficient into efficiency and redistribution. Let $V(a, h, y, s; \phi_r)$ denote the welfare of a household that starts the transition in state (a, h, y, s) under rule (25), measured as in (26): its expected discounted utility from consumption and housing, net of the labor disutility and wage-adjustment cost common to all households, plus a steady-state continuation. Aggregate welfare is $\mathcal{W}(\phi_r) = \sum_s \iint V(a, h, y, s; \phi_r) g_{ss}(a, h, y, s) da dy$. We express each household's gain in units of perpetual consumption, one unit of consumption at every date, which the household values at

$$\lambda(a, h, y, s) = \sum_{t \geq 1} \tilde{\beta}_t \mathbb{E}_1[u_c(c_{it}, h_{it}) \mid (a_{i1}, h_{i1}, y_{i1}, s_{i1}) = (a, h, y, s)], \quad (\text{D.2})$$

again with a steady-state continuation, evaluated along the transition at the assessed coefficient. Let $\bar{\lambda} = \sum_s \iint \lambda(a, h, y, s) g_{ss}(a, h, y, s) da dy$ be its population average and $\omega(a, h, y, s) = \lambda(a, h, y, s) / \bar{\lambda}$ the implied welfare weight, which averages to one. In units of perpetual consumption, the marginal welfare effect then splits into

$$\begin{aligned} \frac{1}{\bar{\lambda}} \frac{d\mathcal{W}}{d\phi_r} = & \underbrace{\sum_s \iint \frac{1}{\lambda(a, h, y, s)} \frac{dV(a, h, y, s; \phi_r)}{d\phi_r} g_{ss}(a, h, y, s) da dy}_{\text{efficiency}} \\ & + \underbrace{\sum_s \iint (\omega(a, h, y, s) - 1) \frac{1}{\lambda(a, h, y, s)} \frac{dV(a, h, y, s; \phi_r)}{d\phi_r} g_{ss}(a, h, y, s) da dy}_{\text{redistribution}}. \end{aligned} \quad (\text{D.3})$$

Efficiency adds up households' willingness to pay for a higher coefficient and does not depend on the planner's welfare weights. Redistribution is the covariance between the weights and that willingness to pay: it is positive when a higher coefficient favors the households that value consumption most at the margin, mostly renters and levered owners.

Let $dV^\lambda(a, h, y, s; \phi_r) / d\phi_r$, equal to $\lambda(a, h, y, s)^{-1} dV(a, h, y, s; \phi_r) / d\phi_r$, denote that willingness to pay. For any two functions $x(a, h, y, s)$ and $z(a, h, y, s)$ of the initial state, define the population average and covariance

$$\mathbb{E}_{ss}[x] = \sum_s \iint x(a, h, y, s) g_{ss}(a, h, y, s) da dy, \quad \text{Cov}_{ss}(x, z) = \mathbb{E}_{ss}[xz] - \mathbb{E}_{ss}[x] \mathbb{E}_{ss}[z].$$

Because the weights average to one, $\mathbb{E}_{ss}[\omega] = 1$, the decomposition (D.3) reads

$$\frac{1}{\bar{\lambda}} \frac{d\mathcal{W}}{d\phi_r} = \mathbb{E}_{ss} \left[\frac{dV^\lambda(a, h, y, s; \phi_r)}{d\phi_r} \right] + \text{Cov}_{ss} \left(\omega(a, h, y, s), \frac{dV^\lambda(a, h, y, s; \phi_r)}{d\phi_r} \right). \quad (\text{D.4})$$

We evaluate the derivatives in (D.3) by centered differences across neighboring points of the ϕ_r grids, holding $\lambda(a, h, y, s)$ at its value at the assessed coefficient, and obtain household welfare and the numeraire (D.2) by integrating the stored household policies forward from $g_{ss}(a, h, y, s)$. The total crosses zero within half a grid step of each optimum in Section 6.2.1; the coefficients that each component alone favors, quoted in Section 6.3, are the zeros of its line in Figure 9, interpolated between grid points.

E Data

This appendix documents the data sources and construction of the calibration targets and untargeted moments. We use Spanish observations unless otherwise stated. Sample

periods differ across sources according to availability; averages are computed over the observations specified below.

E.1 House Values and Rents

We construct the house-value-to-income ratio from the public statistical tables of the ECB Household Finance and Consumption Survey (HFCS), using the 2010, 2014, 2017, and 2021 waves. For each wave, we divide the mean value of the household’s main residence among homeowners (HFCS statistical table B4) by mean annual gross household income across all households (HFCS statistical table I2). Both series are expressed in the same monetary units. The calibration target is the simple average of these four ratios, equal to 5.25. This is a ratio of group means, rather than an average of household-level value-to-income ratios.

In the model, the target applies to a reference dwelling of unit quality. The two housing qualities, 0.905 and 1.095, have midpoint one; this normalization does not impose a population-weighted mean quality of one. Because owner-occupiers live in the higher-quality dwelling (1.095), an owner-occupier’s dwelling has a value of approximately 5.75 times mean annual income.

We calibrate the reference rent to capture the substantial cost of rental housing relative to household resources in Spain. Eurostat EU-SILC (`ilc_mdcd02`) reports rental payments averaging 30.0 percent of tenant households’ disposable income over 2004–2025. Guided by this benchmark, we set rent equal to 30 percent of mean household income in the model. Renters’ lower endogenous income then implies a rental burden of about 38 percent of their own income. The model also reproduces, without targeting it, the tenant income gap behind that number: renters’ mean income is 0.78 of the population mean, against 0.74 in EU-SILC.

This burden is close to the 38.7 percent average reported by [Khametshin et al. \(2024\)](#), who assess rental affordability by assigning local average rents to households that do not own their main residence, including those in rent-free or reduced-rent accommodation, and measuring the burden against gross household income. EU-SILC instead measures actual rental payments against disposable income, so the two are complementary affordability benchmarks rather than the same statistic. Together they support a calibration in which rental payments absorb a substantial share of renters’ resources and influence their consumption and tenure choices. The gap between these two empirical measures reflects differences in coverage and in the income denominator; it is not informative about the lease block, which governs how incumbent rental payments adjust over time.

E.2 Tenure Shares

We obtain Spanish tenure shares for the same four waves from the ECB HFCS statistical tables and the HFCS International Key Figures Dashboard hosted by the Oesterreichische Nationalbank (OeNB). The renter proxy is the published “Renters / other” category: households that do not own their main residence, including rent-free and other arrangements. The landlord share is the fraction of households reporting income from renting real estate, irrespective of whether they own their main residence. We approximate the owner-occupier share by subtracting the landlord share from the share of households owning their main residence.

The simple averages across waves are 21.7 percent for the renter proxy, 9.5 percent for landlords, and 68.8 percent for owner-occupiers. The published aggregates do not identify the joint distribution of main-residence ownership and rental income, so the owner-occupier measure is approximate. It can also include households owning additional properties that generate no rental income. These empirical categories provide counterparts to the model’s three mutually exclusive tenure states, with the qualifications above.

E.3 Construction and Household Wealth

The residential investment share is gross fixed capital formation in dwellings divided by GDP at market prices. The numerator comes from Eurostat national accounts, dataset nama_10_an6, asset code N111G; the denominator comes from nama_10_gdp, item B1GQ. Both are annual current-price series in millions of euros. The average Spanish ratio over 1995–2025 is 6.9 percent. This measure covers investment in dwellings rather than total construction activity.

Using the same national accounts, we construct the nonresidential investment share as total gross fixed capital formation (N11G) less investment in dwellings (N111G), divided by GDP. The average over 1999–2025 is 15.6 percent. Comparing the residential and nonresidential shares helps assess the model’s residential investment share when housing is its only produced illiquid asset.

Household wealth comes from the ECB Distributional Wealth Accounts (DWA). We sum the financial-asset components F2M, F3, F5, and F62, add housing wealth (NUN), and subtract liabilities. This construction excludes nonfinancial business wealth and therefore differs from the DWA’s broader published net-wealth aggregate. The income denominator is the four-quarter rolling sum of gross disposable income (B6G) from the ECB Quarterly Sector Accounts.

We average the quarterly wealth-to-income ratios over 2011Q4–2025Q4 for Spain and

2009Q1–2025Q4 for the euro area, obtaining 7.33 and 6.13, respectively. The corresponding housing-wealth-to-income ratios are 5.66 and 4.37. The wealth statistics cover households (S14), whereas the available income denominator also includes nonprofit institutions serving households (S1M).

The top-decile net-wealth share is taken from the HFCS 2021 wave. The Spanish hand-to-mouth benchmarks are taken from [Kaplan et al. \(2014\)](#), who distinguish households with little liquid wealth according to whether they hold illiquid assets.

E.4 Mortgage Market

We measure the mortgage spread using monthly ECB interest-rate data. The mortgage rate covers new Spanish loans to households for house purchase with a floating rate or an initial rate-fixation period of up to one year (MIR series M.ES.B.A2C.F.R.A.2250.EUR.N). We subtract the monthly average one-year Euribor rate (FM series M.U2.EUR.RT.MM.EURIBOR1YD_.HSTA). Both rates are quoted in annual percent, so their difference is measured in percentage points.

The mean spread over May 2001–April 2026 is 1.38 percentage points. The calibrated mortgage wedge implies an annual spread of 1.5 percentage points at the steady-state real interest rate. The loan-to-value and loan-to-income limits and the amortization requirement are set as described in Section [3.1](#).

E.5 Tenure-Transition Matrix

We estimate tenure transitions using the Banco de España’s Encuesta Financiera de las Familias (EFF), linking households across the 2014–2017, 2017–2020, and 2020–2022 wave pairs. These pairs contain 3,634, 3,831, and 3,974 matched households, respectively. We use all five imputed datasets supplied for each wave.

We classify a household as a landlord if it reports rental income ($p7_1=1$ or $p7_2>0$), giving this classification priority over main-residence ownership. Among the remaining households, those owning their main residence ($p2_1=2$) are owner-occupiers; all others are renters, including rent-free and other nonownership arrangements. Properties that generate no reported rental income do not by themselves imply landlord status.

We build the matrix in three steps. First, for each wave pair j and each of the five imputed datasets, we weight matched households by the panel weight $pesopan_1$ (reported in the later wave and representative of the earlier wave’s population), divide the weighted origin-to-destination counts by the weighted count in each origin state, and average the five resulting matrices. This gives P_j , the transition matrix over the pair’s

horizon. Second, we convert each P_j into a quarterly matrix $Q_j = P_j^{1/q_j}$, its principal matrix root, where q_j is the number of quarters between the two waves: $(q_1, q_2, q_3) = (12, 12, 8)$. The roots are nonnegative and row-stochastic, and raising Q_j to the power q_j recovers P_j . This step assumes a time-homogeneous quarterly Markov process between waves, so quarterly transitions are inferred rather than observed. Third, we average the three quarterly matrices row by row, weighting pair j 's row h by n_{jh} , the unweighted number of matched households that start pair j in state h :

$$Q_{hk} = \frac{\sum_j n_{jh}(Q_j)_{hk}}{\sum_j n_{jh}}.$$

The resulting matrix describes transitions among continuing households. It excludes household formation and dissolution, so its implied stationary distribution need not reproduce cross-sectional tenure shares. Direct renter–landlord transitions also rely on relatively few observations. We use the matrix to assess the model's tenure mobility without targeting its entries in the calibration.

E.6 Actual Rents in the HICP

We obtain annual expenditure weights from Eurostat's HICP item-weight dataset (`prc_hicp_inw`), using actual rentals for housing (CP041) over 2000–2025. Published weights are expressed per thousand; we divide them by ten to report percentages. The euro-area series uses changing composition.

Figure E.1 reports the expenditure weight of actual rents in the euro-area HICP. Over 2000–2025, this weight averages approximately 6.3 percent, compared with 10.4 percent in Germany and 2.8 percent in Spain. We use a rounded euro-area weight of 6 percent in the policy experiment of Section 5.

In the calibrated stationary equilibrium, rental expenditure represents

$$\omega_r^{\text{model}} = \frac{p_{ss}^r \mu_{ss}^0}{C_{ss} + p_{ss}^r \mu_{ss}^0} \simeq 7.2 \text{ percent.}$$

This share follows from the calibrated rent and renter share, together with nondurable consumption. It describes the model's simplified consumption basket, whereas the empirical HICP weight reflects expenditure across a broader basket and heterogeneous rental arrangements. The policy experiment uses the empirical euro-area weight; the model-implied share provides a comparison.

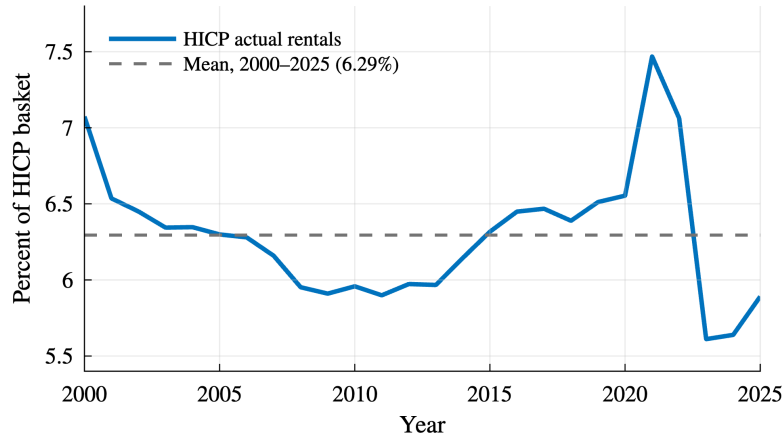


Figure E.1. Euro-Area HICP Weight of Actual Rents

Note. The figure reports the expenditure weight of actual rents paid by tenants in the euro-area HICP basket since 2000. The 2021 spike reflects pandemic reweighting, with no discrete change in rental-market coverage: an ECB Economic Bulletin box documents pandemic-induced shifts in HICP expenditure weights, with food and housing services increasing especially strongly (Gonçalves et al., 2021).

F Additional Figures and Tables

F.1 Tenure-Transition Responses

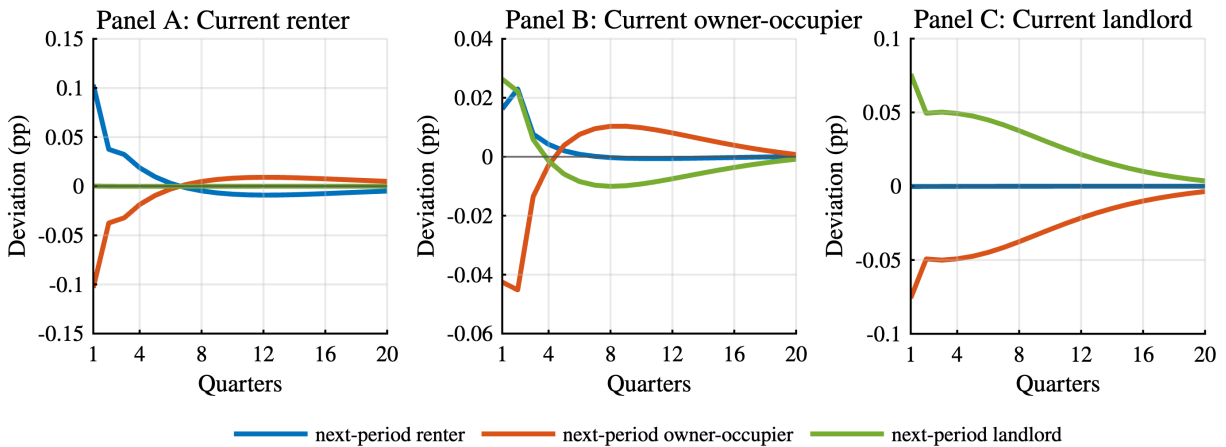


Figure F.1. Tenure-Transition Response to a Contractionary Monetary Shock

Note. Each panel conditions on the household's current tenure state and reports deviations in next-period tenure-choice probabilities from the stationary equilibrium, in percentage points. Conditional probabilities are computed by normalizing expected transition masses within each current tenure state. Current and next-period tenure states are renter, owner-occupier, and landlord.

The tenure flows behind the rent channels of Section 4.3 are visible in the transition probabilities. Figure F.1 plots, for each current tenure state $h \in \{0, 1, 2\}$, the deviation $\mathbb{P}_t(h' | h) - \mathbb{P}_{ss}(h' | h)$ for next-period choices h' , where $\mathbb{P}_t(h' | h)$ is the share of households in tenure state h at date t that choose h' . The homebuyer mortgage-cost channel appears in the renter panel: on impact, current renters are about 0.10 percentage points more likely to remain renters and about 0.10 percentage points less likely to become owner-occupiers, so the tightening raises rental demand partly by keeping marginal renters in the rental market, while also slightly pushing existing owners back into renting (Panel B).³⁷

The landlord channels appear in the owner and landlord panels, and balance sheets determine who enters. Current owners become more likely to enter landlord status, while current landlords become less likely to exit into owner-occupation: the owner-to-landlord transition rises by 0.03 percentage points on impact, while the landlord-to-owner transition falls by 0.08 percentage points. Both margins push the landlord share up, as clearing requires. The additional owner-to-landlord entrants are high-asset households, typically unlevered before entry—households whose current balance sheets expose them to the portfolio channel rather than to the mortgage-cost channels, and who enter because expected higher future rents compensate them for the forgone return on financial savings. The increase in the landlord share is thus a net reallocation: expected higher rents draw unconstrained owners into rental supply and reduce exit among incumbent landlords.³⁸

F.2 Consumption Incidence of a Monetary Shock across Households

We fix each household group, defined below, at the date of the shock and follow its members forward as they move across tenure and balance-sheet states. We compare each initial cohort's per-capita consumption along the transition with that of the identical initial population propagated under steady-state policies. This comparison avoids conflating the consumption response of a given cohort with changes in the composition of contemporaneous tenure groups.

We report two partitions: borrowers and savers, defined by net financial assets; and renters, mortgagor-owners, unlevered owners, mortgagor-landlords, and unlevered landlords, defined by tenure and leverage. Table F.1 reports the impact and cumulative consumption contractions for both partitions. Figure F.2 reports impact responses across

³⁷ The model holds the household unit fixed, so all adjustment runs through tenure choices. In the data, downturns can also reshape household composition: household size is countercyclical because young adults move back into, or delay leaving, the parental home (Dyrda et al., 2024).

³⁸ The discrete housing ladder shapes these flows: households cannot adjust the scale of a rental portfolio, so a landlord who reduces rental-property holdings must exit landlord status, and rental supply that existing landlords would otherwise provide by expanding their portfolios comes here through entry by new landlord households.

net-financial-asset bins in Panel A and the consumption paths of three representative initial cohorts in Panel B.

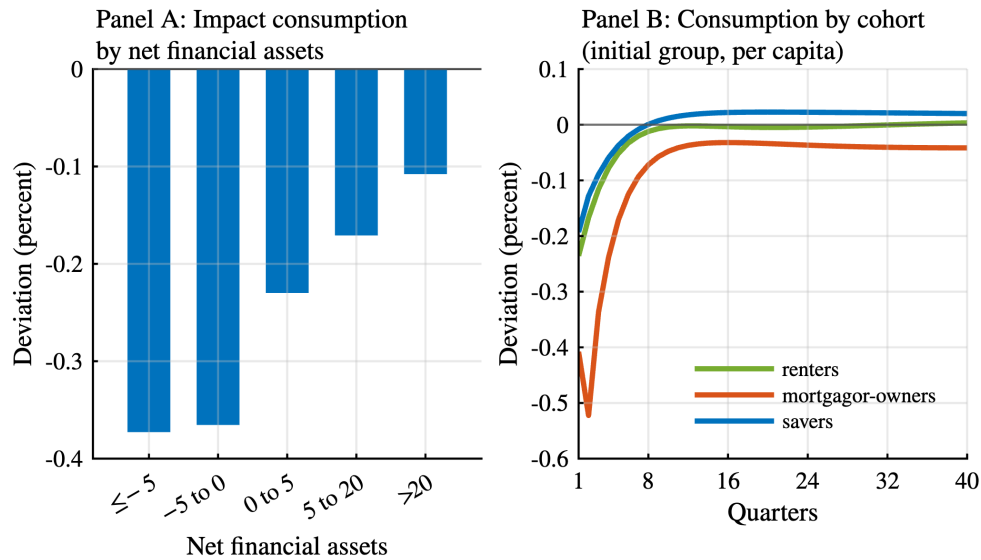


Figure F.2. Consumption Incidence of the Monetary Shock

Note. Panel A reports impact consumption responses by date-of-shock net-financial-asset bin (in units of quarterly steady-state income), in percent deviations from steady state. Panel B reports per-capita consumption paths for three representative initial cohorts (renters, mortgagor-owners, and savers), measured as percent deviations from each cohort's no-shock counterfactual; these span the exposure range but are not a partition, since savers ($a \geq 0$) include the renters and the unlevered owners and landlords.

Balance-sheet exposure. Mortgagor-owners experience an impact consumption contraction more than twice that of unlevered owners, while the contraction among mortgagor-landlords is approximately twice that of unlevered landlords.³⁹ Tenure also matters within net-financial-asset bins: renters holding small positive balances contract by roughly as much as the average borrower.

Mortgage repricing connects these consumption responses to the landlord mortgage-cost and homebuyer mortgage-cost channels of Section 4.3. Higher borrowing costs discourage the provision of rental housing and entry into homeownership; they also reduce the resources available for consumption among households already carrying mortgage debt. The cohort responses reported here incorporate all equilibrium channels, rather than isolating these mortgage effects.

³⁹ The impact response is anticipatory rather than contemporaneous. The real return on assets carried into the first period is set before the shock, so debt service at $t = 1$ remains at its steady-state rate. Indebted households adjust consumption in anticipation of higher debt service from the following quarter onward.

	SS share	Impact	Cumulative
<i>By net financial assets</i>			
Borrowers ($a < 0$)	0.263	−0.368	−4.251
Savers ($a \geq 0$)	0.737	−0.193	−0.535
<i>By housing tenure</i>			
Renters	0.217	−0.236	−0.794
Mortgagor-owners	0.162	−0.408	−5.553
Unlevered owners	0.404	−0.187	−0.528
Mortgagor-landlords	0.101	−0.324	−2.630
Unlevered landlords	0.116	−0.166	−0.396

Table F.1. Consumption Incidence of the Monetary Shock across Initial Cohorts

Note. Each cohort is fixed at the date of the shock and tracked forward; entries are per-capita consumption deviations from the cohort’s no-shock counterfactual, in percent. *Impact* is the $t=1$ response; *Cumulative* sums negative deviations over the full 100-quarter transition, so it measures the contraction phase only: unlevered cohorts later rise above their counterfactual, and those quarters do not enter. Owner-occupiers and landlords are split by net financial assets into mortgagor ($a < 0$) and unlevered ($a \geq 0$) groups. Renters are not split by net financial assets: they cannot borrow (the third case of (5) imposes $a' \geq 0$ whenever $h' = 0$), so the renter-borrower cohort has exactly zero mass and every renter is a saver. SS share is the steady-state population share.

Rent exposure. Renters experience a smaller impact contraction than mortgagor-owners, but larger impact and cumulative contractions and a weaker subsequent recovery than unlevered owners and landlords. Higher rental payments reduce the resources available for consumption and offset part of the benefit from higher returns on financial savings.

Lease stickiness determines when the higher rental bill arrives. Households signing new leases face the market rent, while regulated renters face the more slowly adjusting regulated rent $\tilde{p}_t^r(0)$ described in Section 2.2. The would-be buyers who remain renters therefore generally absorb the increase gradually. Both rents remain at their steady-state values on impact, so the initial consumption response reflects changes in other equilibrium conditions and anticipation of future payments. As rents subsequently rise, rental exposure adds a burden that a classification based only on financial assets does not capture.

F.3 Distributional Incidence of the Rent-Inclusive Rule

Table F.2 applies the fixed-cohort method of Appendix F.2 to the two monetary policy rules. The last two columns measure the additional consumption loss generated by including rent inflation in the policy index. The impact contraction increases for every cohort. Although the real rate chosen on impact is slightly lower under the rent-inclusive rule, the return on assets carried into that period is predetermined. Households instead adjust to

	$\omega_r = 0$	$\omega_r = 0.06$	Δ	% Δ
<i>Impact response</i>				
Borrowers ($a < 0$)	-0.368	-0.388	0.020	5.3%
Savers ($a \geq 0$)	-0.193	-0.205	0.012	6.1%
Renters	-0.236	-0.252	0.017	7.1%
Mortgagor-owners	-0.408	-0.430	0.022	5.3%
Unlevered owners	-0.187	-0.198	0.011	6.0%
Mortgagor-landlords	-0.324	-0.342	0.017	5.3%
Unlevered landlords	-0.166	-0.175	0.008	5.1%
<i>Cumulative contraction</i>				
Borrowers ($a < 0$)	-4.251	-4.413	0.162	3.8%
Savers ($a \geq 0$)	-0.535	-0.603	0.068	12.7%
Renters	-0.794	-0.865	0.072	9.0%
Mortgagor-owners	-5.553	-5.847	0.293	5.3%
Unlevered owners	-0.528	-0.596	0.068	12.9%
Mortgagor-landlords	-2.630	-2.627	-0.003	-0.1%
Unlevered landlords	-0.396	-0.445	0.048	12.2%

Table F.2. Distributional Incidence of the Rent-Inclusive Rule

Note. The first two columns report the impact and cumulative consumption drops associated with a one-percentage-point annualized monetary policy innovation, equivalent to 25 basis points at quarterly frequency, under the two alternative policy rules. Cohorts and statistics are defined as in Table F.1. Δ and % Δ report the change in the magnitude of each loss from $\omega_r = 0$ to $\omega_r = 0.06$; positive values indicate a larger loss under the rent-inclusive rule.

the anticipated path: indebted households face higher future debt service, while renters anticipate higher rental payments.

The additional cumulative contraction is largest for mortgagor-owners. Measured relative to each cohort's no-shock consumption, it is about four times that of renters. Mortgage exposure therefore remains central to the incidence of a rule that responds to rent inflation. Renters also lose through higher rental payments and the broader contraction. The cumulative loss increases for every tenure cohort except mortgagor-landlords, whose response is essentially unchanged. These consumption comparisons describe exposure to the policy-induced equilibrium adjustment; welfare additionally depends on households' marginal utilities.

F.4 Demand Shock

Figure F.3 compares the responses to the demand shock under the benchmark (solid lines) and the optimized rent-inflation rule (dotted lines). The negative rent coefficient eases

the endogenous monetary tightening and limits the initial shift from homeownership to renting.

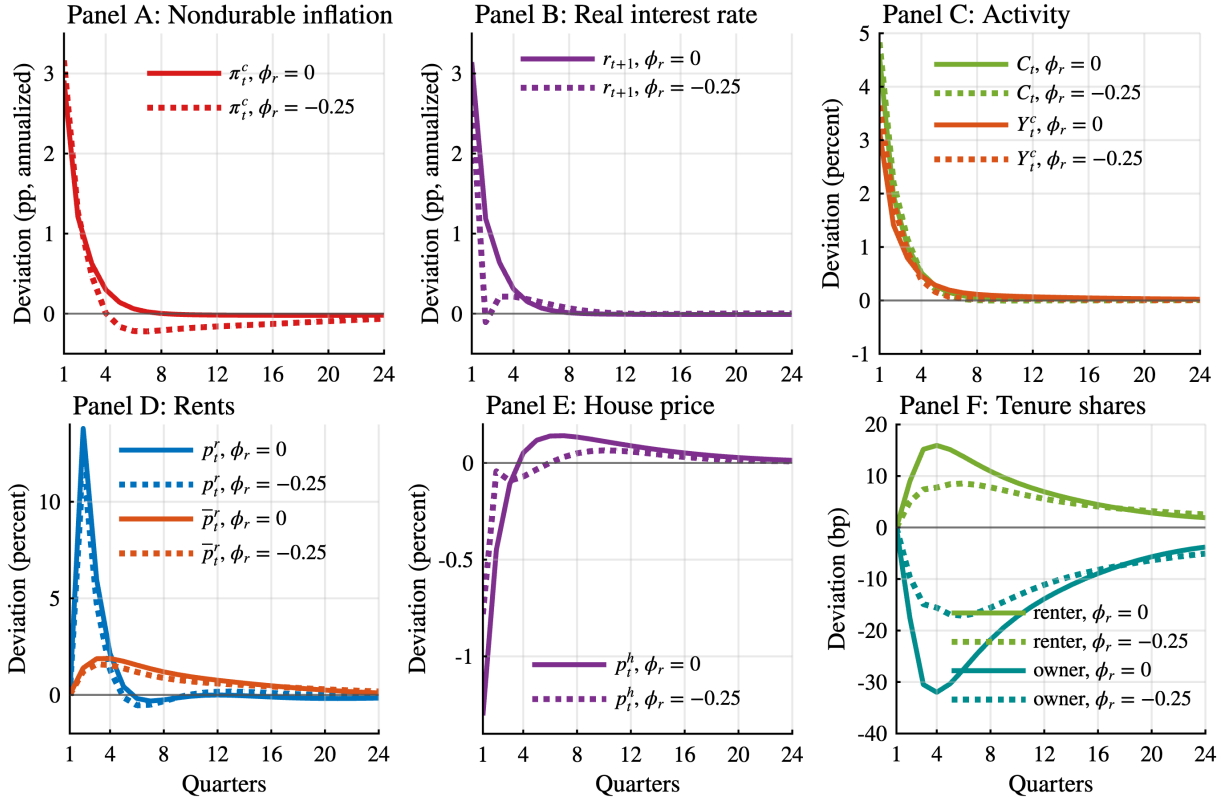


Figure F.3. Demand Shock under Alternative Policy Rules

Note. Responses under $\phi_r = 0$ (solid lines) and the optimized rent-inflation rule $\phi_r = -0.25$ (dotted lines). Panels A and B report nondurable inflation and the real interest rate in annualized percentage points. Panel C reports consumption and nondurable output, Panel D real market and average rents, and Panel E the real house price, all in percent deviations from steady state. Panel F reports renter and owner-occupier share deviations in basis points.

Under the rent-inflation rule, fiscal rebates contribute more to renters' welfare gains than direct rent relief (Figure F.4). Lower mortgage payments also benefit indebted owners and landlords, while lower returns on savings offset these gains for households with greater financial wealth.

F.5 Rental Availability Shock

Figure F.5 compares the responses to the rental availability shock under the benchmark (solid lines) and the optimized rent-inflation rule (dotted lines). The negative rent coefficient lowers the policy rate relative to the benchmark when rent inflation is positive, even though the shock generates little nondurable inflation.

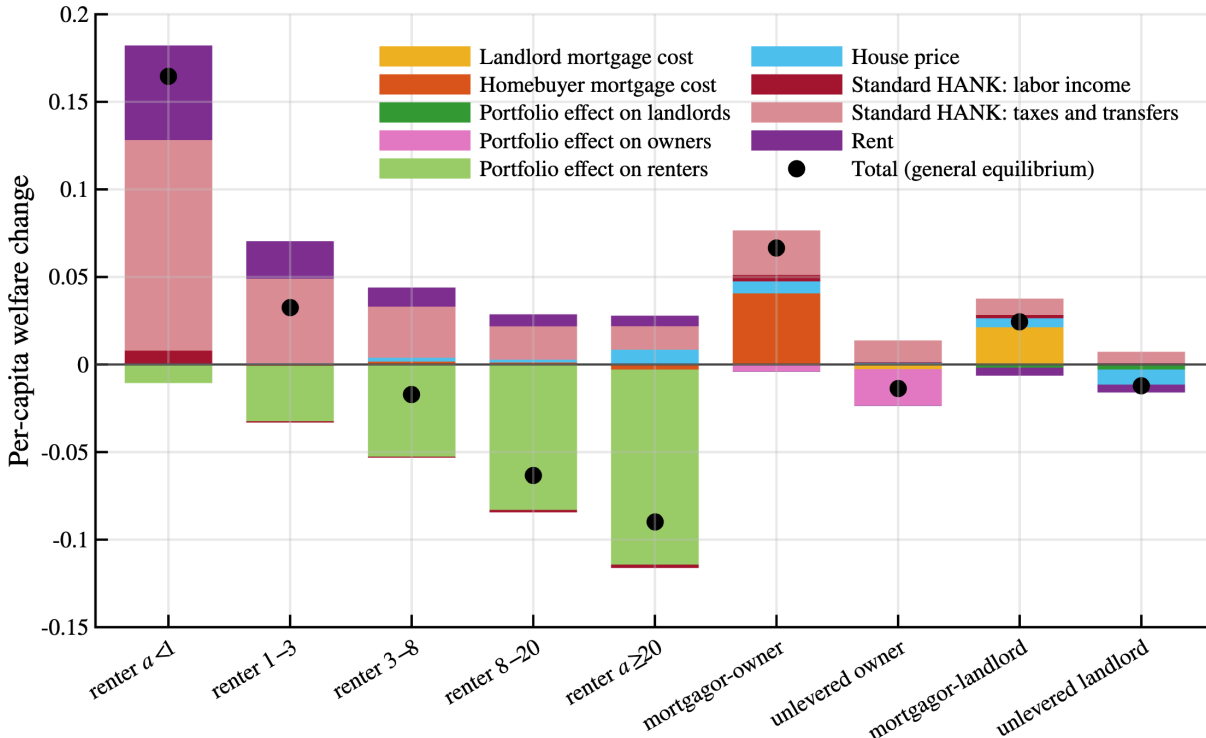


Figure F.4. Distributional Decomposition of the Welfare Gain:
Demand Shock, Rent-Inflation Rule

Note. Per-capita welfare change from moving ϕ_r from 0 to -0.25 , by date-of-shock cohort. Welfare changes are measured in utility units. Cohort definitions, channels, colors, and decomposition conventions follow Figure 8. Black dots report the general-equilibrium total.

Under the rental availability shock, the rent-inflation rule also delivers gains mainly through fiscal rebates and lower mortgage payments, with a smaller contribution from direct rent relief. Wealthier renters lose overall because the decline in returns on their savings outweighs these gains (Figure F.6).

The rent-level rule strengthens the redistribution through rental payments. Its larger reduction in rents raises the rent-channel gains of tenants and the corresponding losses of landlords. Renters with little financial wealth benefit most, while losses on savings continue to outweigh rent relief for the wealthiest renter cohorts (Figure F.7).

The choice of target also matters for rental relief under the rental availability shock. The optimized rent-inflation rule lowers the initial rent increase but leaves the cumulative rental burden close to the benchmark. The rent-level rule delivers a larger and more persistent reduction in rents, essentially eliminating that burden (Figure F.8).

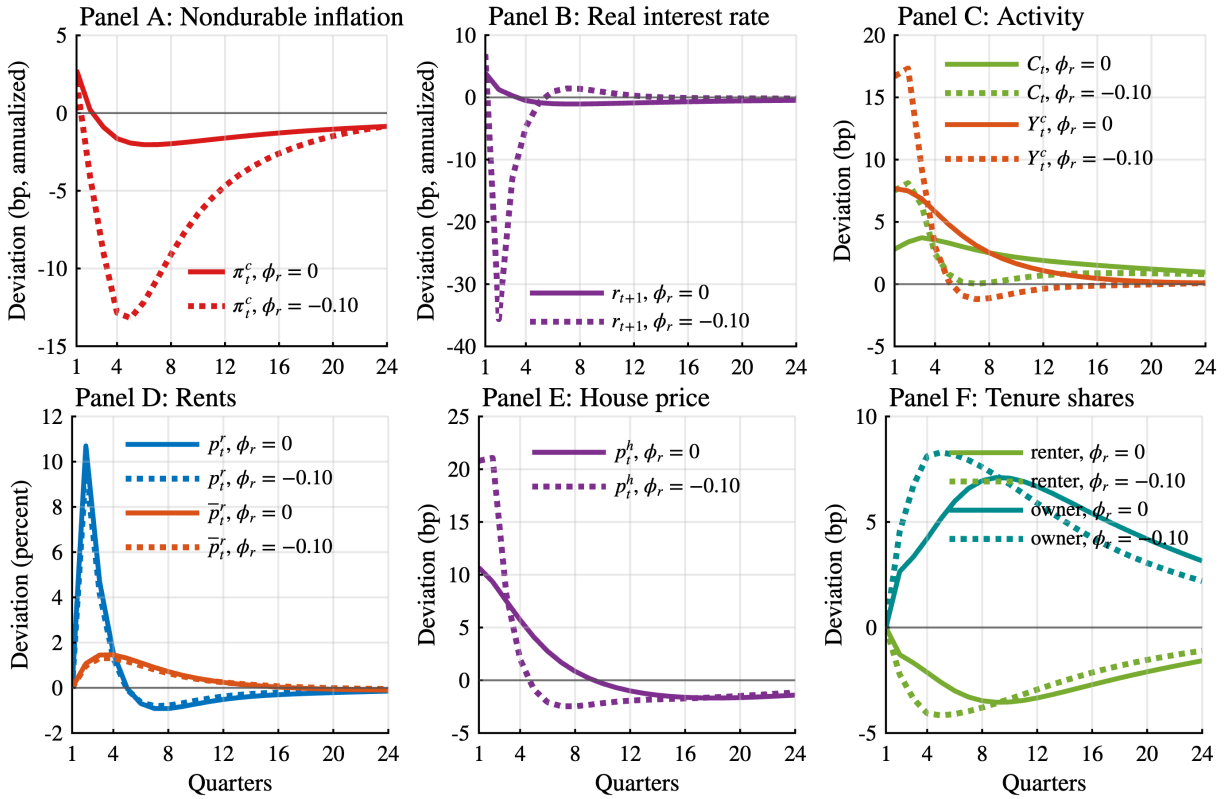


Figure F.5. Rental Availability Shock under Alternative Policy Rules

Note. Responses under $\phi_r = 0$ (solid lines) and the optimized rent-inflation rule $\phi_r = -0.10$ (dotted lines). Panels A and B report nondurable inflation and the real interest rate in annualized basis points. Panel C reports consumption and nondurable output in basis-point deviations from steady state. Panel D reports real market and average rents in percent deviations; Panel E reports relative real house-price deviations in basis points; Panel F reports renter and owner-occupier share deviations in basis points.

F.6 Cost-Push Shock

Figure F.9 compares the responses to the cost-push shock under the benchmark (solid lines) and the optimized rent-inflation rule (dotted lines). The negative rent coefficient dampens the tightening that the rule's response to higher inflation would otherwise impose, which lowers rents and limits the shift from homeownership to renting.

Under the cost-push shock, direct rent relief is the largest source of renters' welfare gains under the rent-inflation rule, ahead of fiscal rebates, unlike under the other two shocks. Wealthier renters still lose overall because the decline in returns on their savings outweighs these gains, while mortgagor-owners gain through lower mortgage costs (Figure F.10).

The rent-level rule strengthens the redistribution through rental payments. Its larger



Figure F.6. Distributional Decomposition of the Welfare Gain: Rental Availability Shock, Rent-Inflation Rule

Note. Per-capita welfare change from moving ϕ_r from 0 to -0.10 , by date-of-shock cohort. Welfare changes are measured in utility units. Cohort definitions, channels, colors, and decomposition conventions follow Figure 8. Black dots report the general-equilibrium total.

reduction in rents raises the rent-channel gains of tenants and the corresponding losses of landlords. Renters with little financial wealth benefit most, while losses on savings continue to outweigh rent relief for the wealthiest renter cohort (Figure F.11).

The choice of target also matters for rental relief under the cost-push shock. The optimized rent-inflation rule lowers the cumulative rental burden by less than half, while the rent-level rule delivers a larger and more persistent reduction in rents, nearly eliminating that burden (Figure F.12).

F.7 Alternative Rules and Fiscal Closures

Figure F.13 decomposes the change in the lump-sum rebate under all three shocks for both optimized rules studied in Section 6. Lower public-debt service provides most of the funding for the subsequent increase in rebates, while lower mortgage-wedge revenues partly offset it. Changes in housing transaction taxes and the rental accounting residual contribute relatively little.

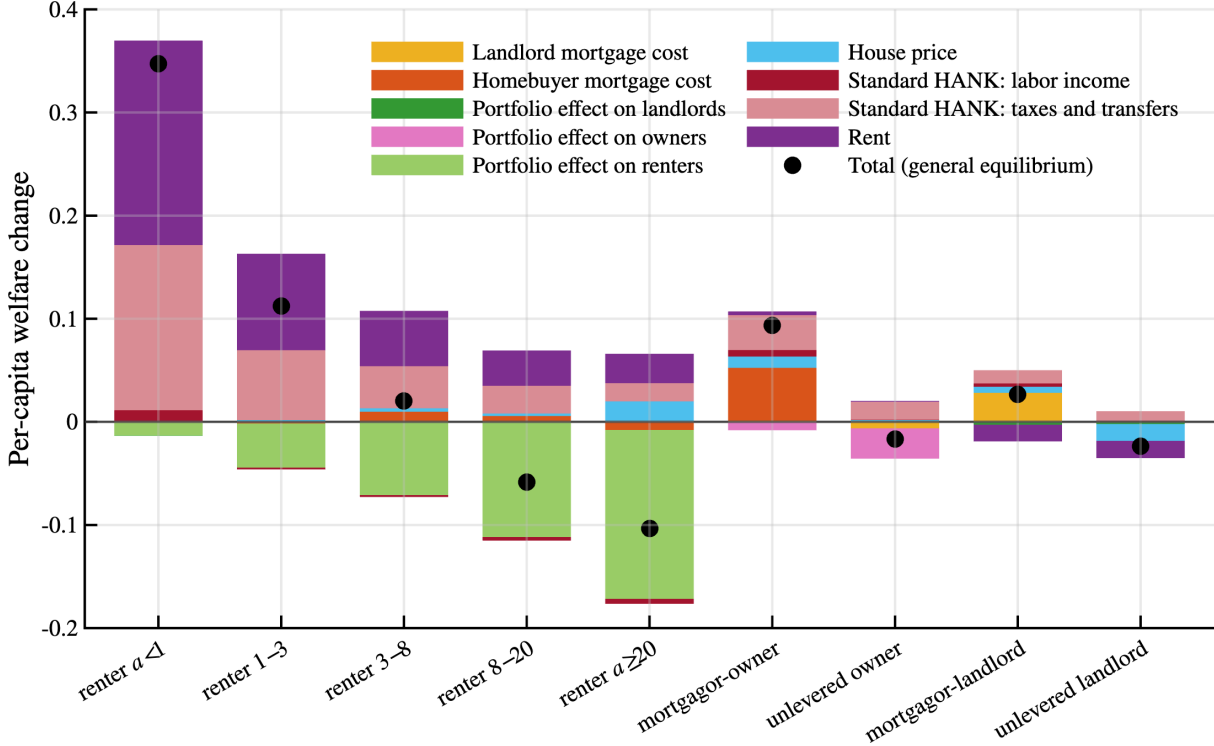


Figure E.7. Distributional Decomposition of the Welfare Gain: Rental Availability Shock, Rent-Level Rule

Note. Per-capita welfare change from moving ϕ_r from 0 to -1.00 , by date-of-shock cohort. Welfare changes are measured in utility units. Cohort definitions, channels, colors, and decomposition conventions follow Figure 8. Black dots report the general-equilibrium total.

This fiscal adjustment redistributes the returns on financial wealth. Interest savings accrue to the government and are returned equally across households, whereas the associated losses of interest income depend on households' asset holdings. The welfare consequences therefore depend on who holds the assets and on the marginal utility of the households receiving the rebate. The figure identifies the sources and timing of the fiscal adjustment; the household welfare decomposition evaluates its incidence.

Alternative fiscal closures. Section 6 checks the optimal rules against two alternative ways of closing the government budget. The first rebates the residual fiscal requirement in proportion to labor productivity rather than lump sum. We re-solve both rules under all three shocks with this closure, and the optimal coefficient remains negative in every case. For the rule on rent inflation, $\phi_r^* = -0.20$ under the demand shock, -0.10 under the rental availability shock, and -0.45 under the cost-push shock; for the rule on the rent level, $\phi_r^* = -0.20$, -0.70 , and below -2.00 . The welfare gains are between 50 and 77 percent of

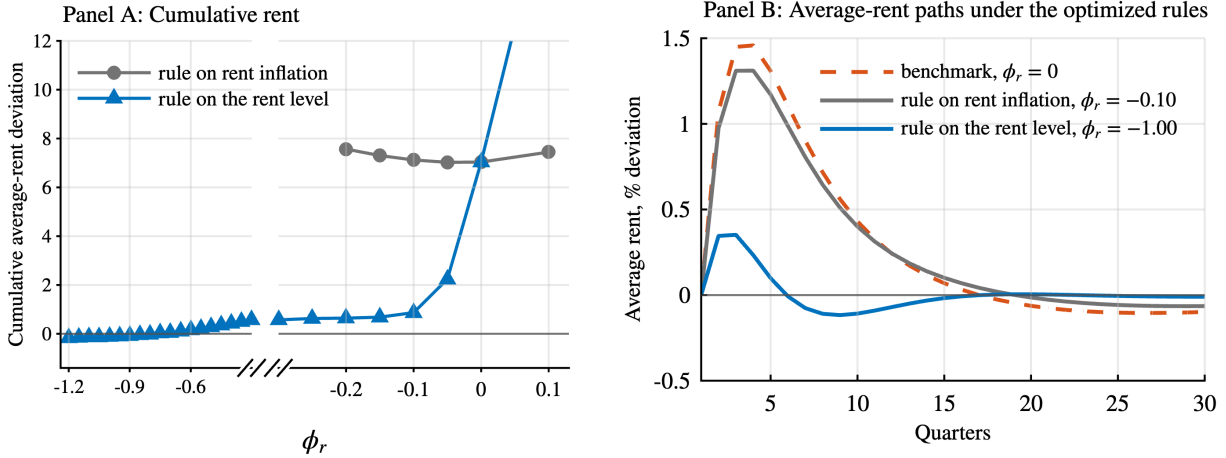


Figure F.8. Average Rents under Alternative Policy Rules: Rental Availability Shock

Note. All rules hold $\phi_\pi = 1.5$. Panel A plots the sum of percentage deviations of the real average rent from steady state over the quarterly transition against the rent coefficient, for rules on nominal rent inflation and on the real rent level. The horizontal axis is broken. Panel B reports the real average rent in percent deviations from steady state under the benchmark ($\phi_r = 0$) and each of the two optimized rules.

their values under the lump-sum rebate under the demand and rental availability shocks, and 91 and 99 percent under the cost-push shock. The second closure holds government spending at its steady-state level and rebates deviations of land income lump sum. At the baseline coefficients of the rule on rent inflation, $\phi_r = -0.25$, $\phi_r = -0.10$, and $\phi_r = -0.45$, the welfare gain remains positive: it is 13 percent larger under the demand shock, 29 percent smaller under the rental availability shock, and 4 percent larger under the cost-push shock than under the baseline closure.

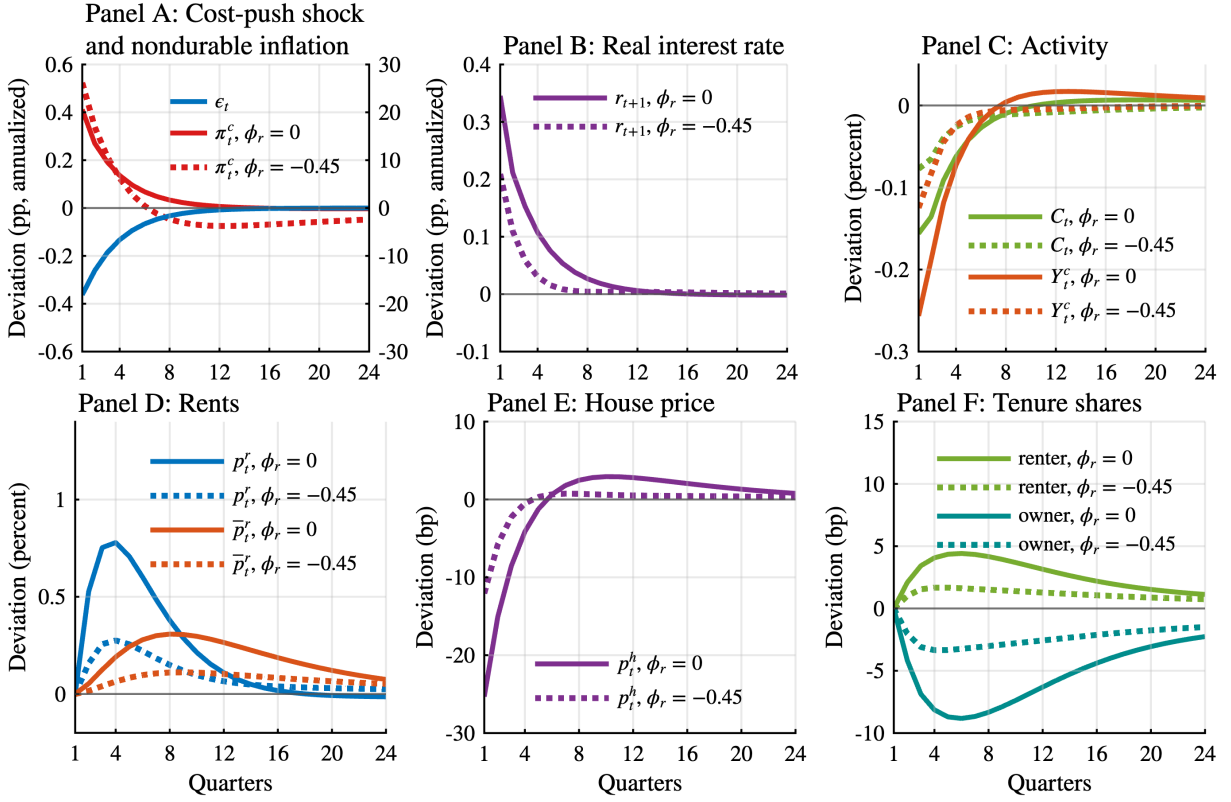


Figure F.9. Cost-Push Shock under Alternative Policy Rules

Note. Responses under $\phi_r = 0$ (solid lines) and the optimized rent-inflation rule $\phi_r = -0.45$ (dotted lines). Panel A reports nondurable inflation in annualized percentage points and, on the right axis, the elasticity ϵ_t in percent deviations from steady state; Panel B reports the real interest rate in annualized percentage points. Panel C reports consumption and nondurable output in percent deviations from steady state. Panel D reports real market and average rents in percent deviations; Panel E reports relative real house-price deviations in basis points; Panel F reports renter and owner-occupier share deviations in basis points.

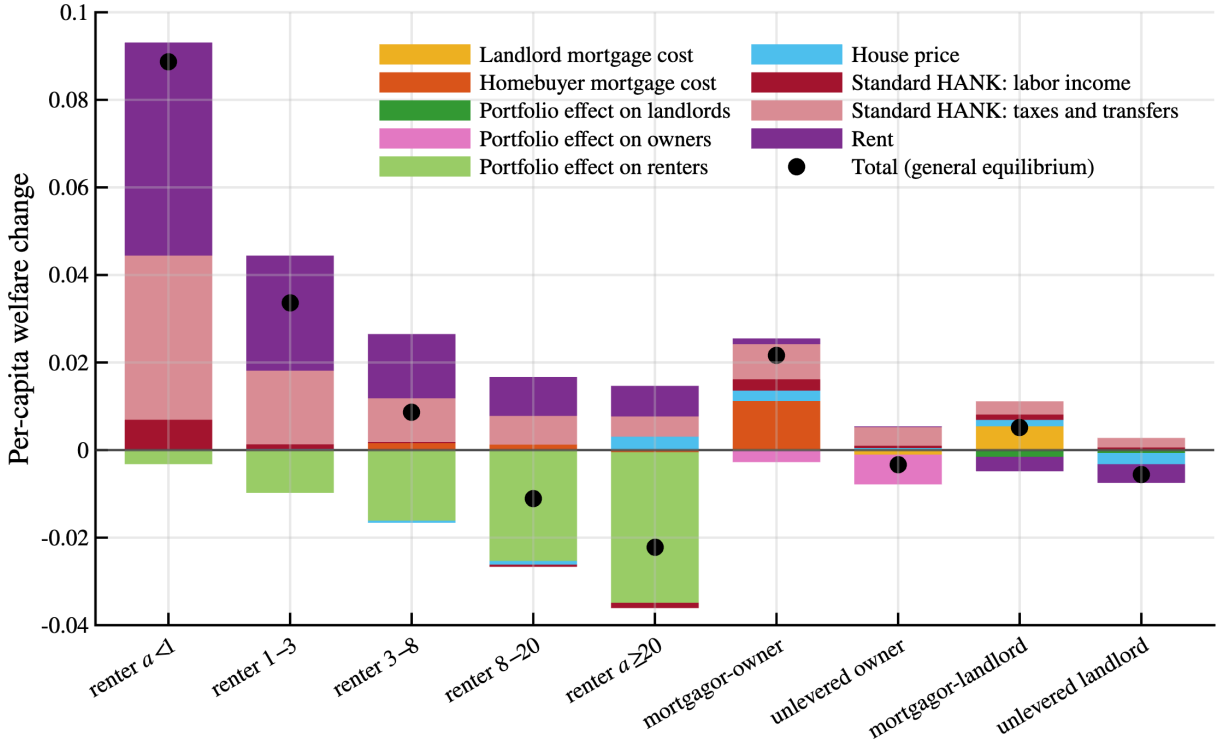


Figure F.10. Distributional Decomposition of the Welfare Gain:
Cost-Push Shock, Rent-Inflation Rule

Note. Per-capita welfare change from moving ϕ_r from 0 to -0.45 , by date-of-shock cohort. Welfare changes are measured in utility units. Cohort definitions, channels, colors, and decomposition conventions follow Figure 8. Black dots report the general-equilibrium total.

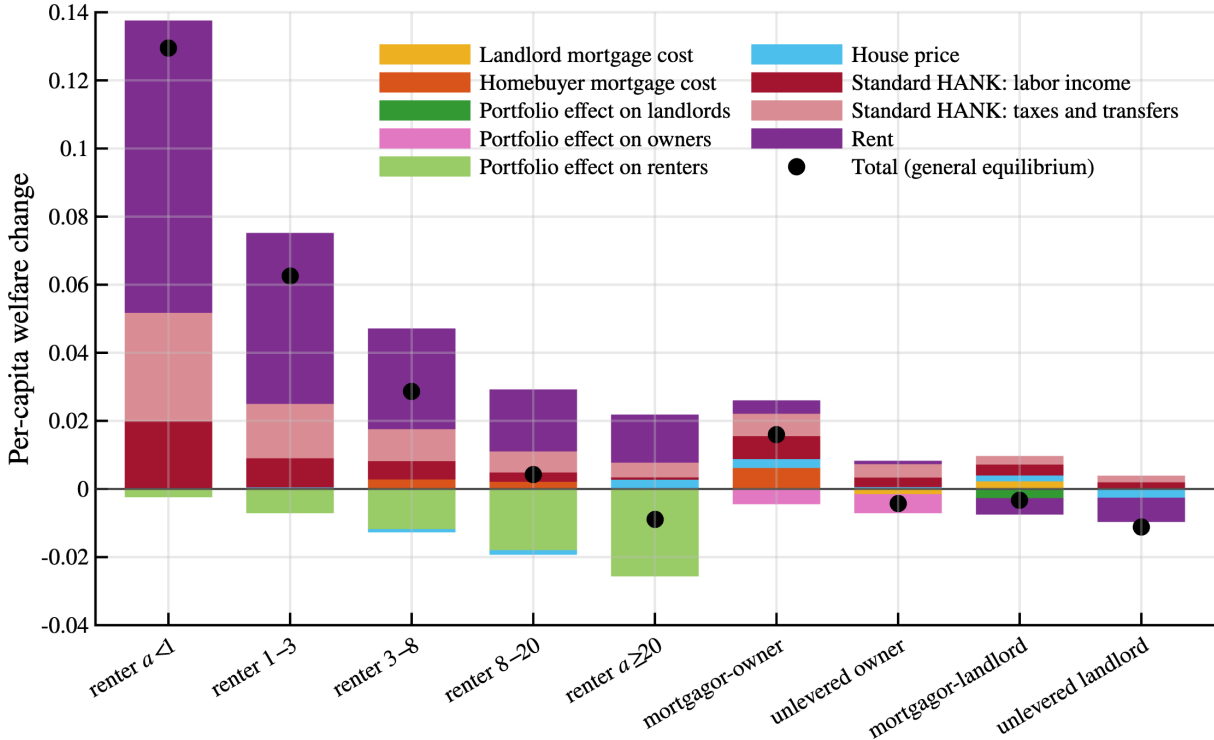


Figure F.11. Distributional Decomposition of the Welfare Gain:
Cost-Push Shock, Rent-Level Rule

Note. Per-capita welfare change from moving ϕ_r from 0 to -2.00 , by date-of-shock cohort. Welfare changes are measured in utility units. Cohort definitions, channels, colors, and decomposition conventions follow Figure 8. Black dots report the general-equilibrium total.

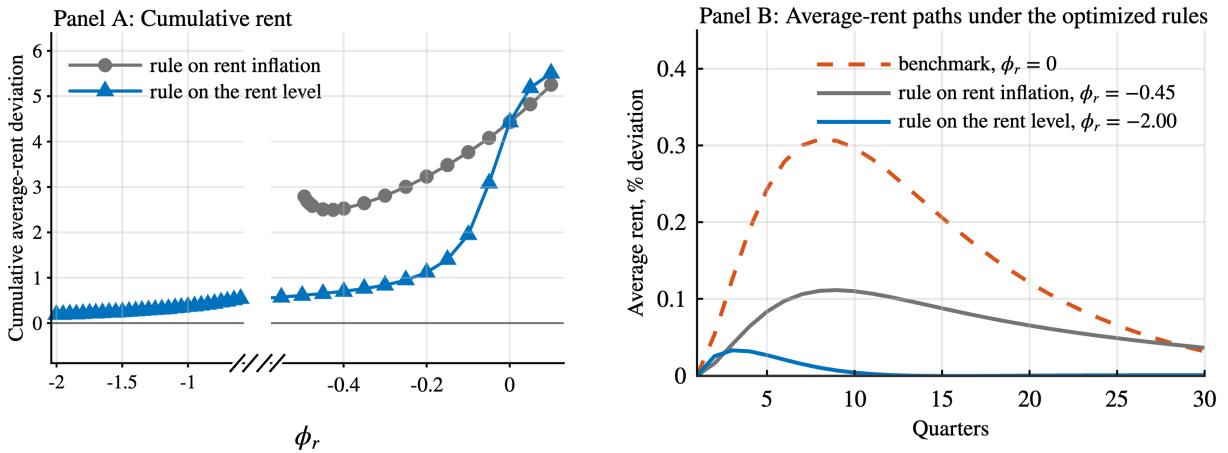


Figure F.12. Average Rents under Alternative Policy Rules: Cost-Push Shock

Note. All rules hold $\phi_\pi = 1.5$. Panel A plots the sum of percentage deviations of the real average rent from steady state over the quarterly transition against the rent coefficient, for rules on nominal rent inflation and on the real rent level. The horizontal axis is broken. Panel B reports the real average rent in percent deviations from steady state under the benchmark ($\phi_r = 0$), the rule on rent inflation at its optimum, and the rule on the rent level at $\phi_r = -2.00$.

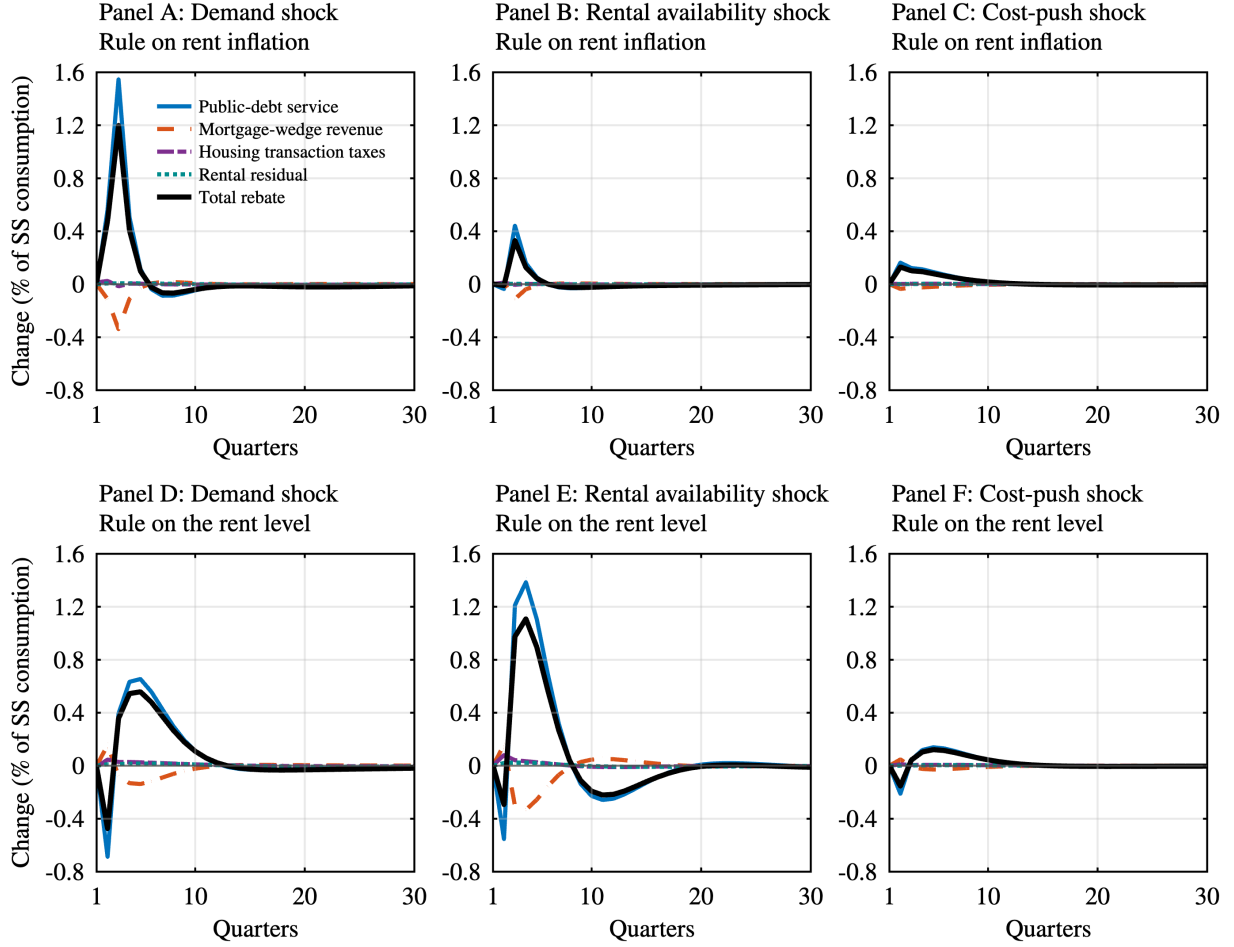


Figure F.13. Decomposition of the Fiscal Adjustment under the Optimized Rules

Note. Differences between each optimized rule and the benchmark $\phi_r = 0$ under the same shock, with lump-sum fiscal adjustment. Panels A–C use the rule on nominal rent inflation, at $\phi_r = -0.25$ (demand), -0.10 (rental availability), and -0.45 (cost push); Panels D–F use the rule on the real rent level, at $\phi_r = -0.20$, -1.00 , and -2.00 . The colored lines report changes in the four components of (A.9): public-debt service $-r_t \bar{B}$, mortgage-wedge revenue $-r_t \kappa A_t^-$ (including the landlord-specific wedge in Panels B and E), housing transaction-tax revenue $\mathcal{T}_t^h$, and the rental residual $-R_t^{\text{net}}$. Positive contributions increase the rebate. The black line reports the change in the total rebate $\bar{T}_t$. All quantities are expressed as percentages of steady-state aggregate consumption. Debt service uses the return on assets carried into each quarter.